

SQANTI3 report

Unique Genes: 4293

Unique Isoforms: 4950

Transcript Classification

Category	Isoforms, count
FSM	4622
ISM	92
NIC	123
NNC	88
Genic Genomic	10
Antisense	4
Fusion	8
Intergenic	3
Genic Intron	0

Gene Classification

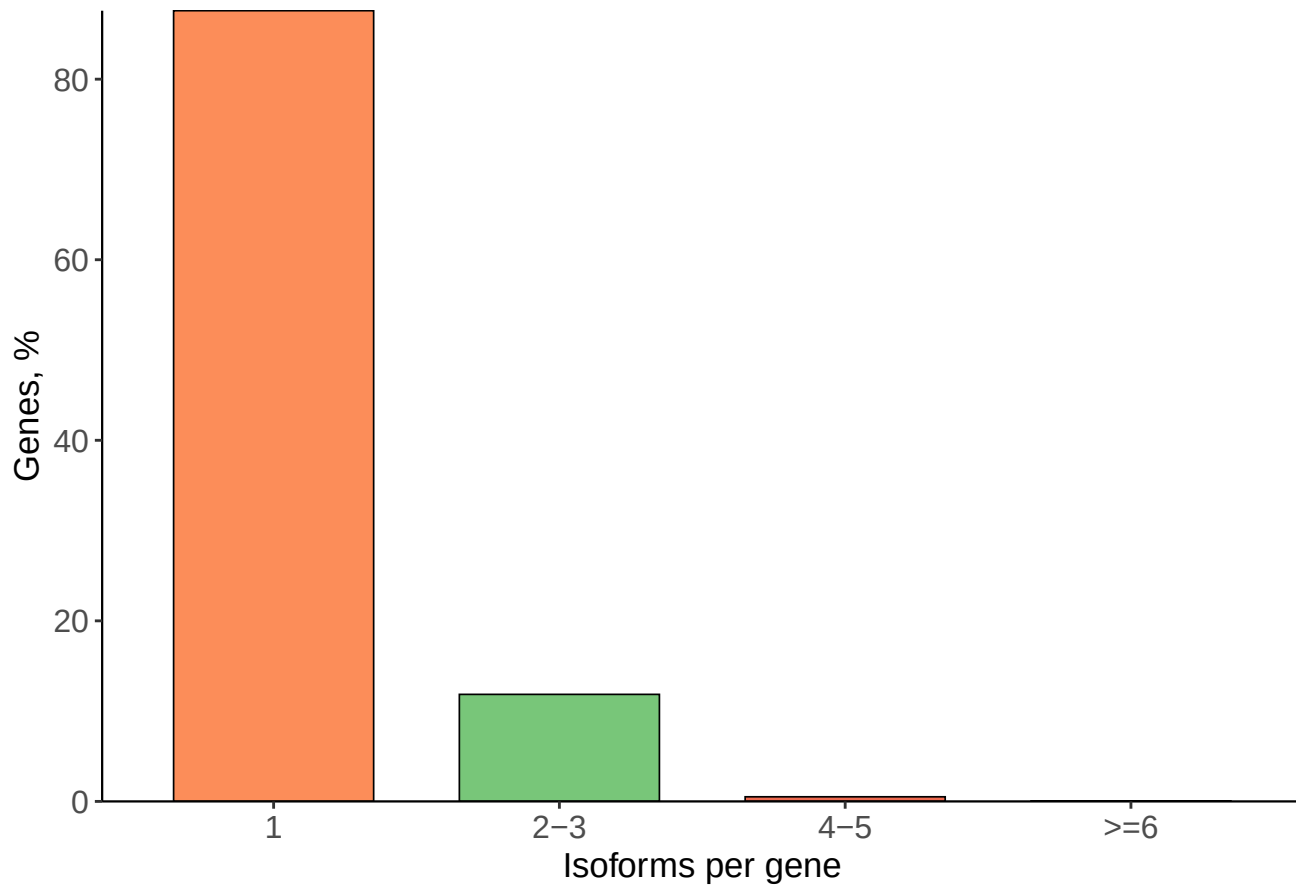
Category	Genes, count
Annotated Genes	4286
Novel Genes	7

Splice Junction Classification

Category	SJs, count	Percent
Known canonical	33333	99.52
Known Non-canonical	7	0.02
Novel canonical	133	0.40
Novel Non-canonical	22	0.07

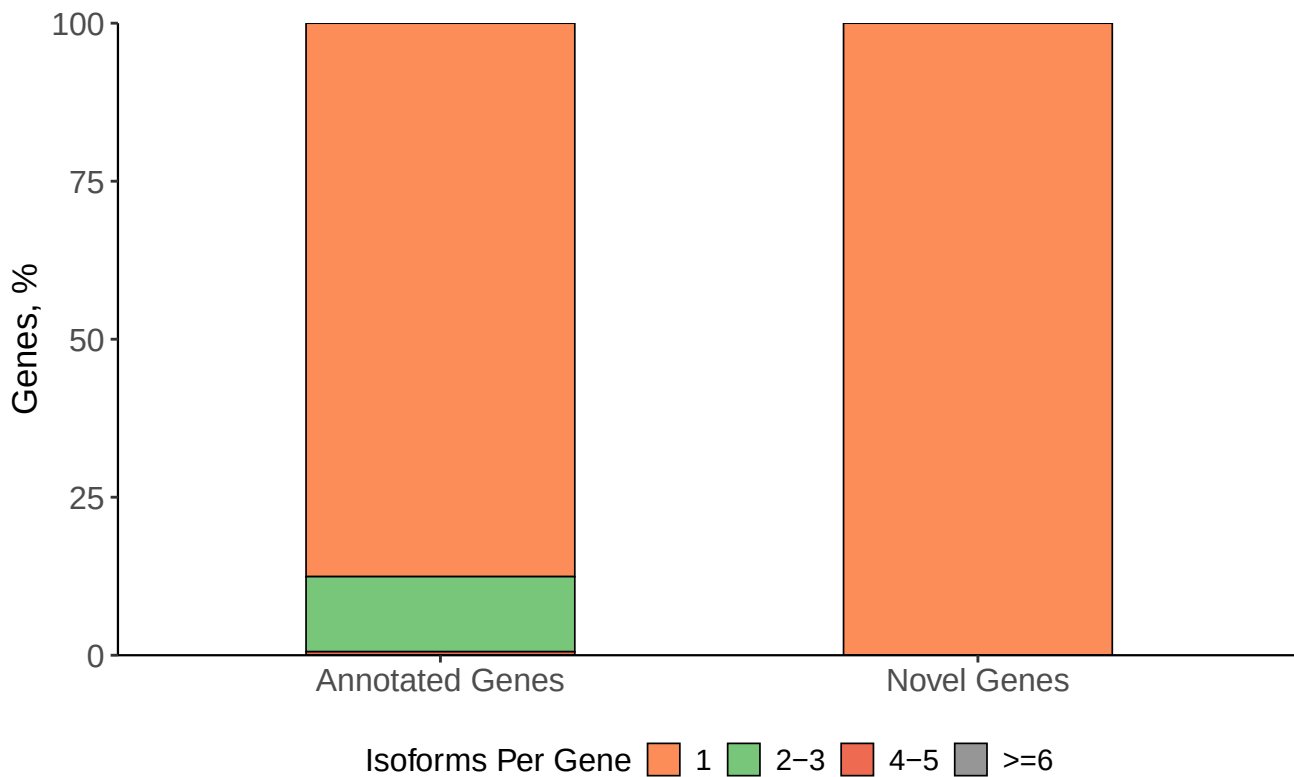
Gene Characterization

Number of Isoforms per Gene

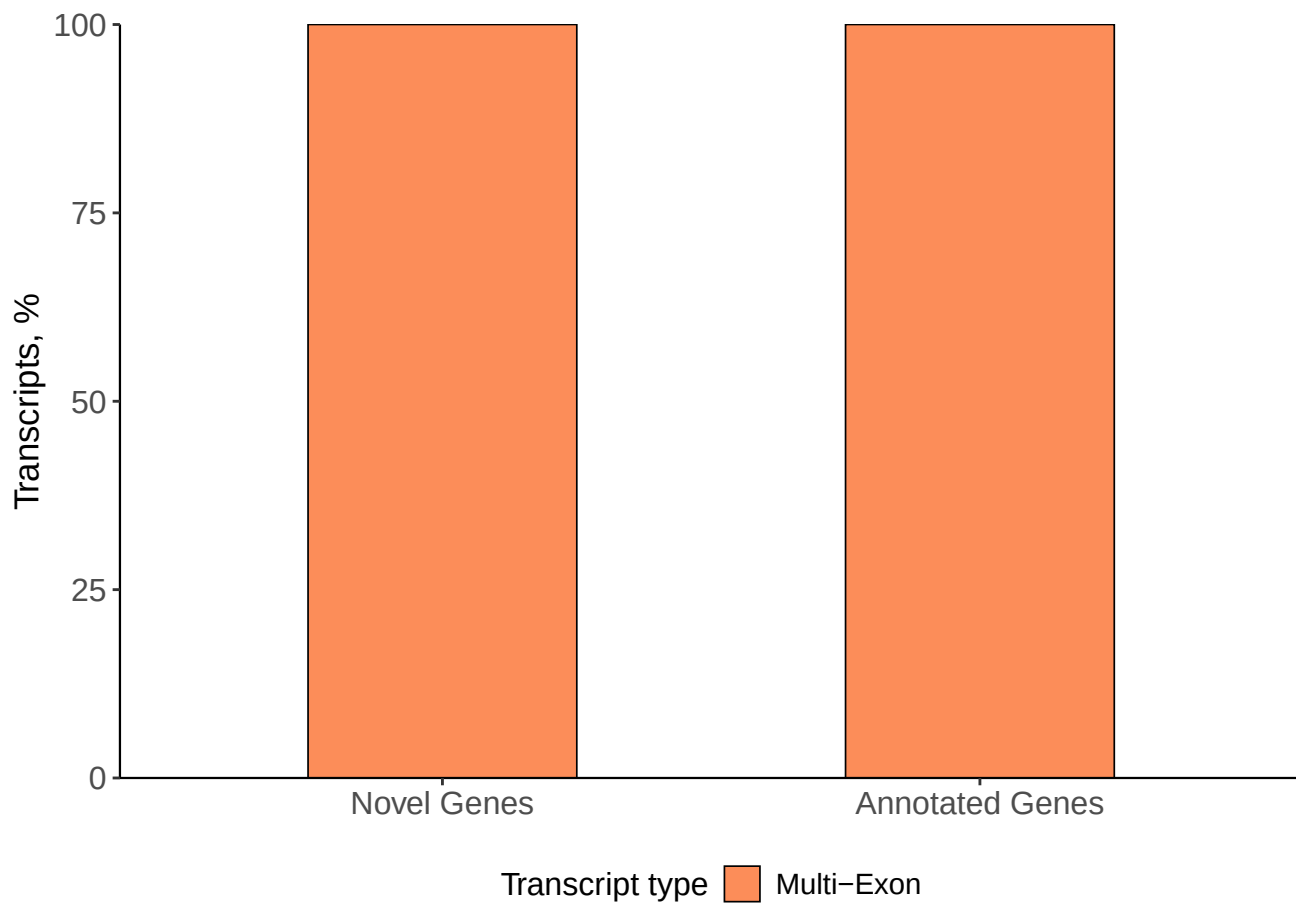


Number of Isoforms per Gene

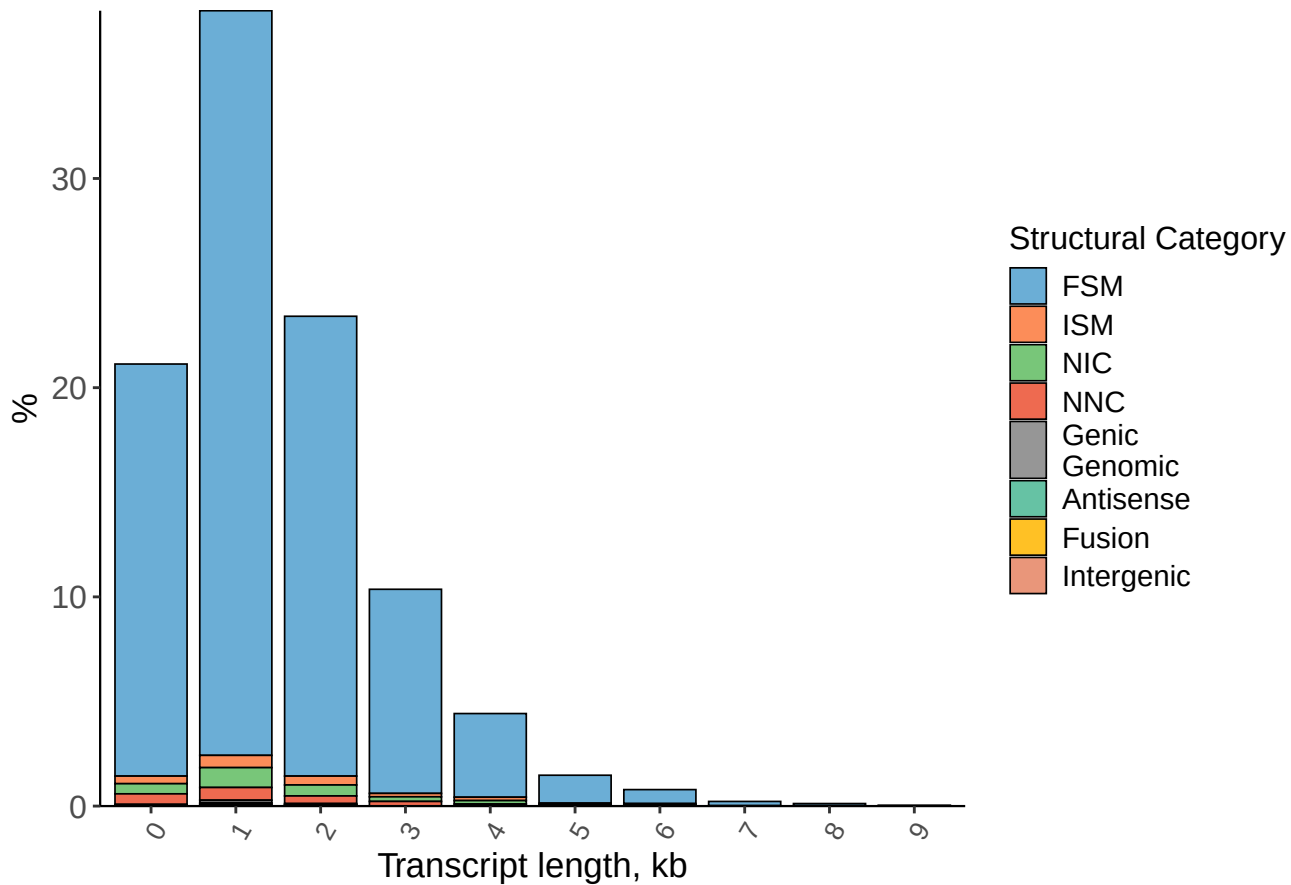
Known vs Novel Genes



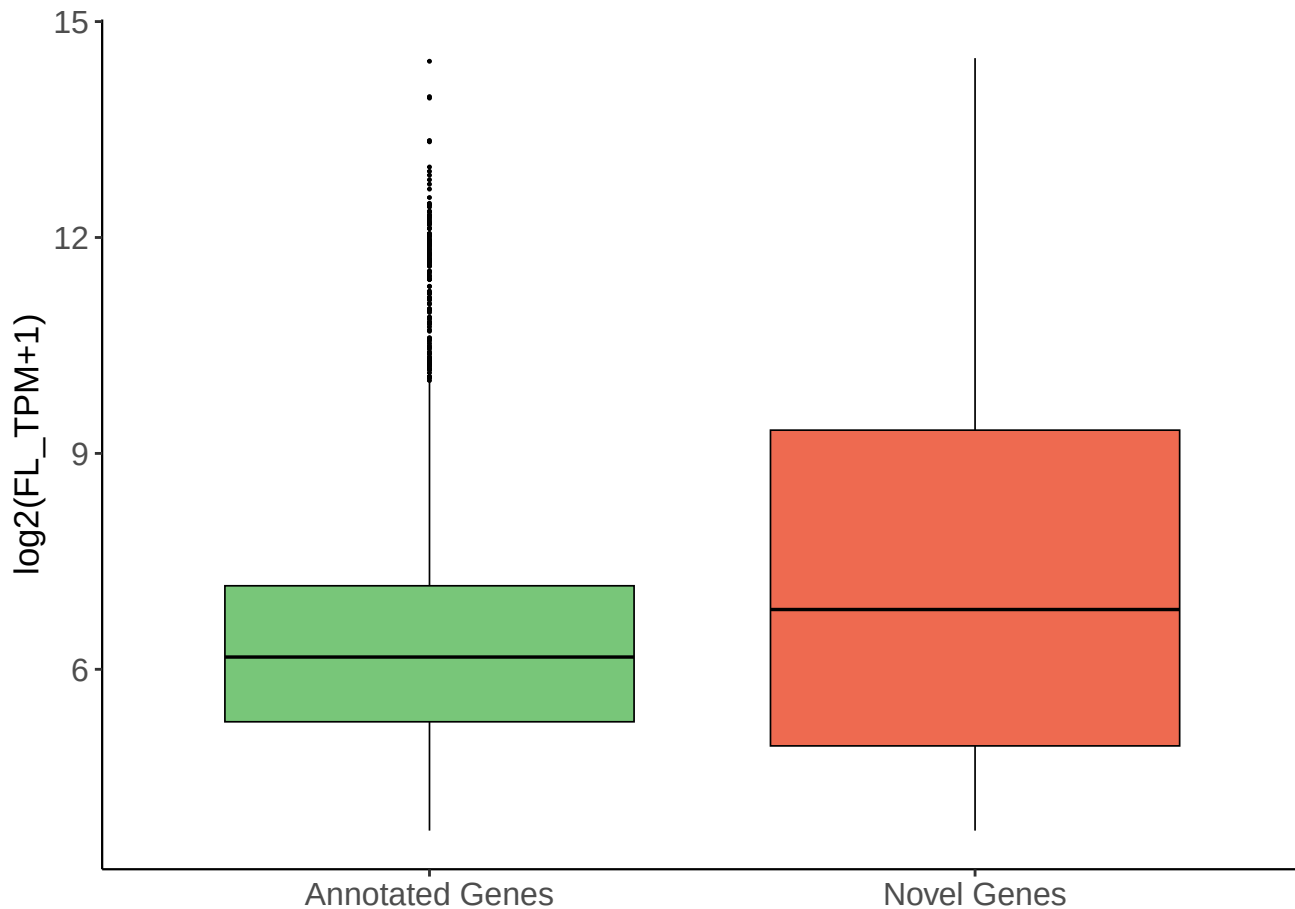
Distribution of Mono- vs Multi-Exon Transcripts



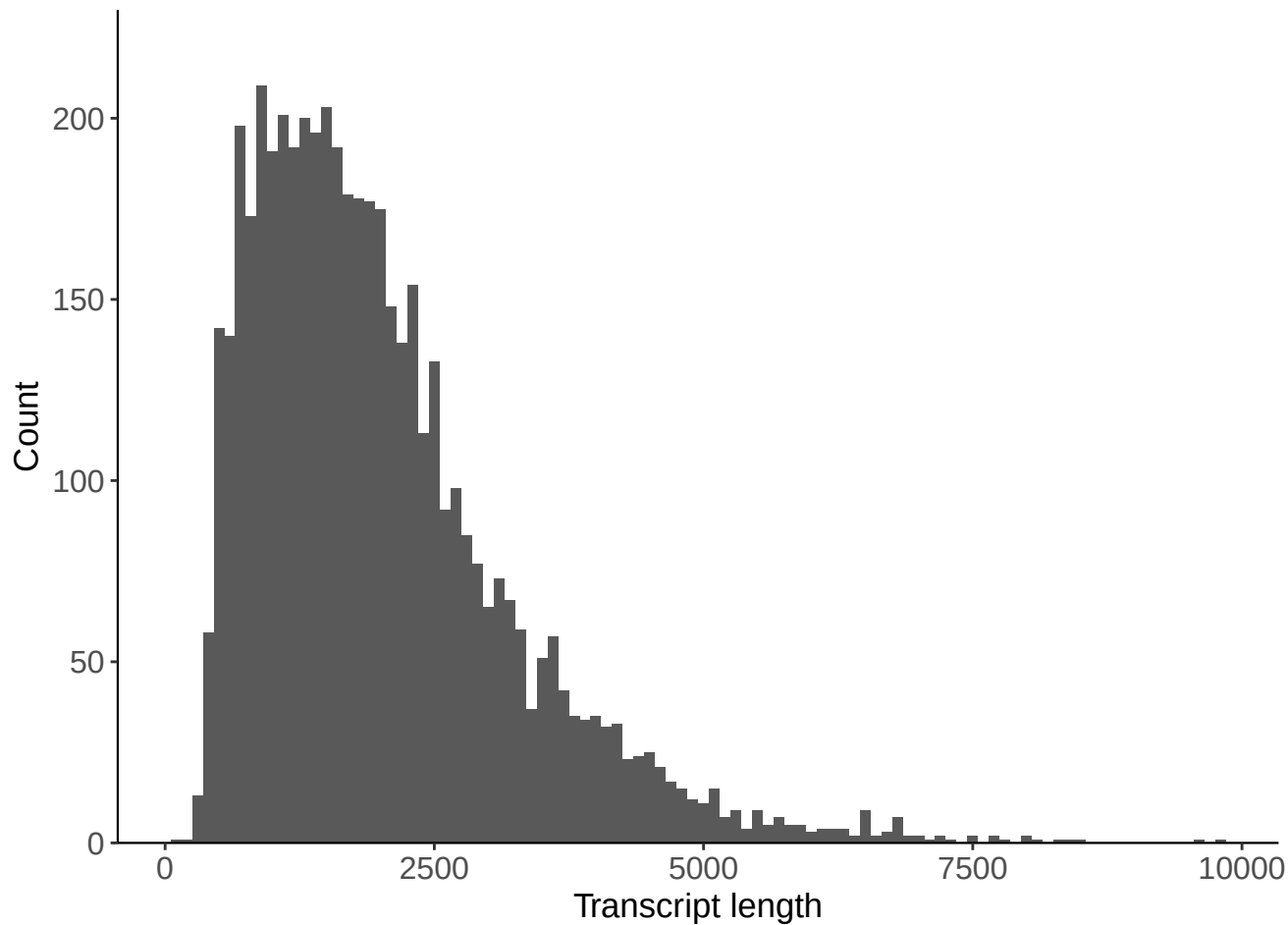
Structural Categories by Transcript Length



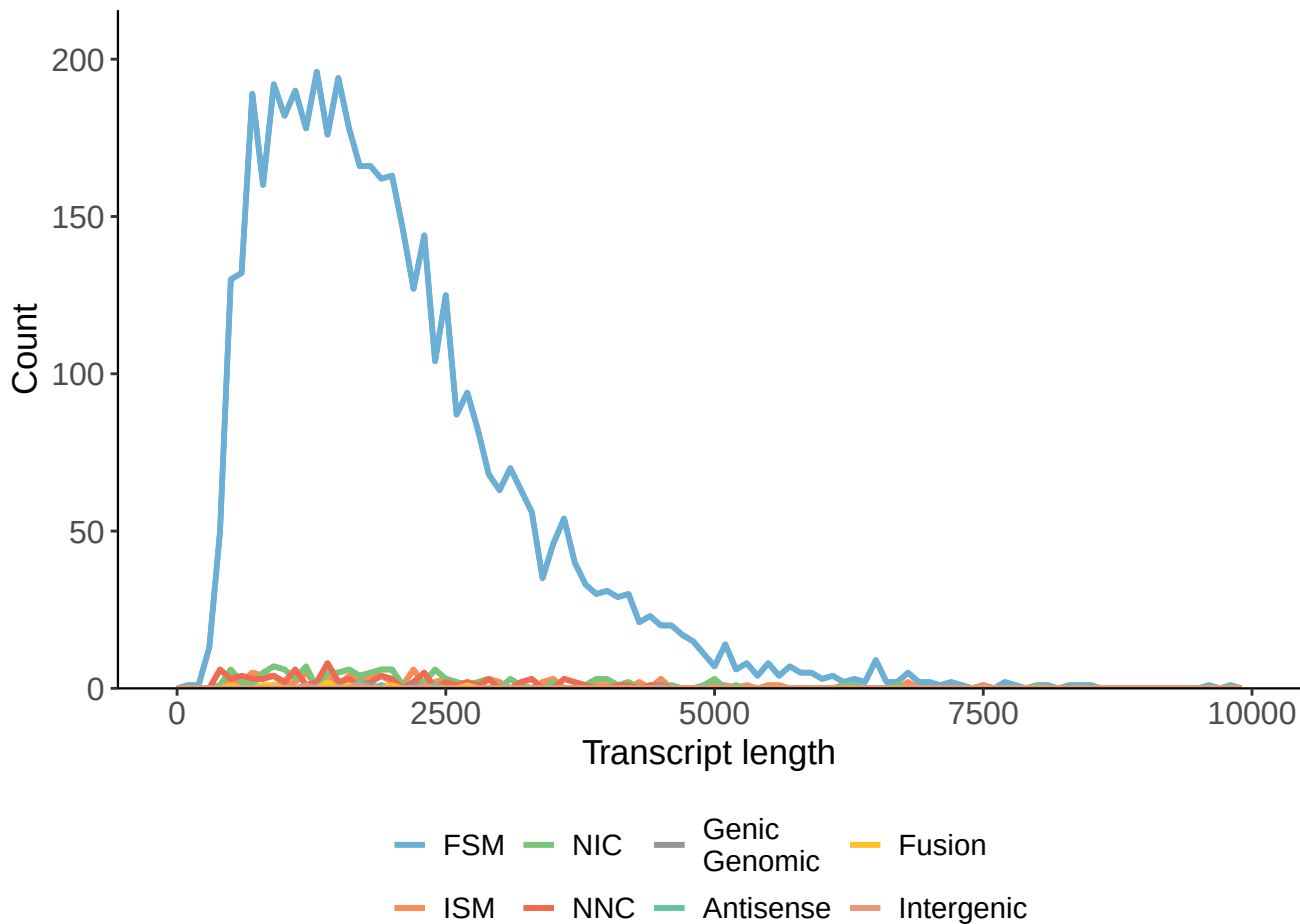
Number of FL reads per Gene by Type of Gene Annotation



All Transcript Lengths Distribution



Transcript Lengths Distribution by Structural Category

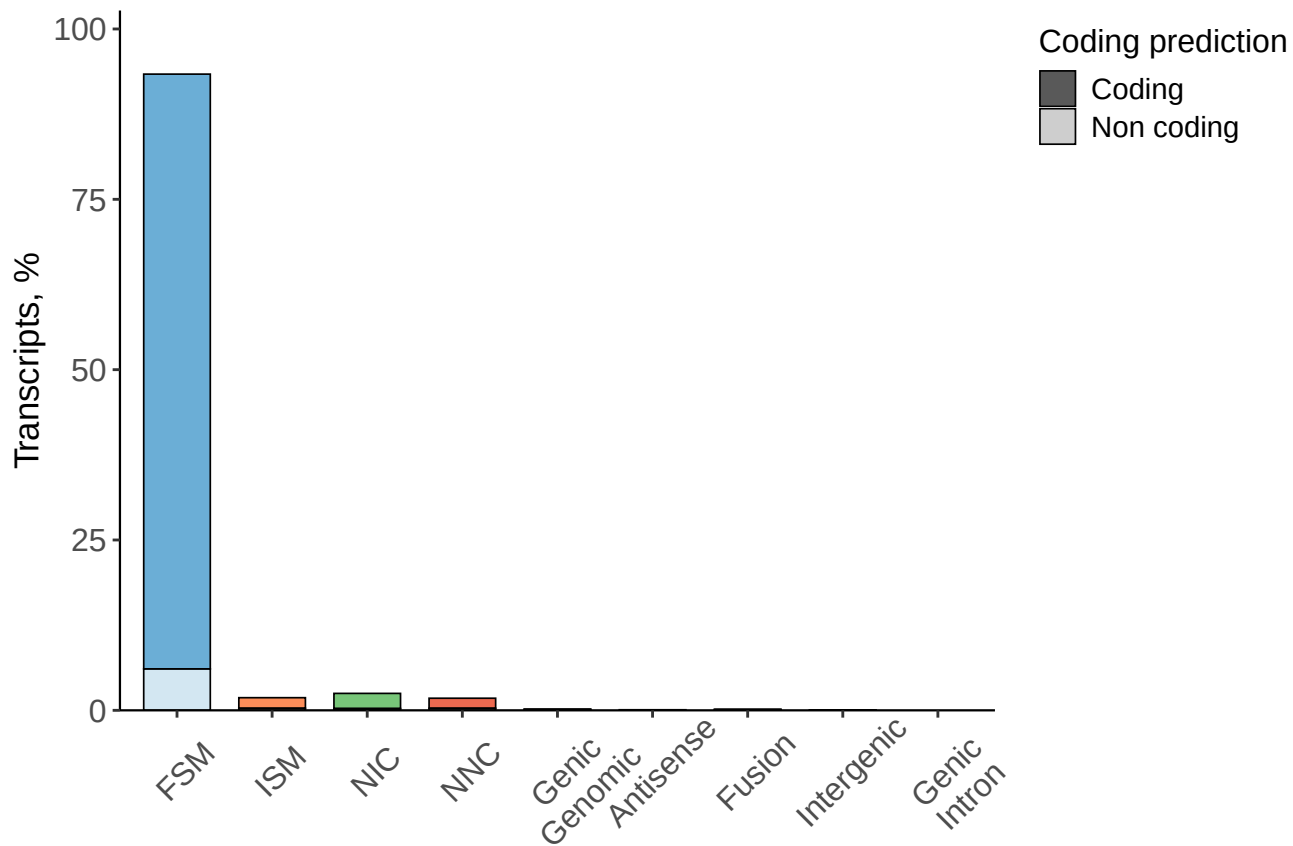


Mono- vs Multi- Exon Transcript Lengths Distribution

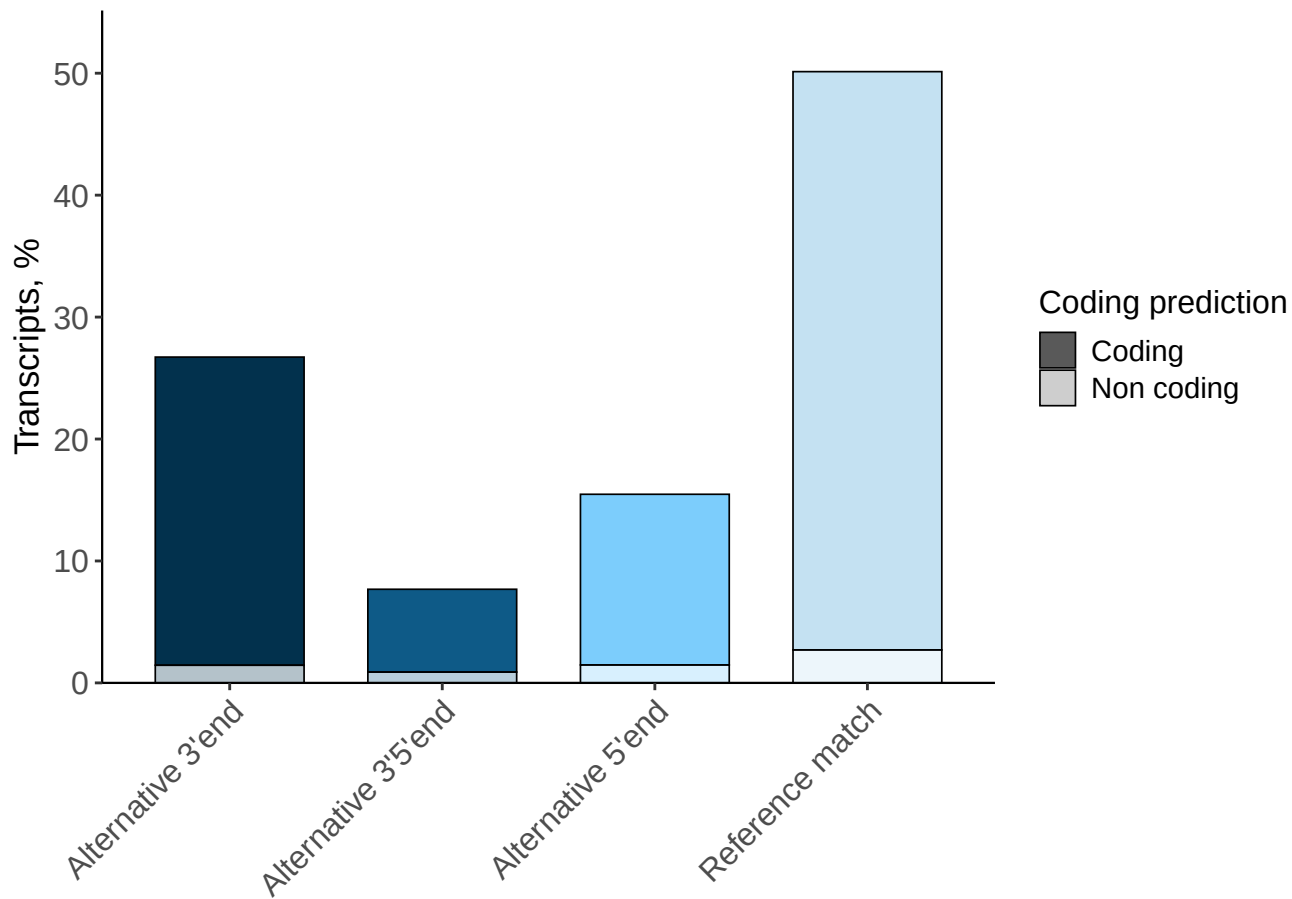


Structural Isoform Characterization

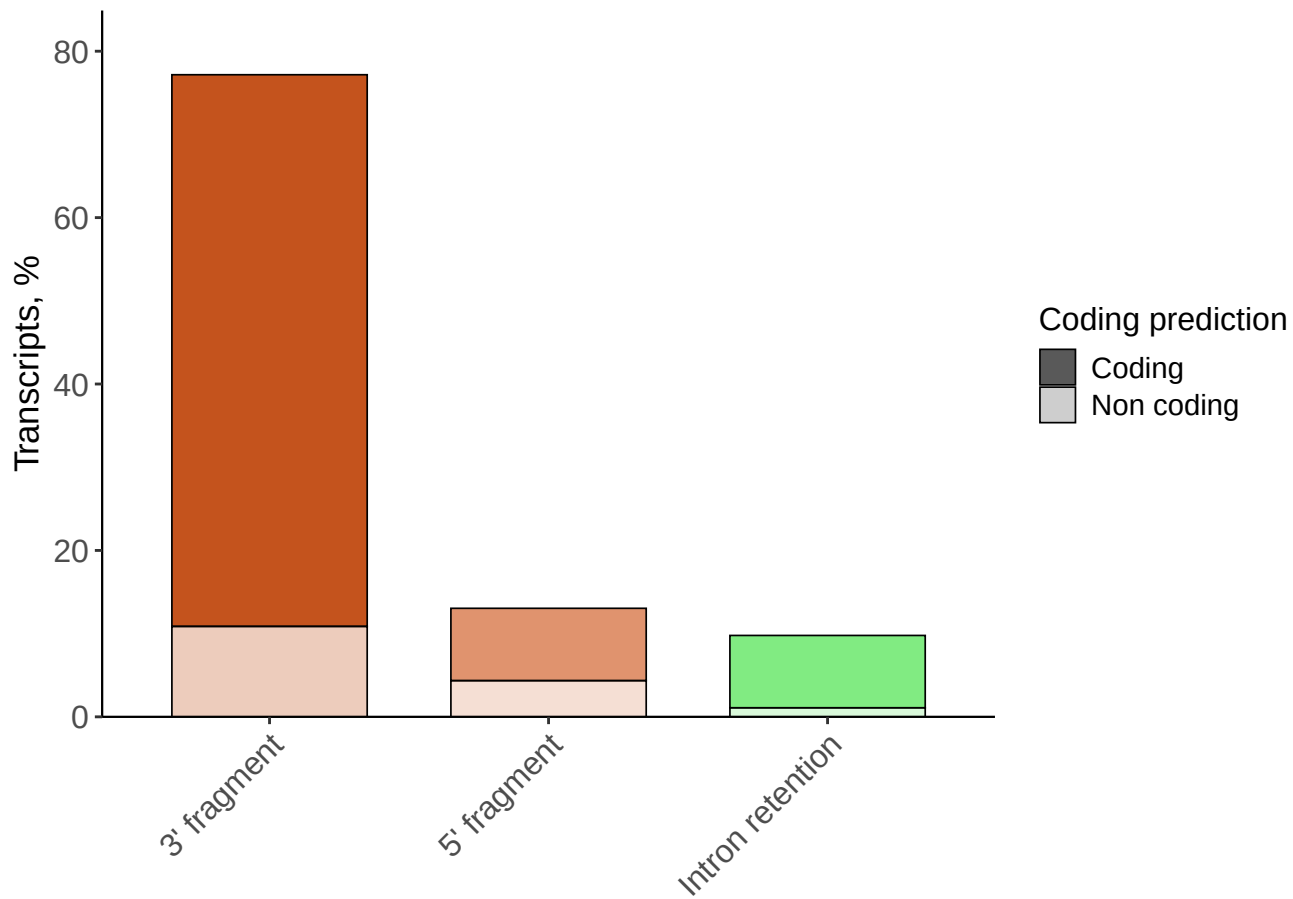
Isoform Distribution Across Structural Categories



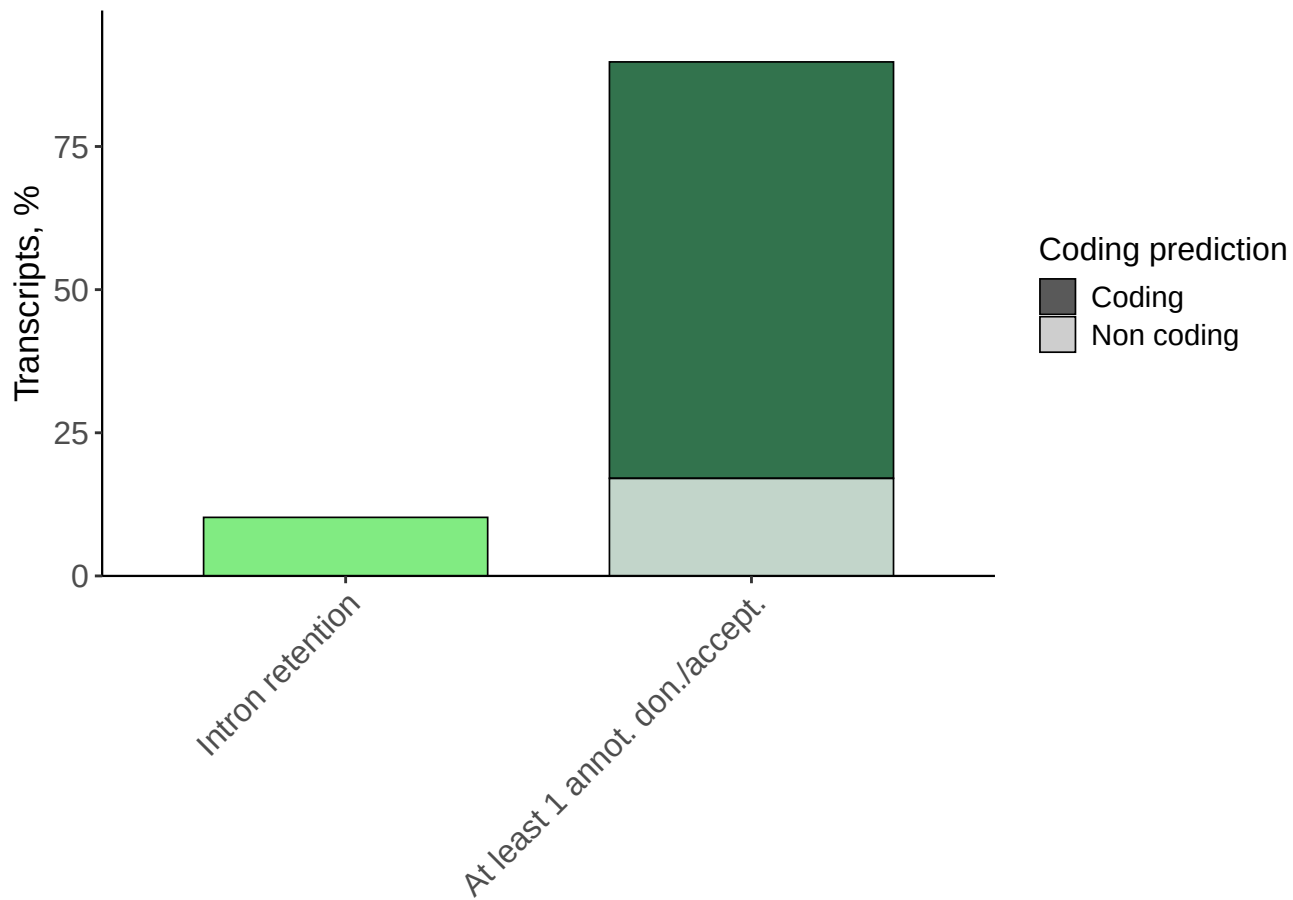
Isoform Distribution Across FSM



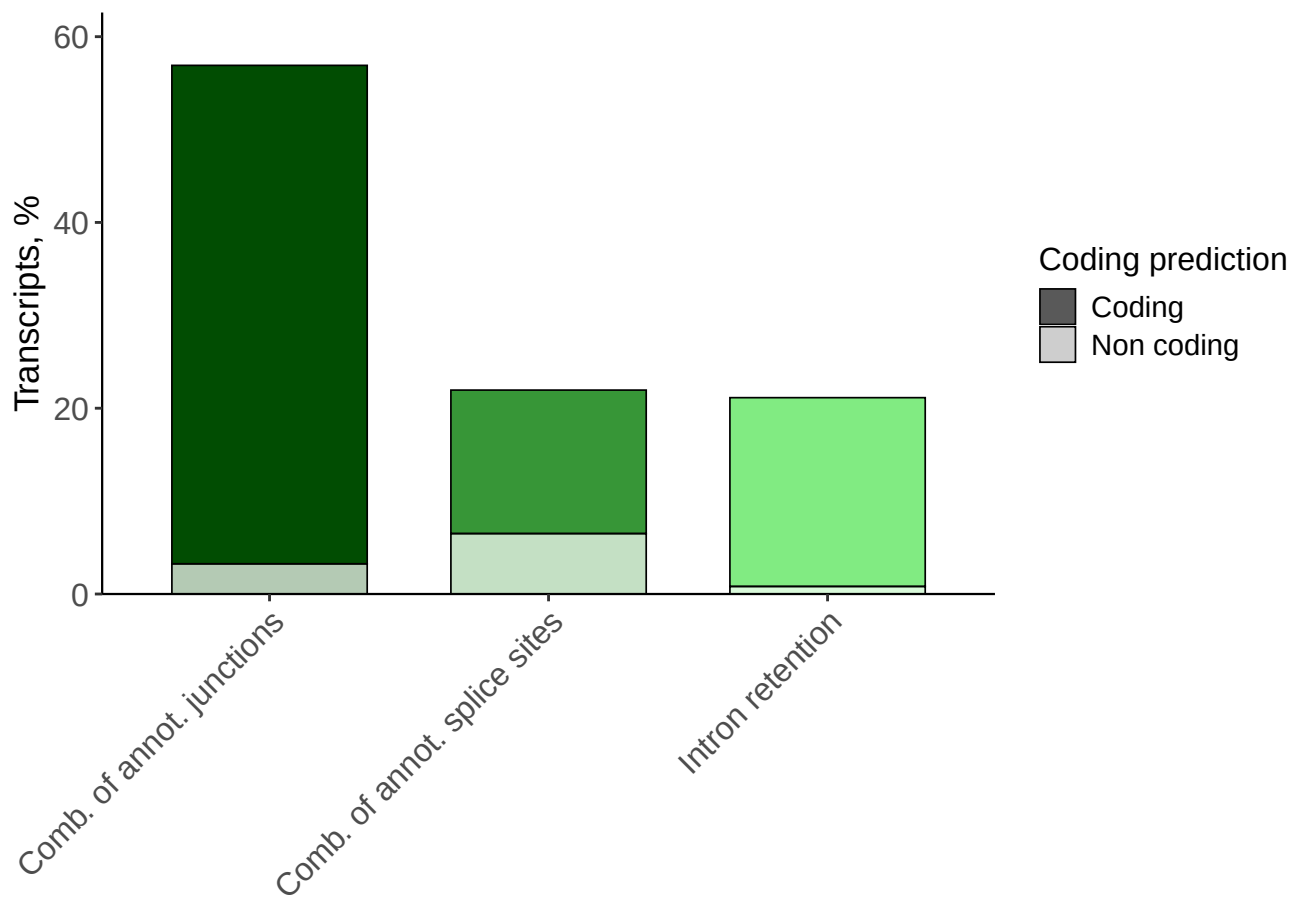
Isoform Distribution Across ISM



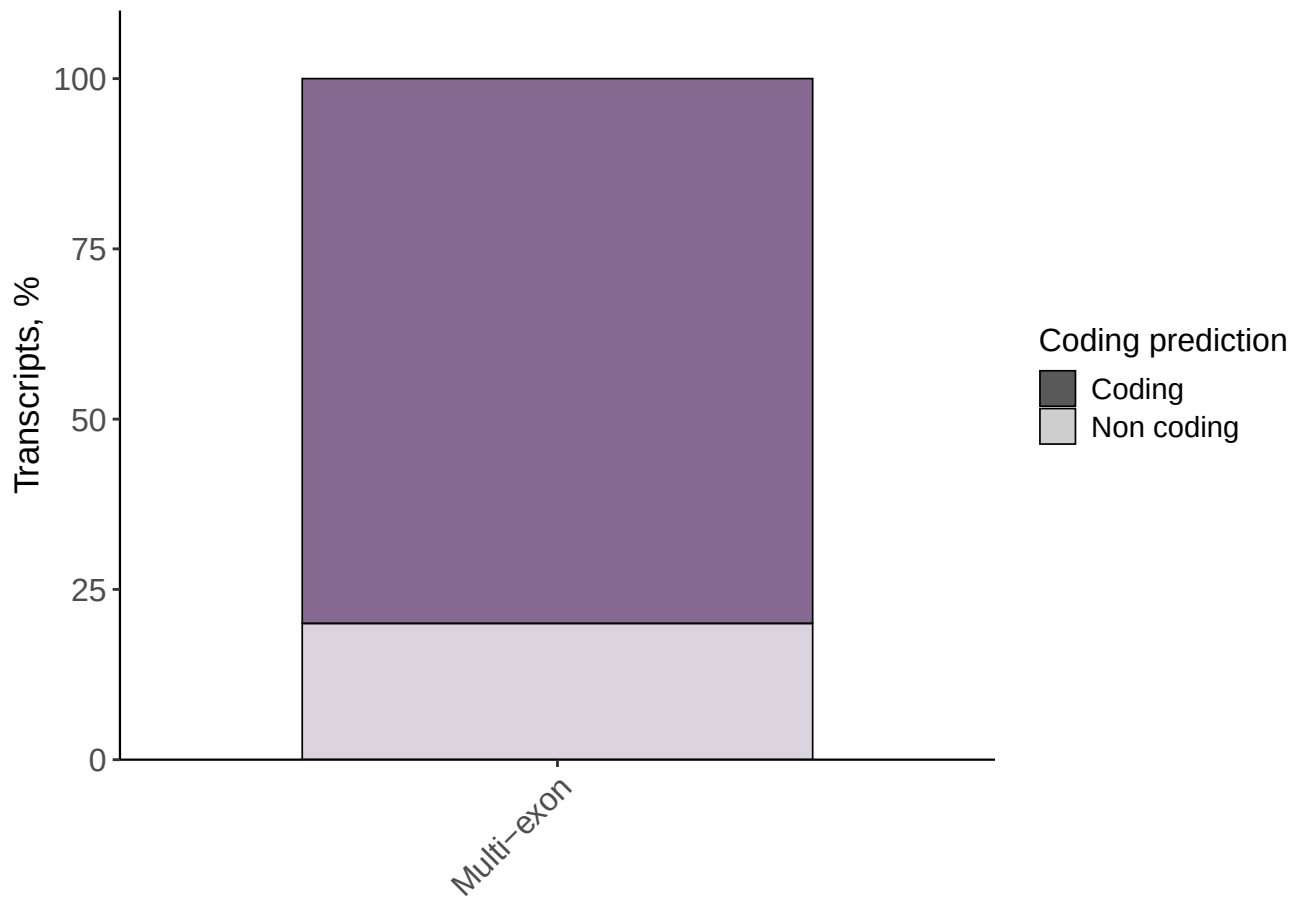
Isoform Distribution Across NNC



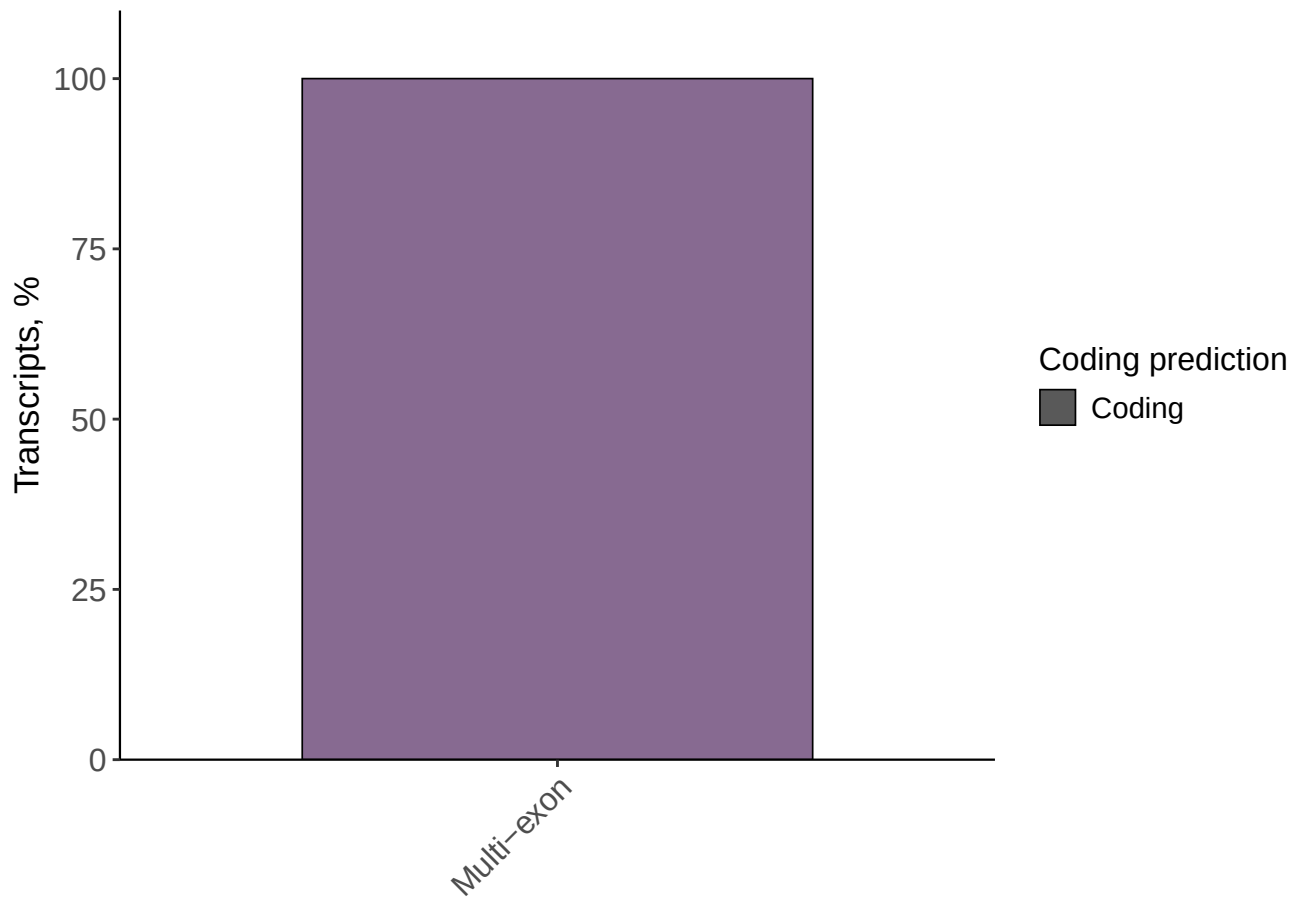
Isoform Distribution Across NIC



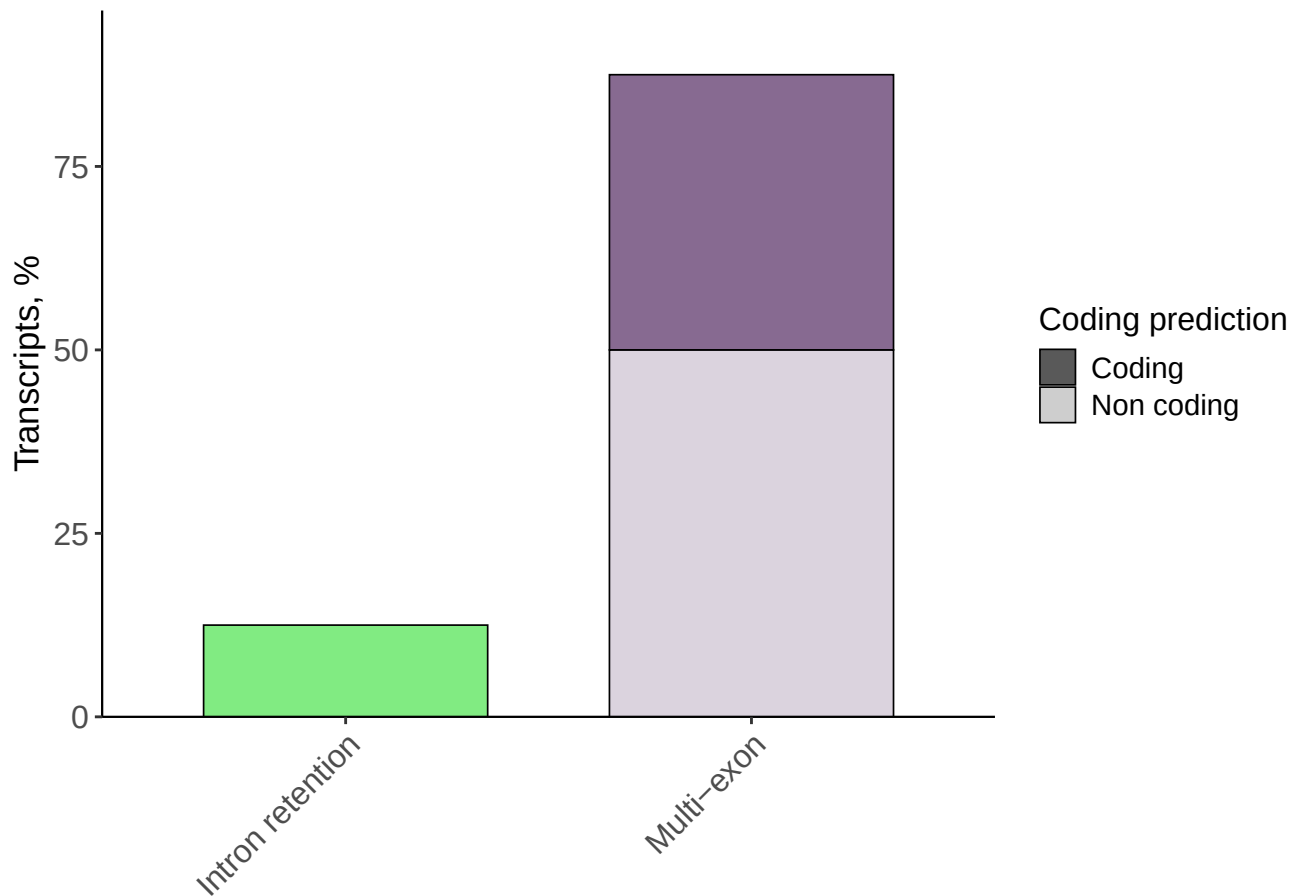
Isoform Distribution Across Genic Genomic



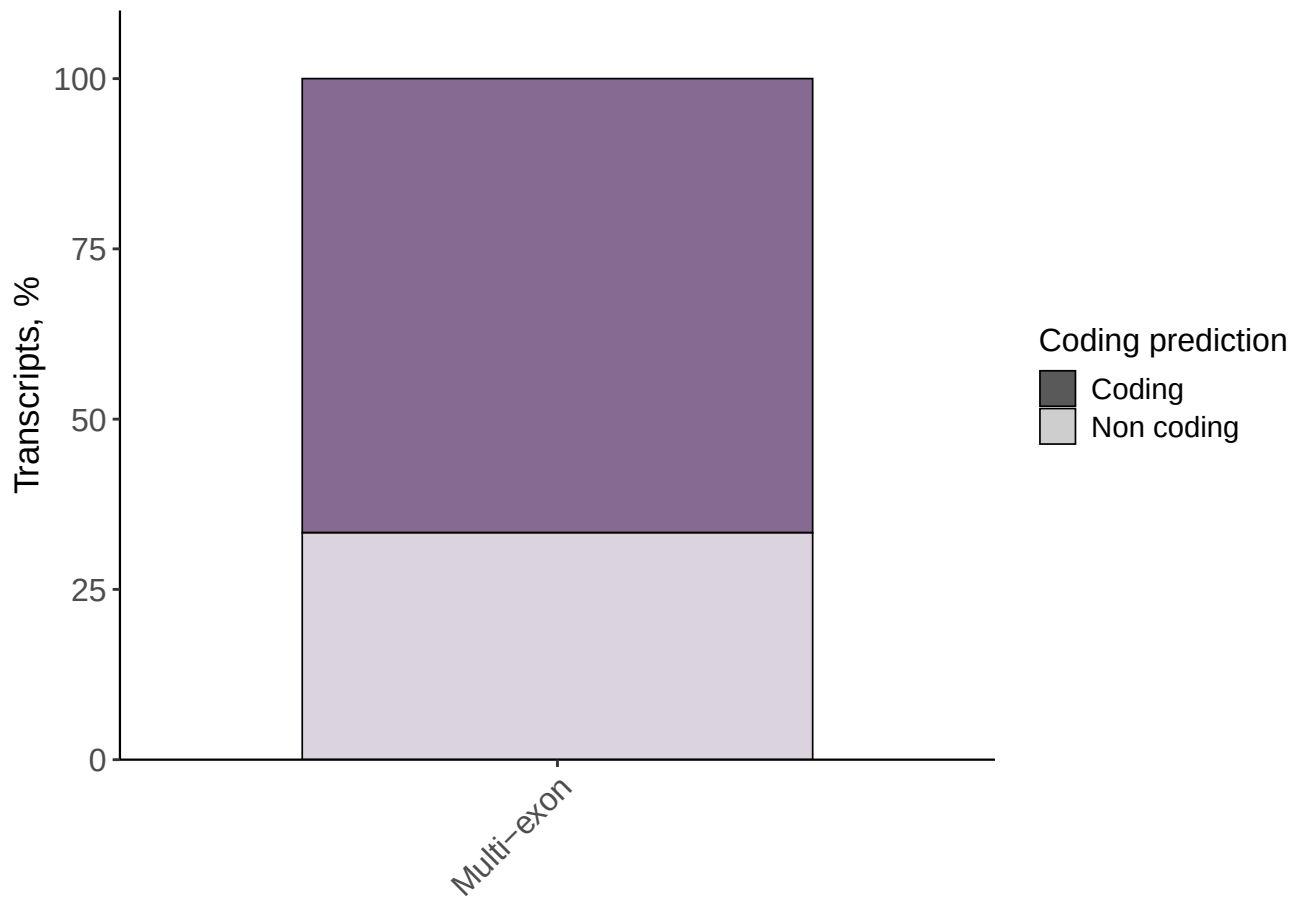
Isoform Distribution Across Antisense



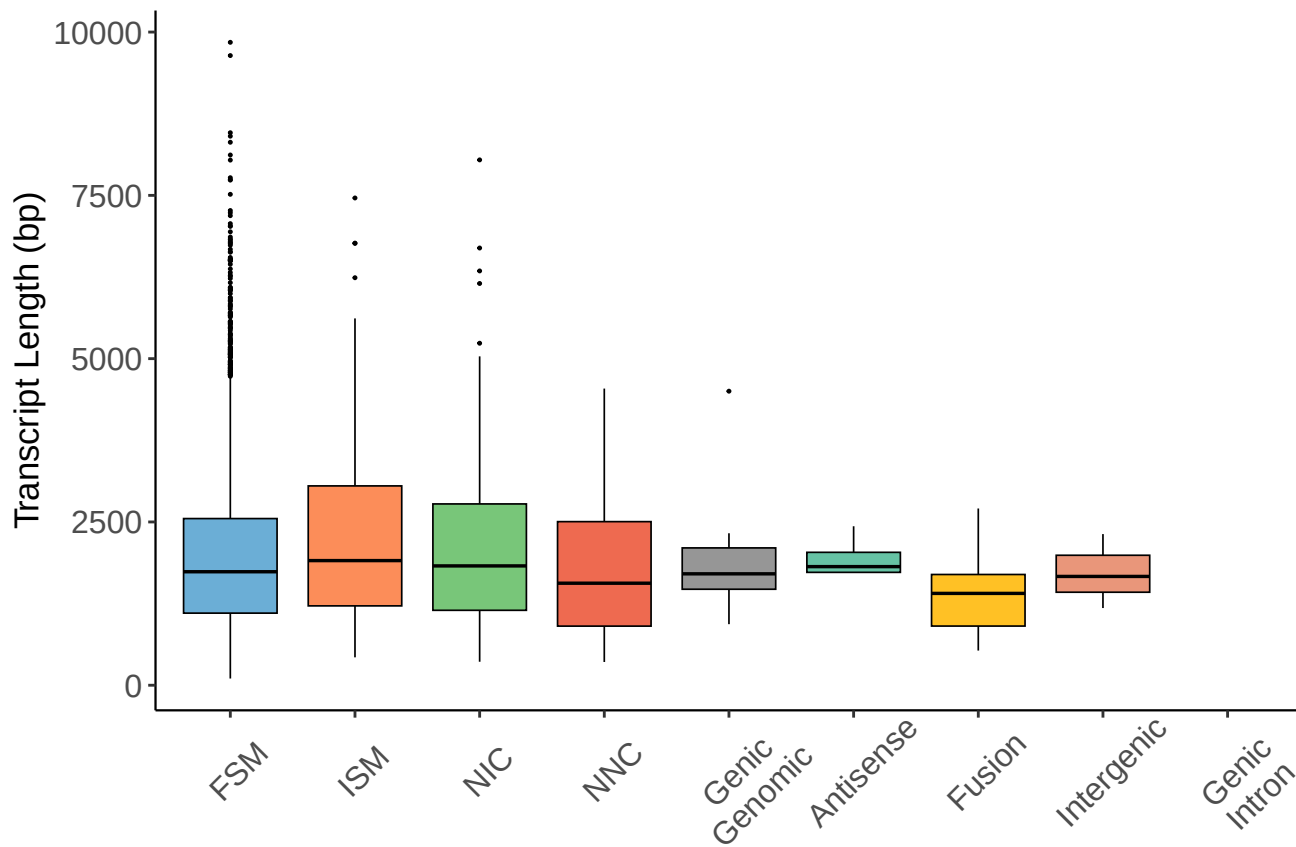
Isoform Distribution Across Fusion



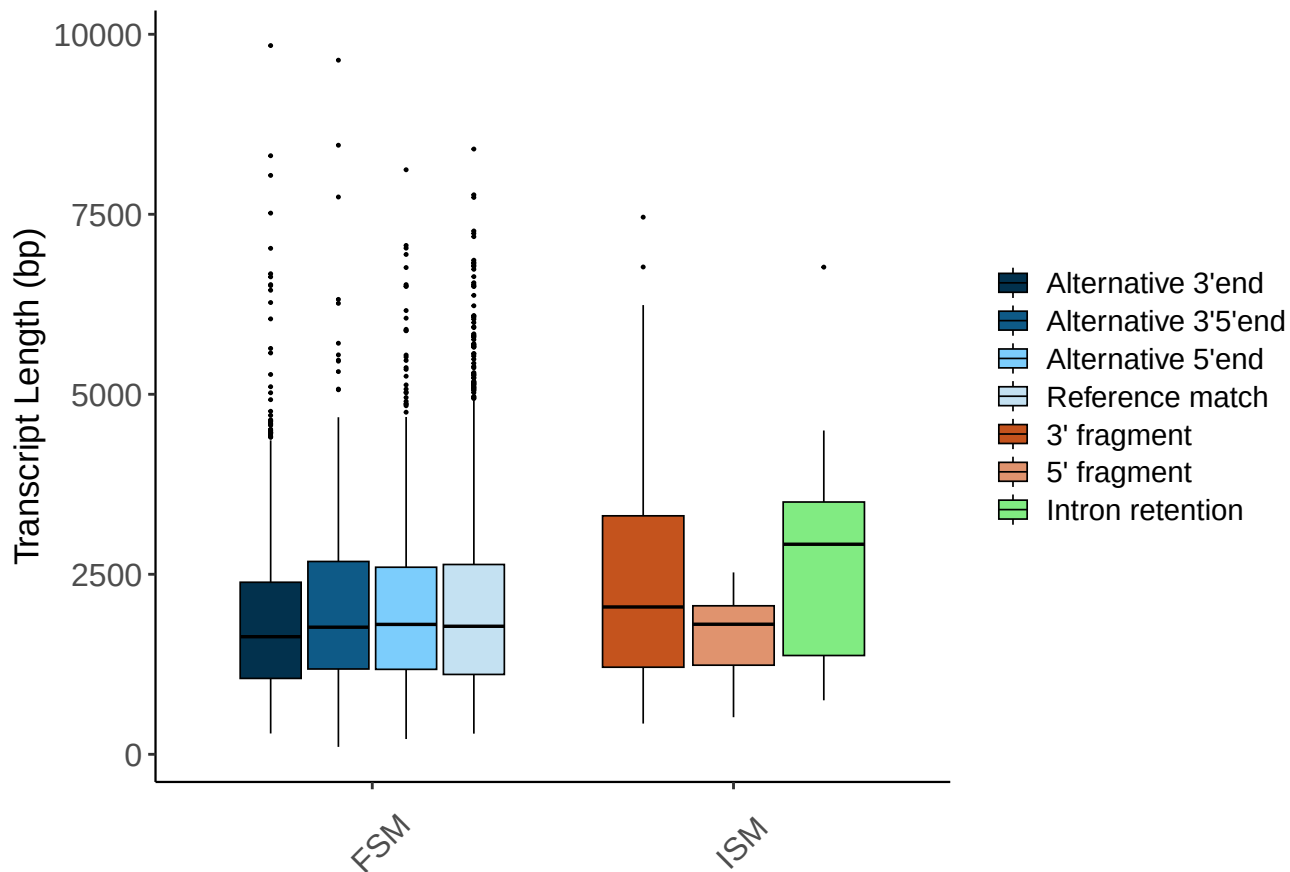
Isoform Distribution Across Intergenic



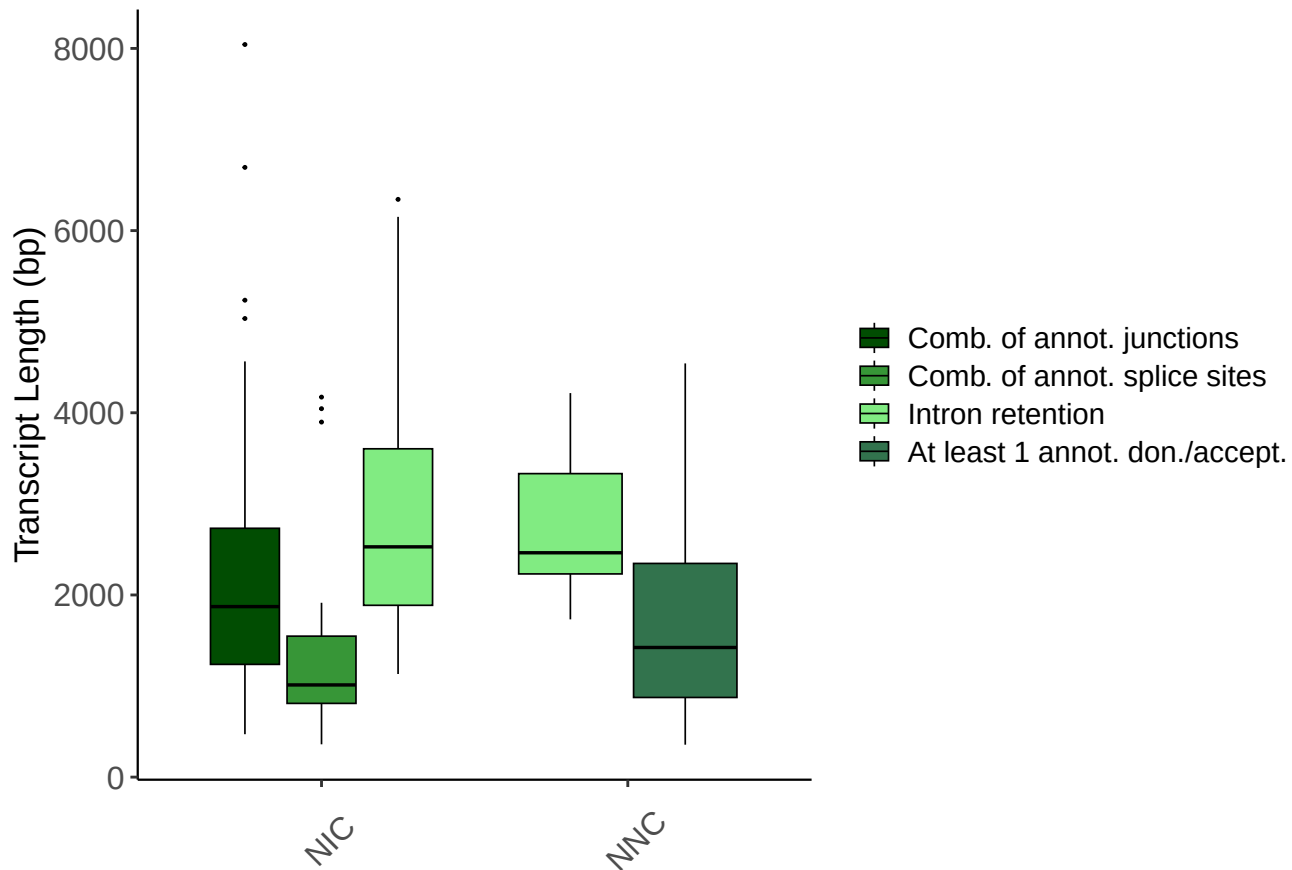
Transcript Lengths by Structural Classification



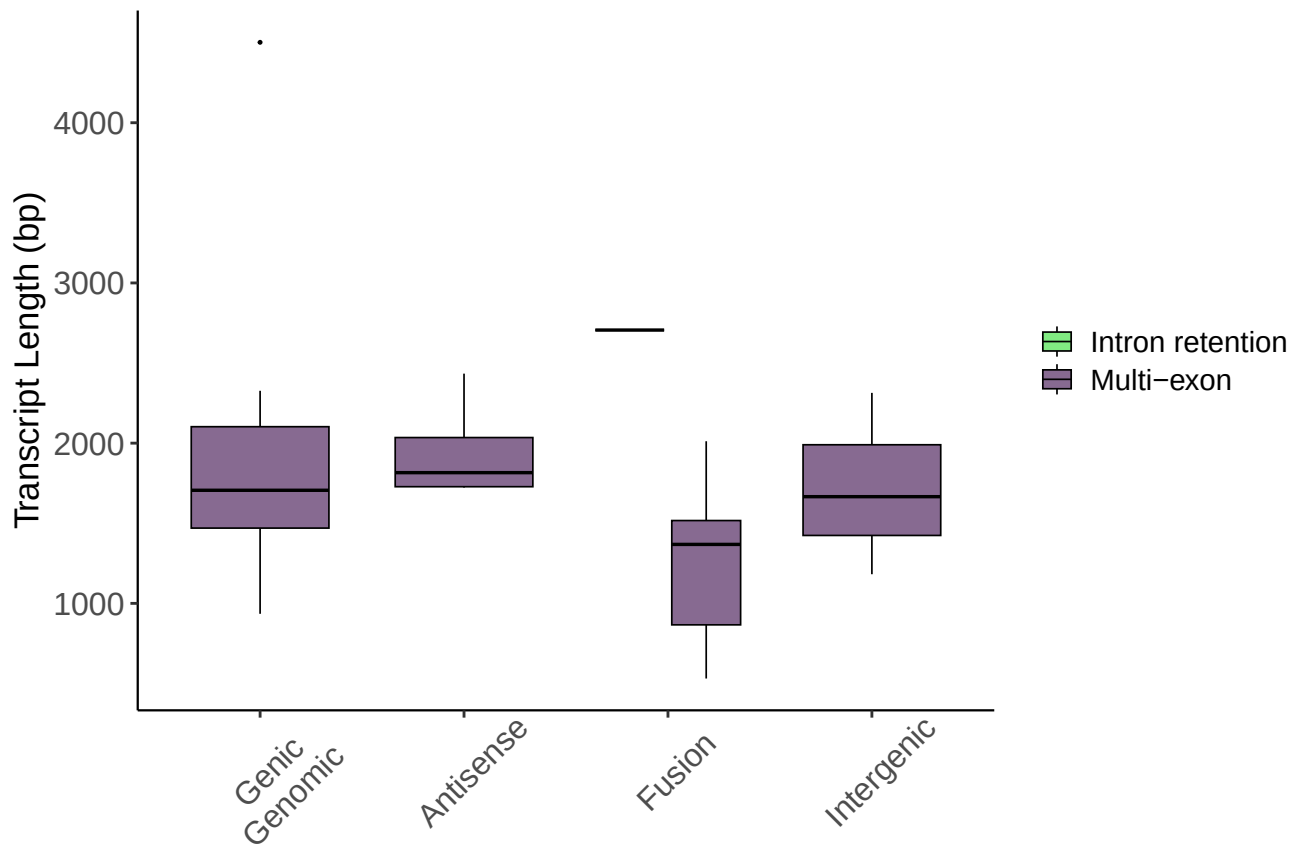
Transcript Lengths by Subcategory



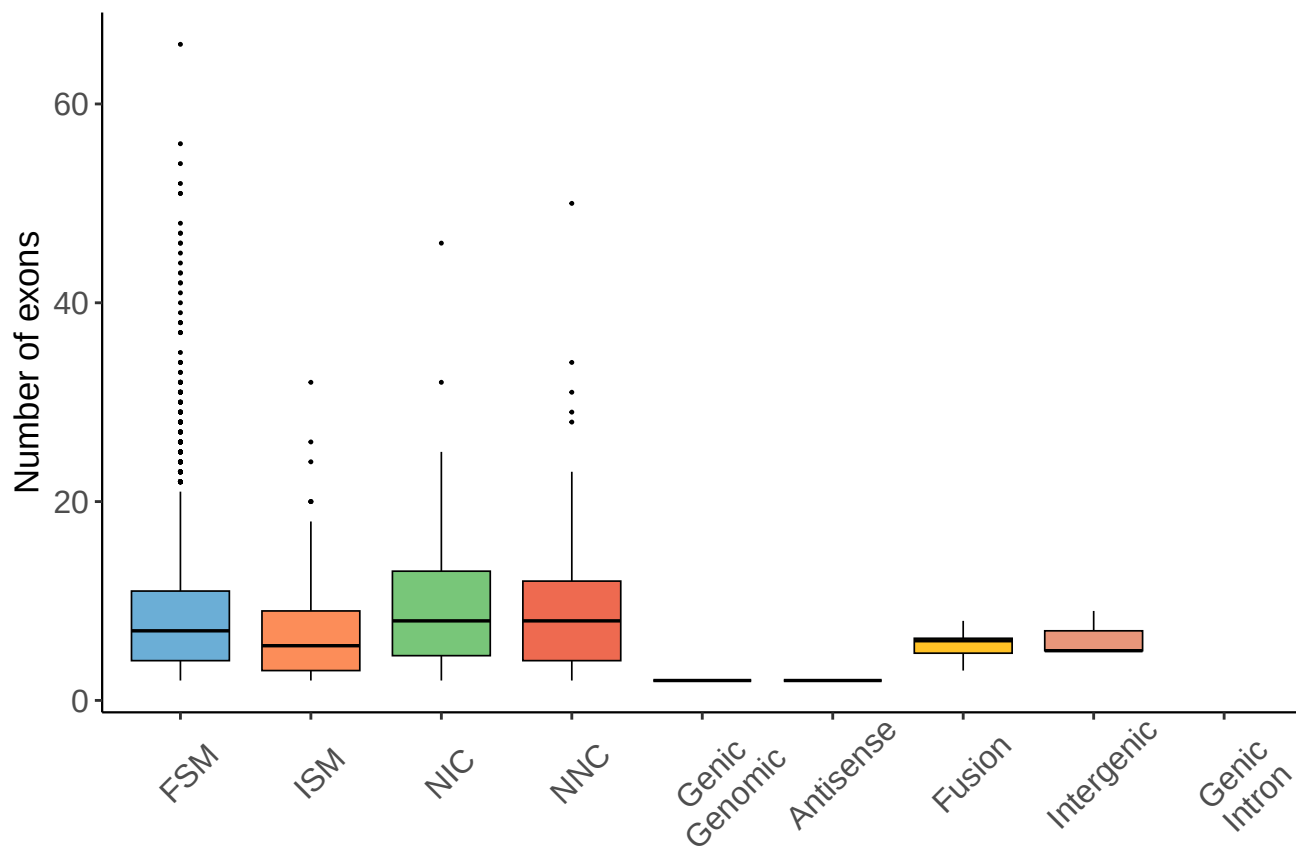
Transcript Lengths by Subcategory



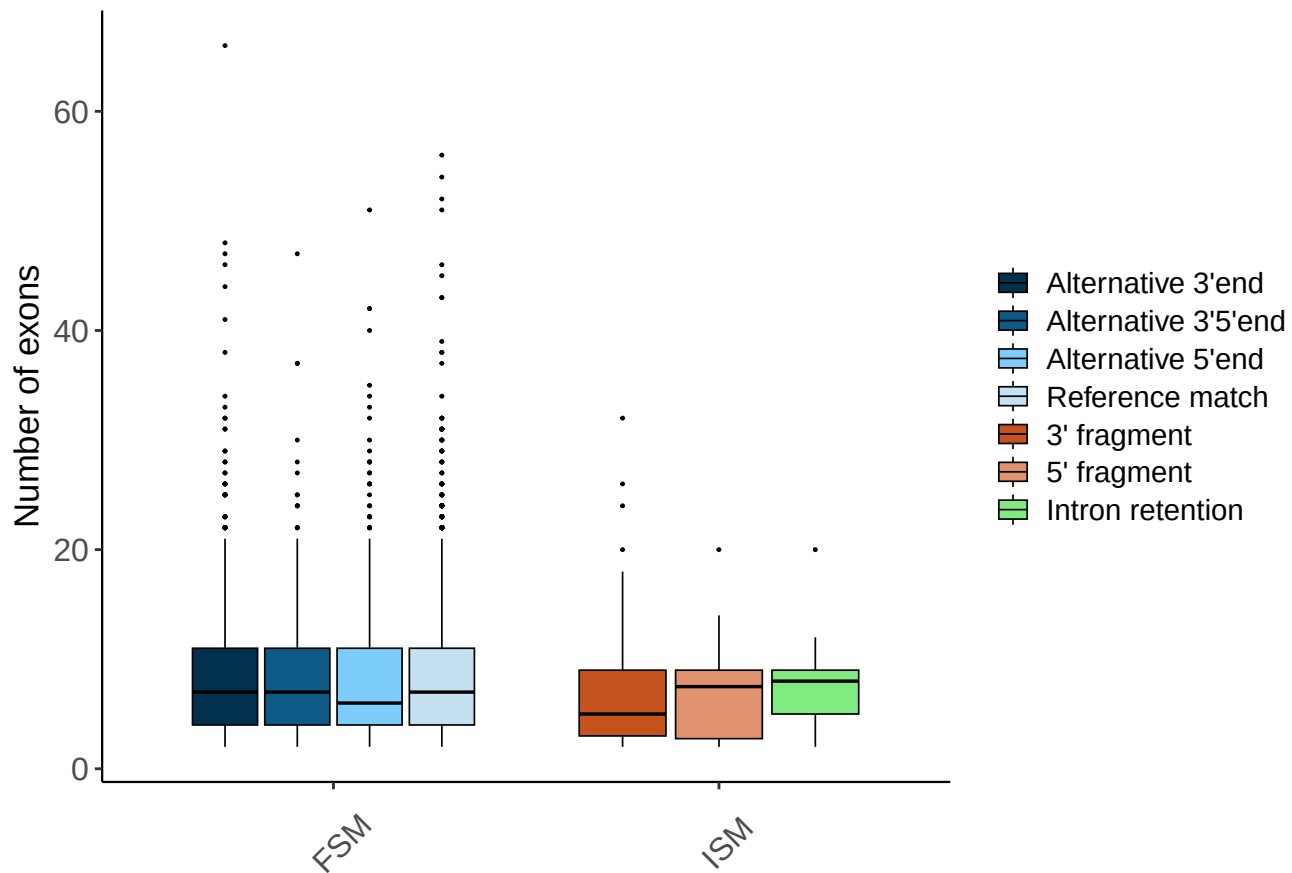
Transcript Lengths by Subcategory



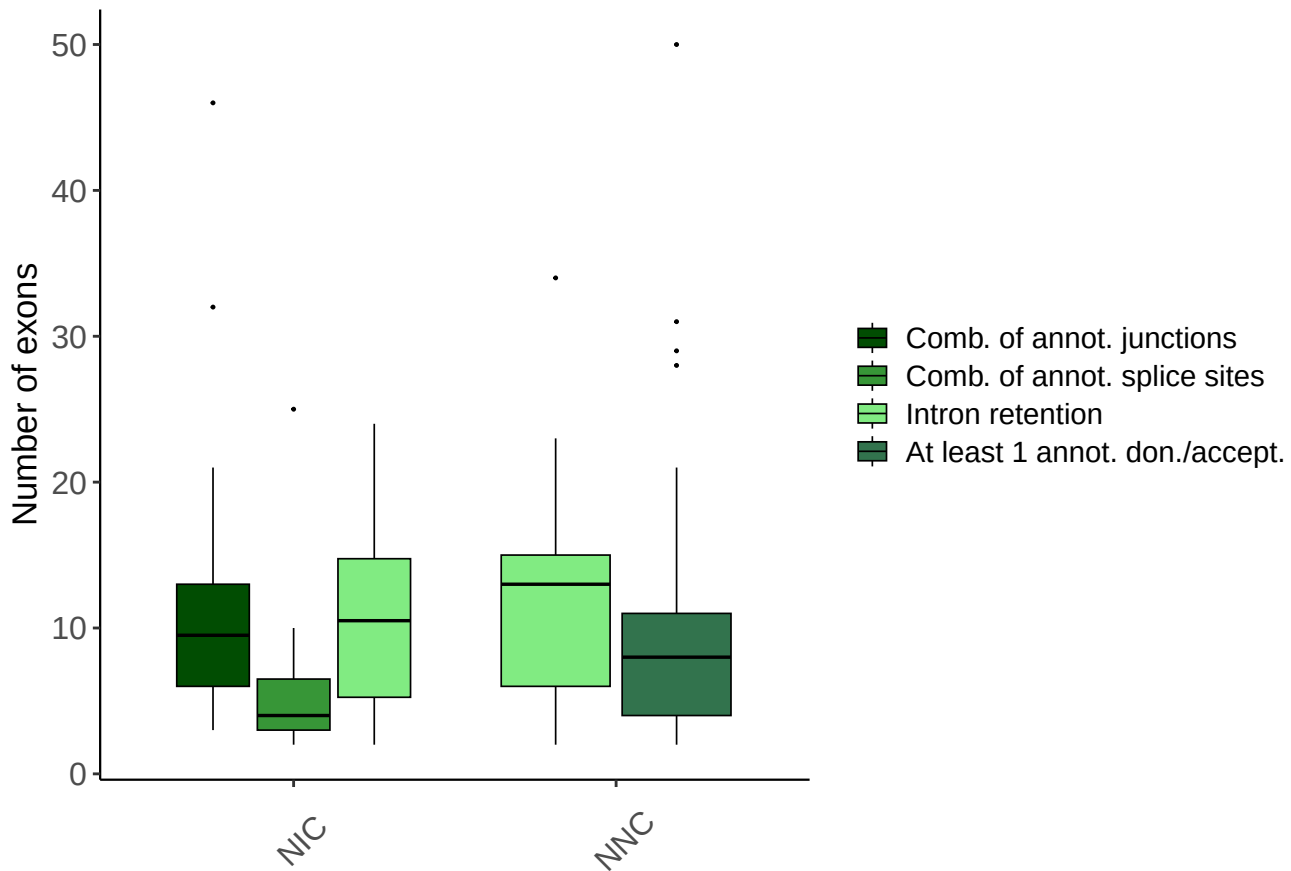
Exon Counts by Structural Classification



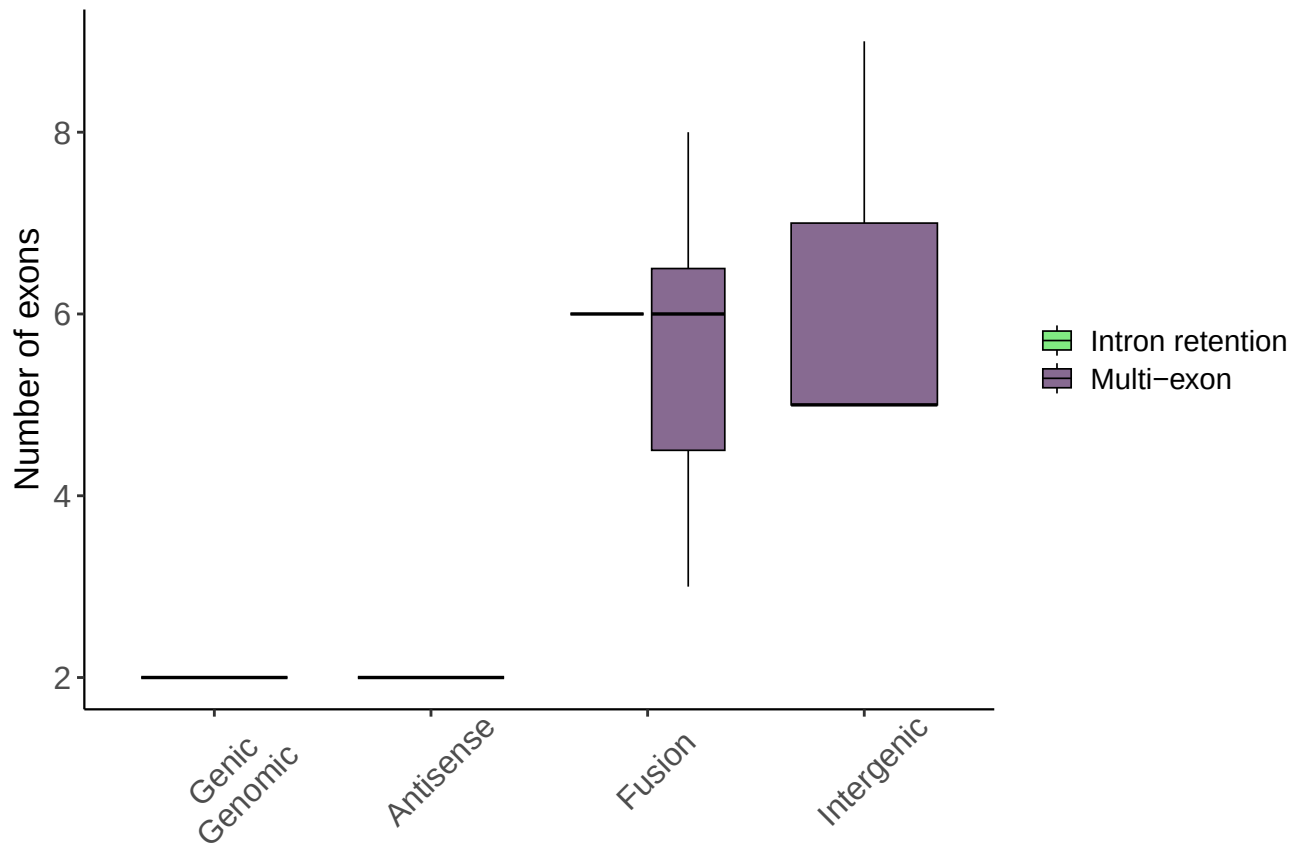
Exon Counts by Subcategory



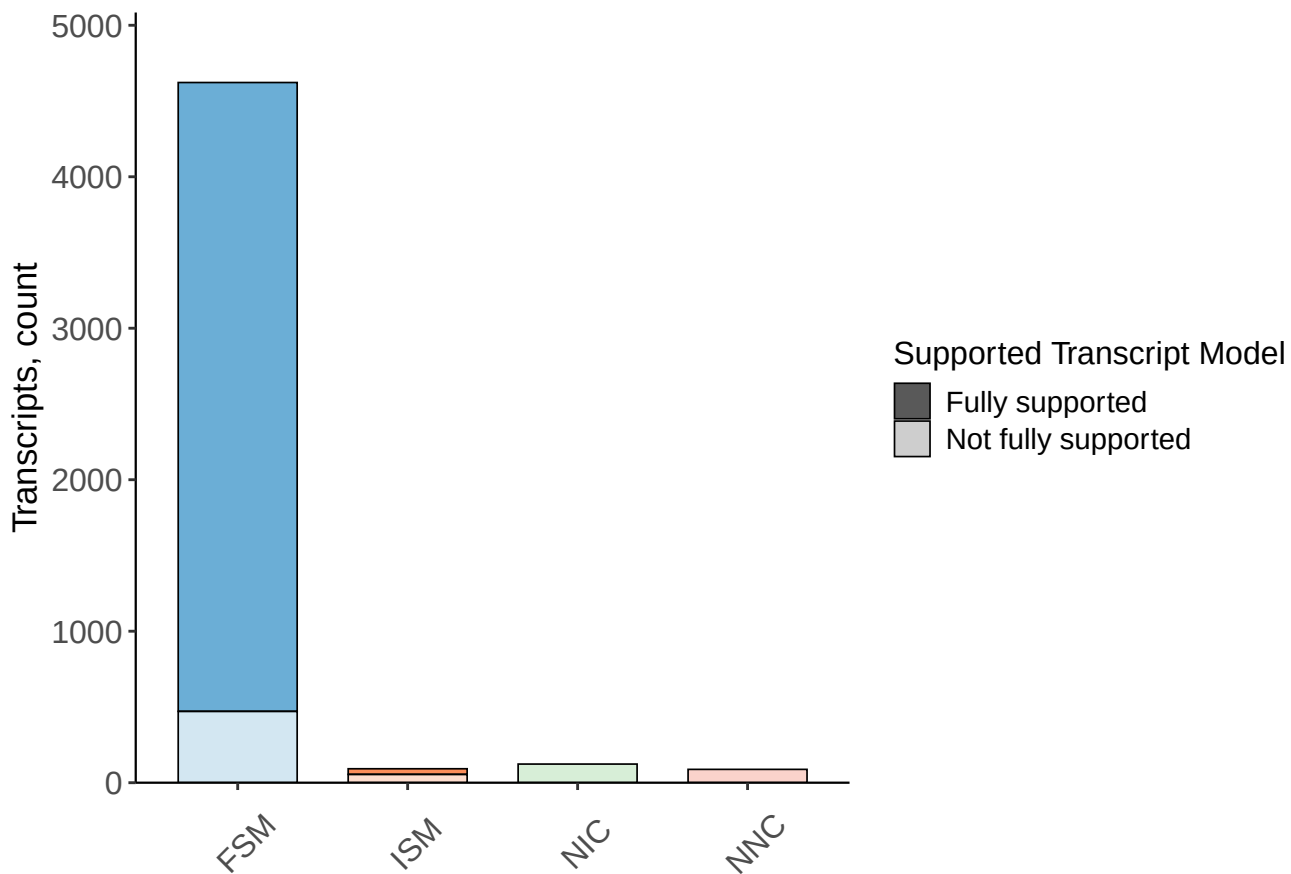
Exon Counts by Subcategory



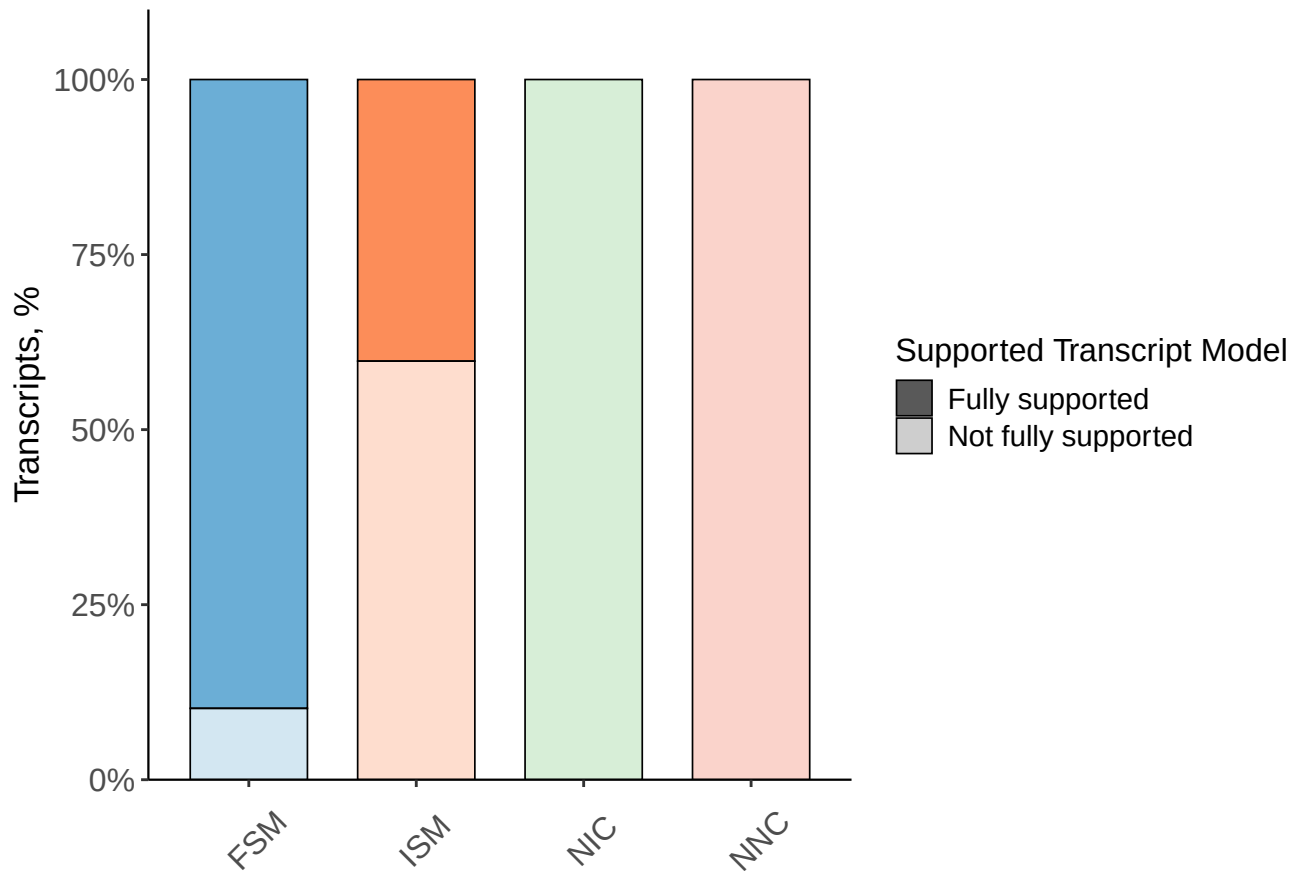
Exon Counts by Subcategory



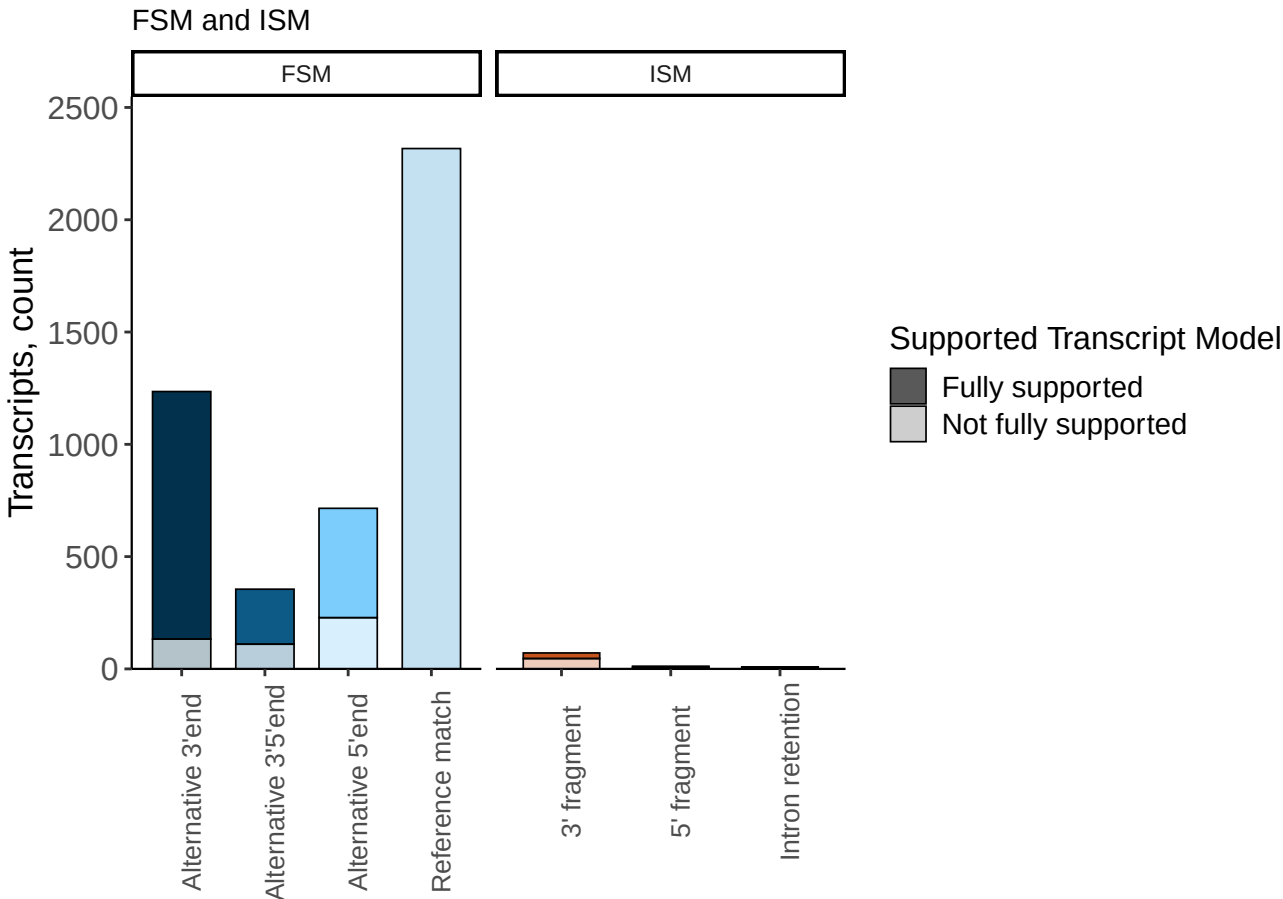
Isoform Distribution Across Structural Categories



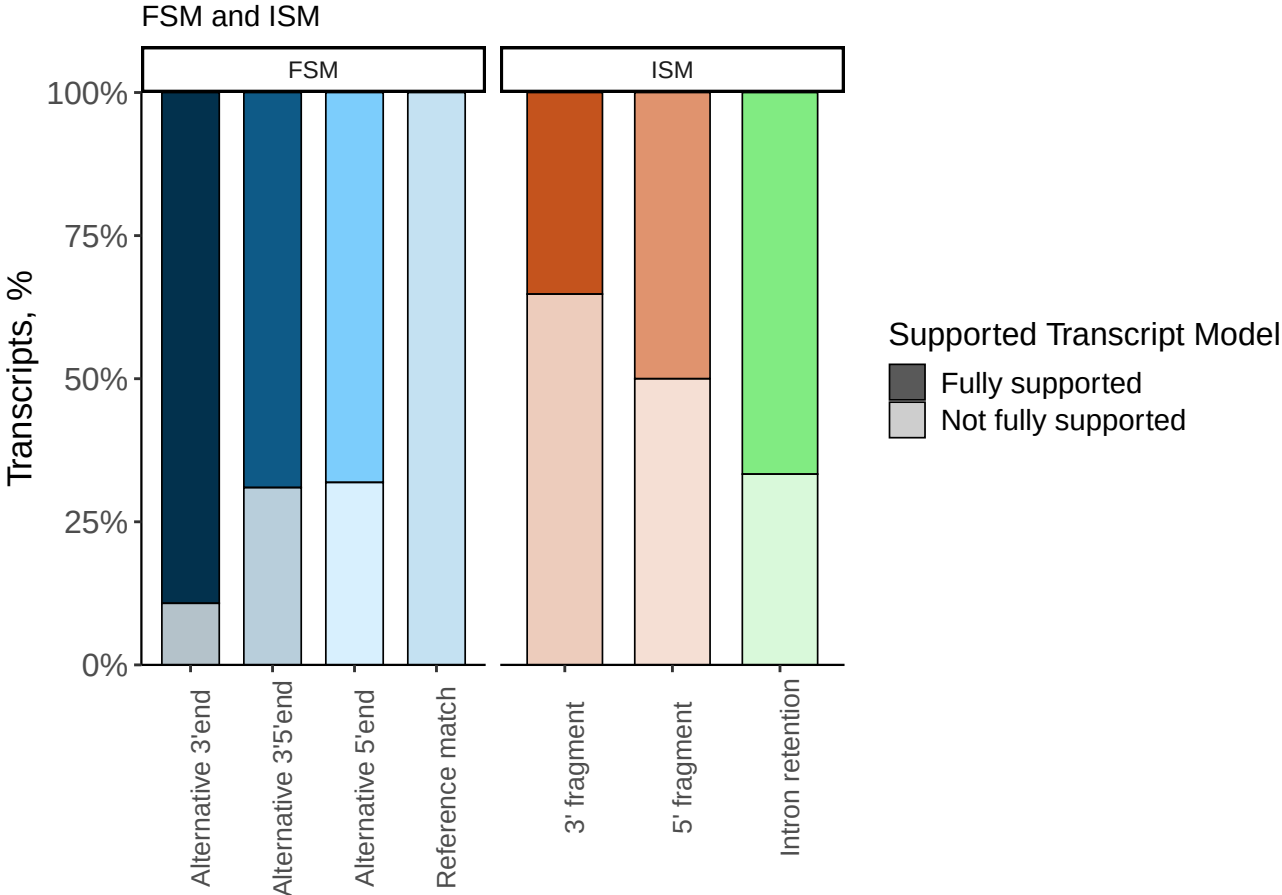
Isoform Distribution Across Structural Categories



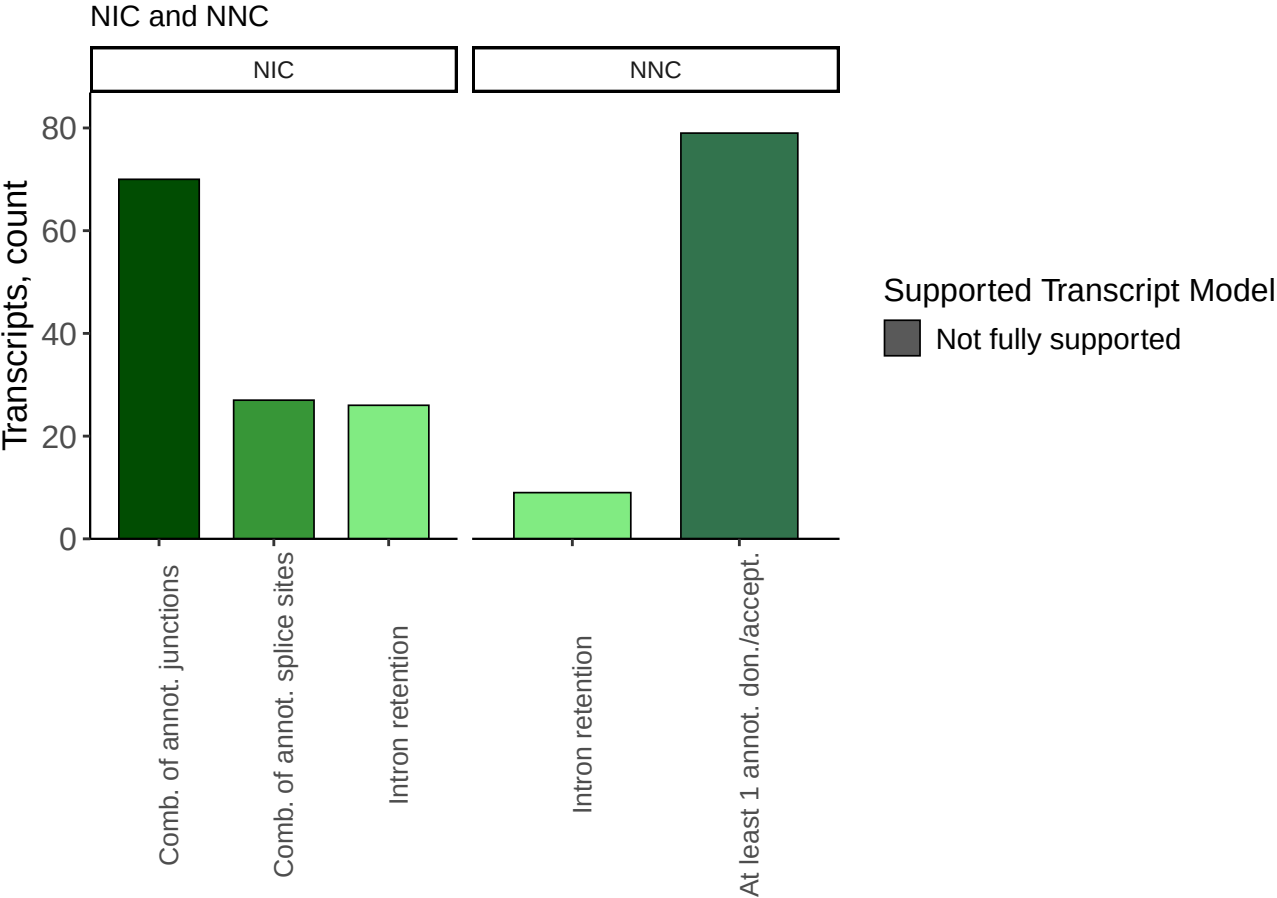
Isoform Distribution Across Structural Subcategories



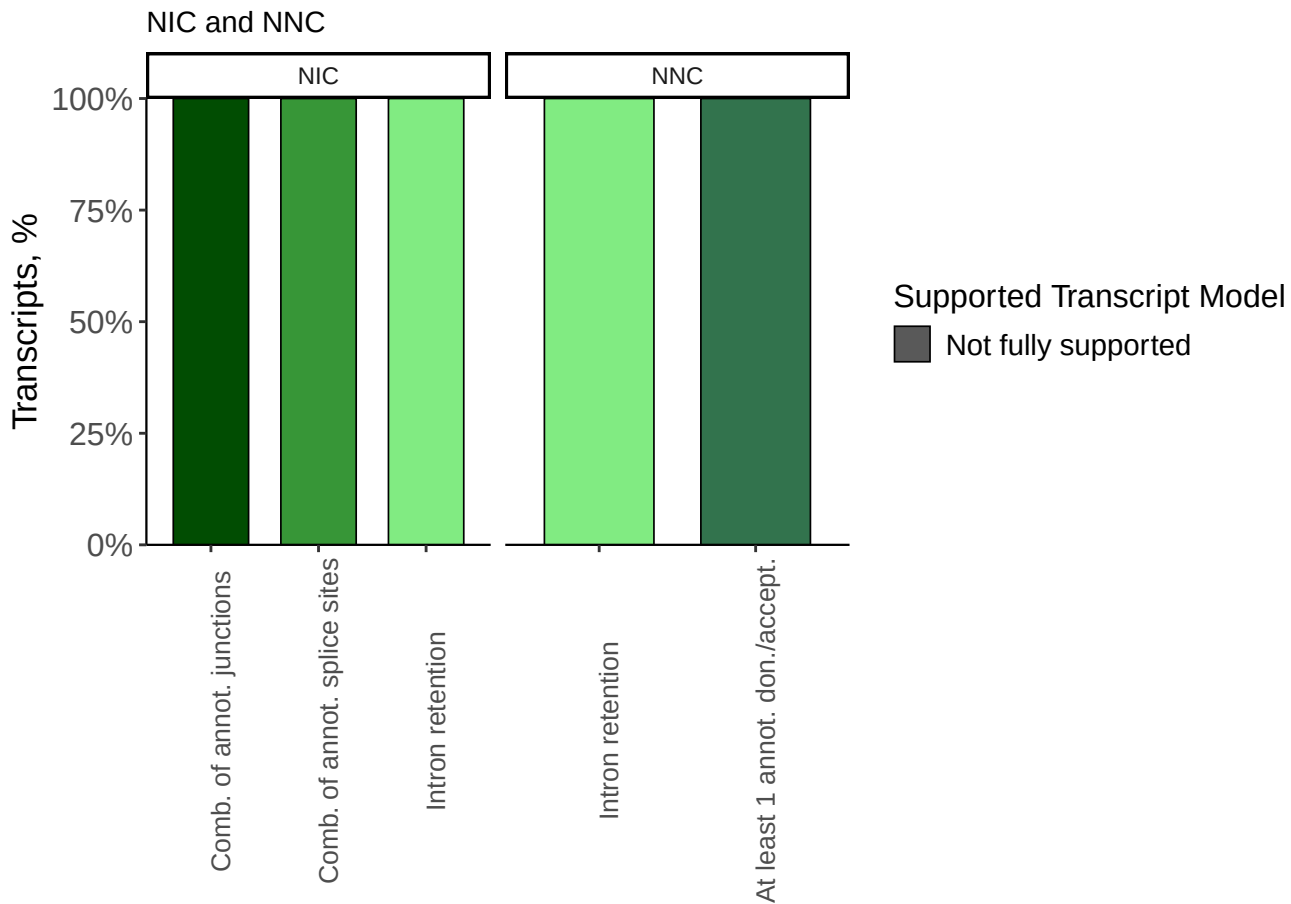
Isoform Distribution Across Structural Subcategories



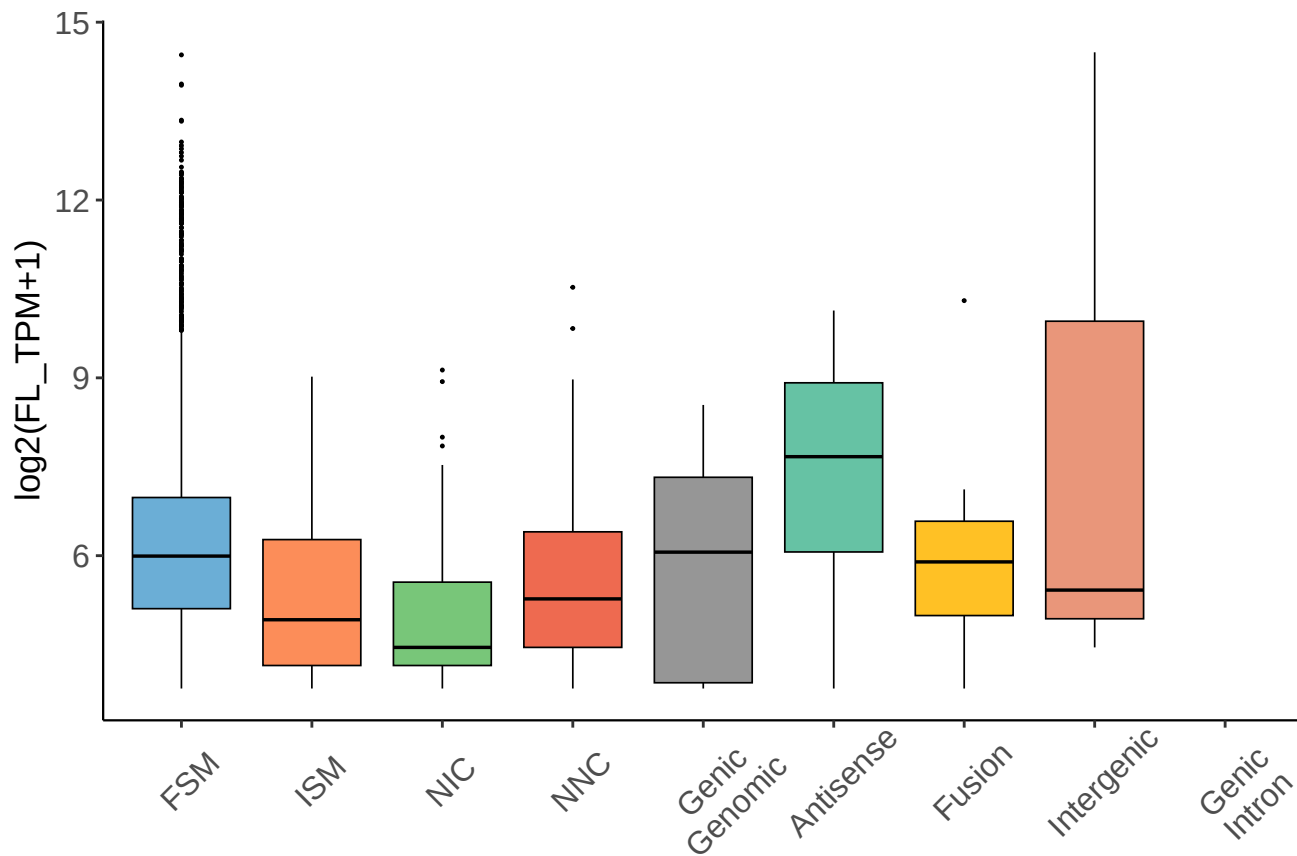
Isoform Distribution Across Structural Subcategories



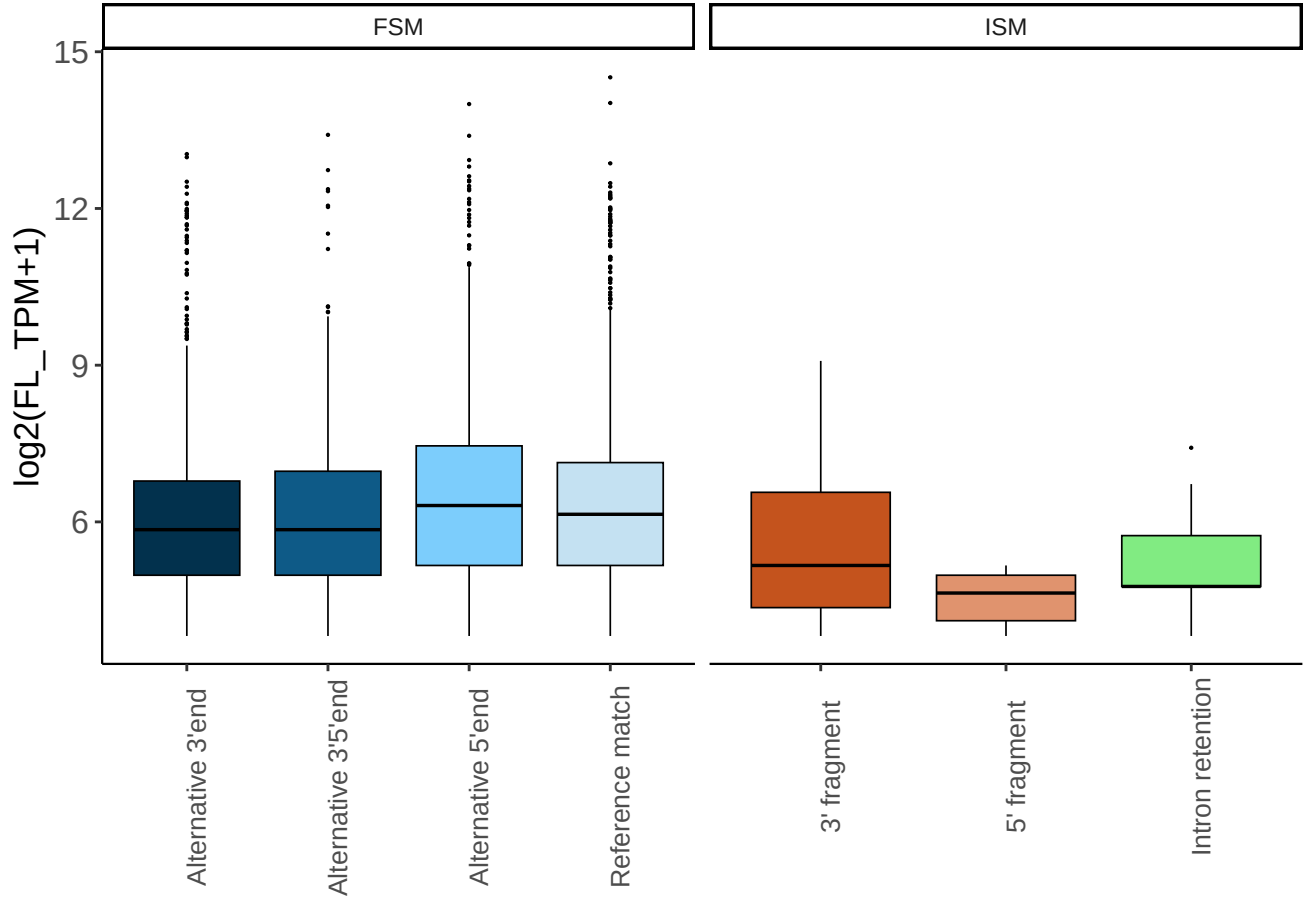
Isoform Distribution Across Structural Subcategories



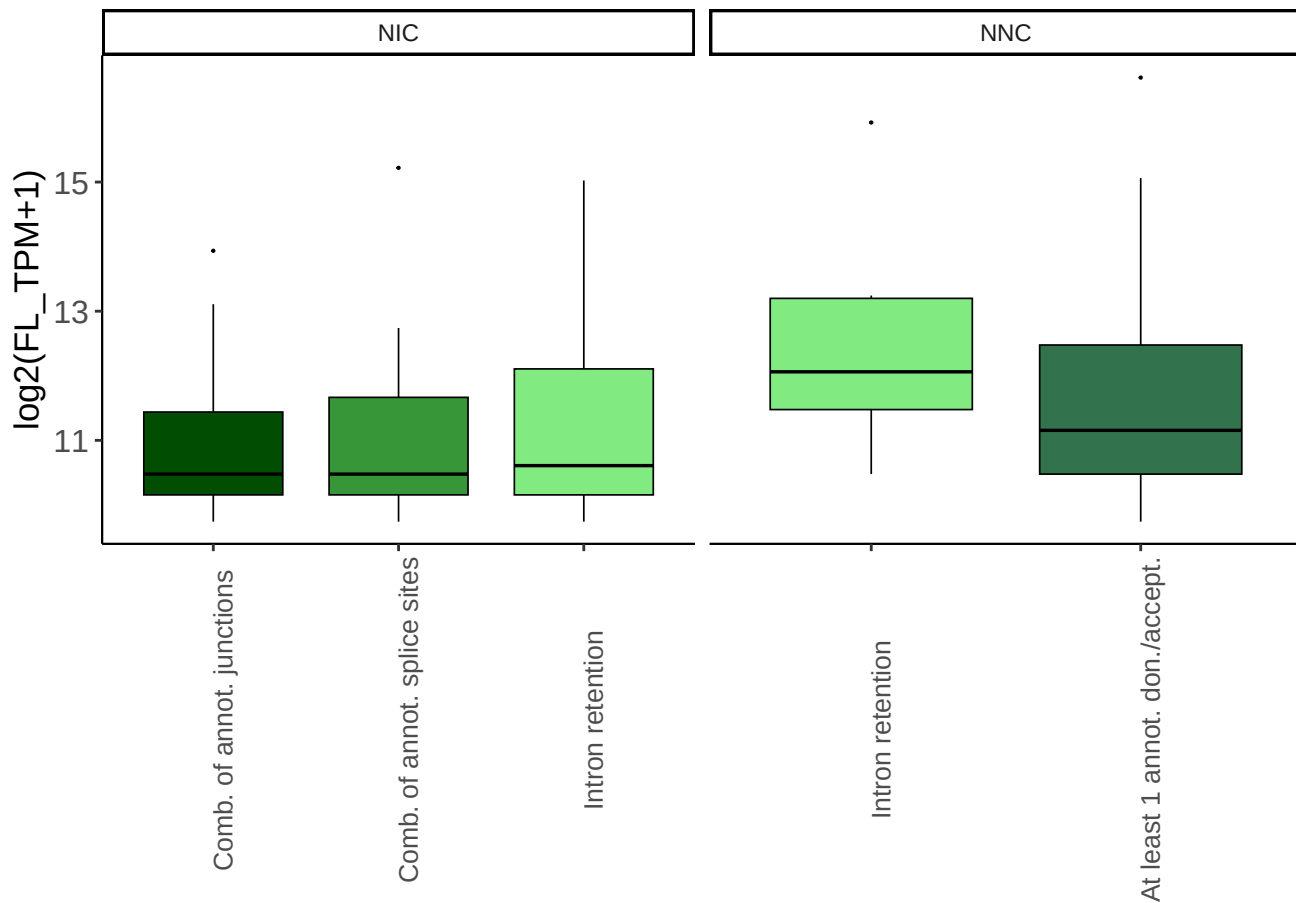
Long Reads Count by Structural Category



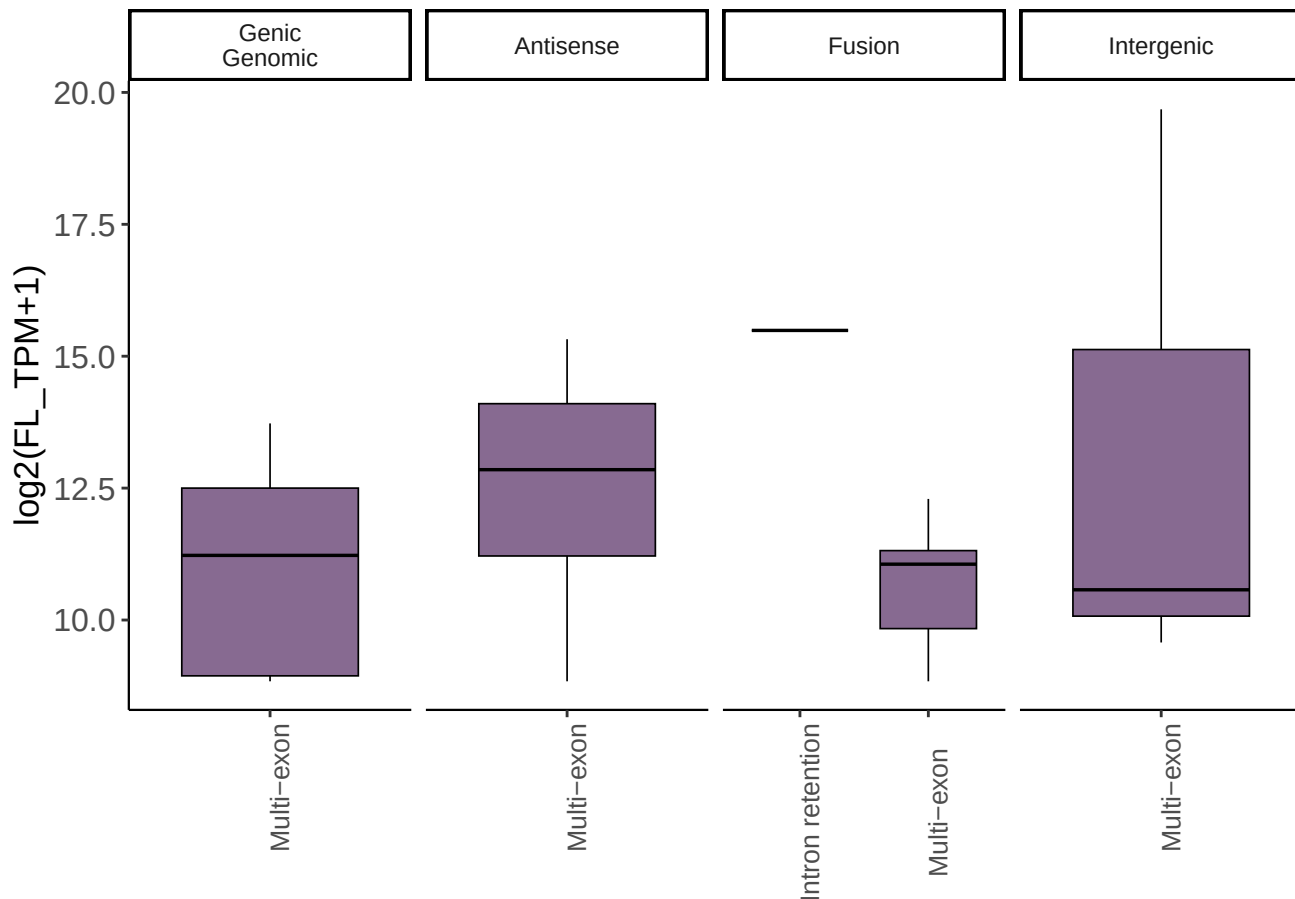
Long Reads Count by Subcategory



Long Reads Count by Subcategory

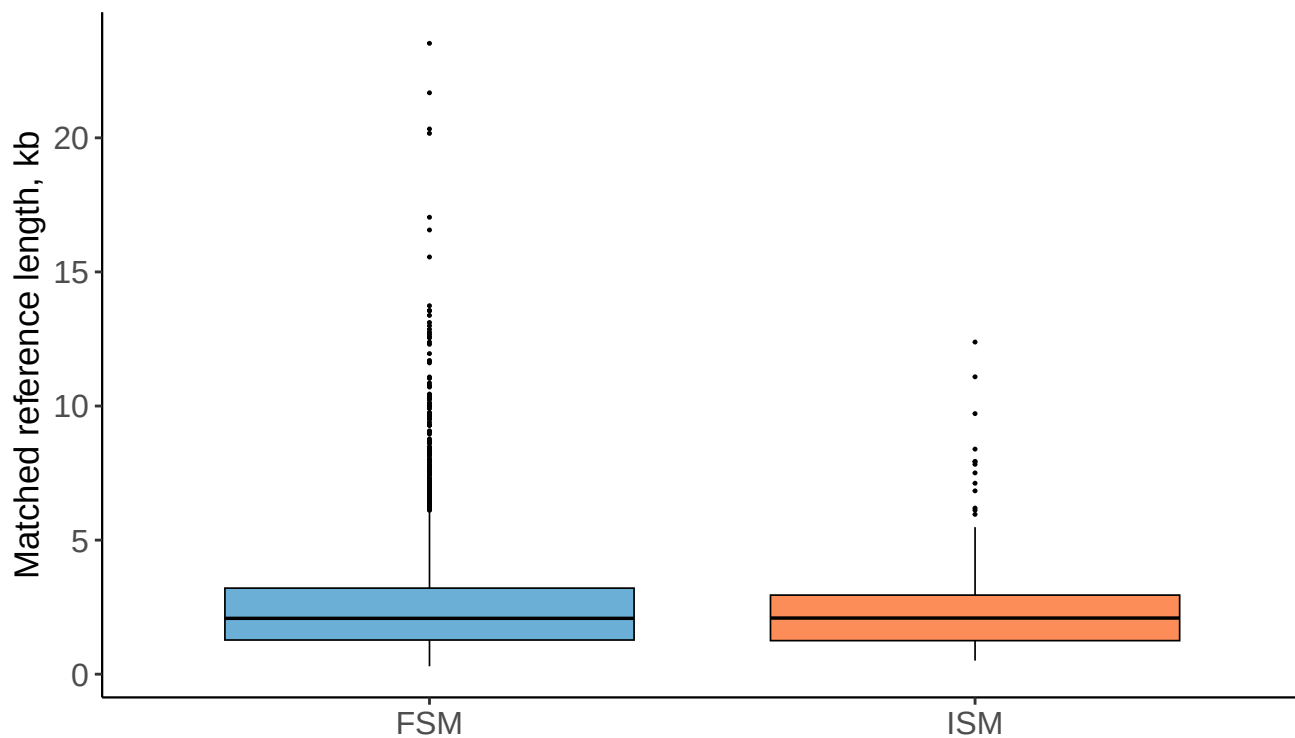


Long Reads Count by Subcategory



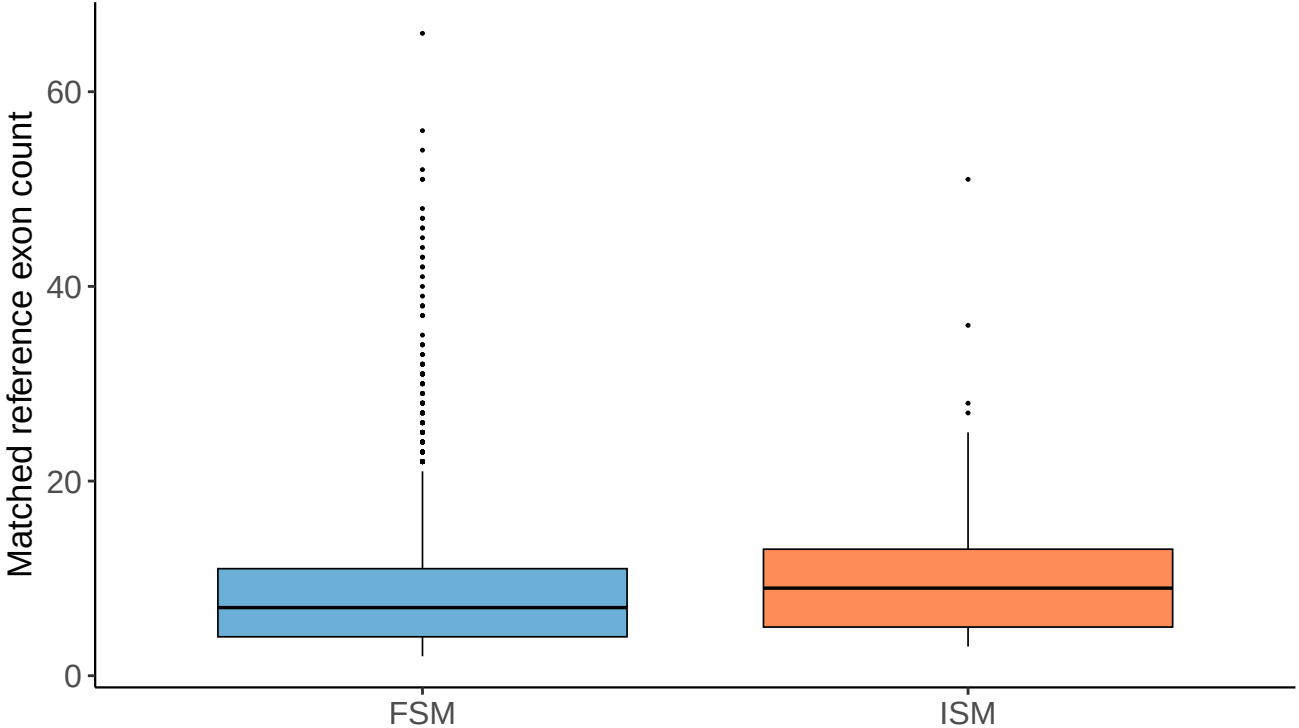
Length Distribution of Matched Reference Transcripts

Applicable Only to FSM and ISM Categories



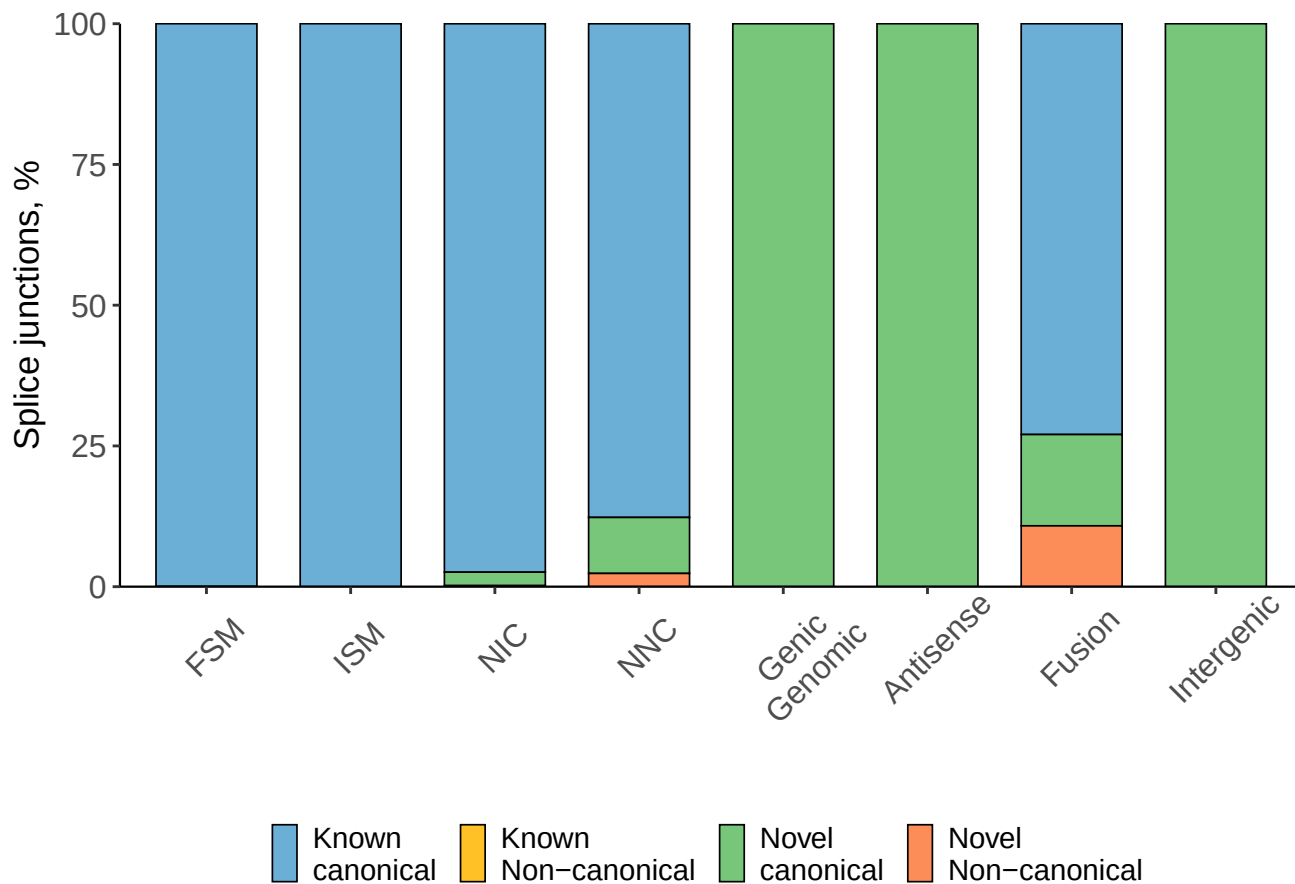
Exon Count Distribution of Matched Reference Transcripts

Applicable Only to FSM and ISM Categories

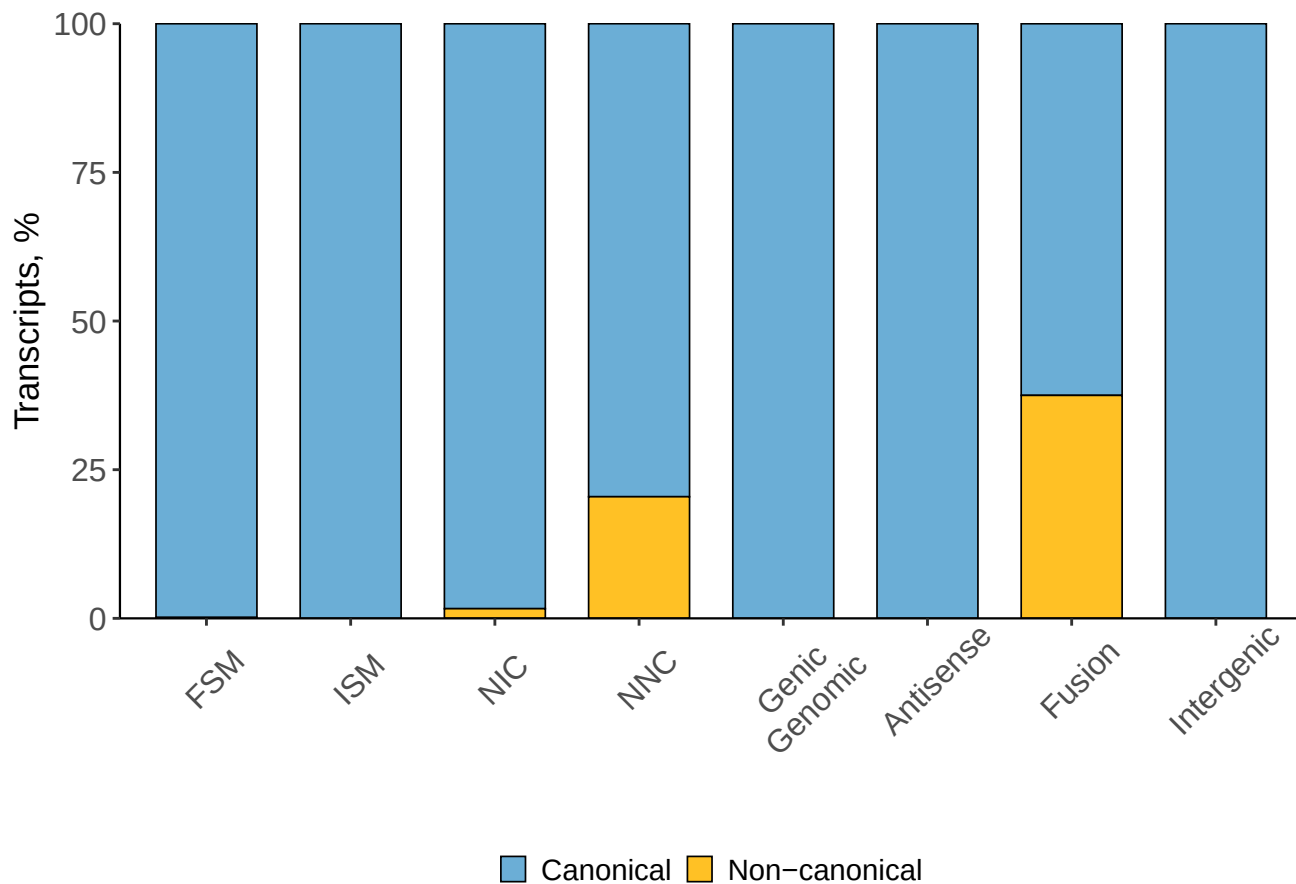


Splice Junction Characterization

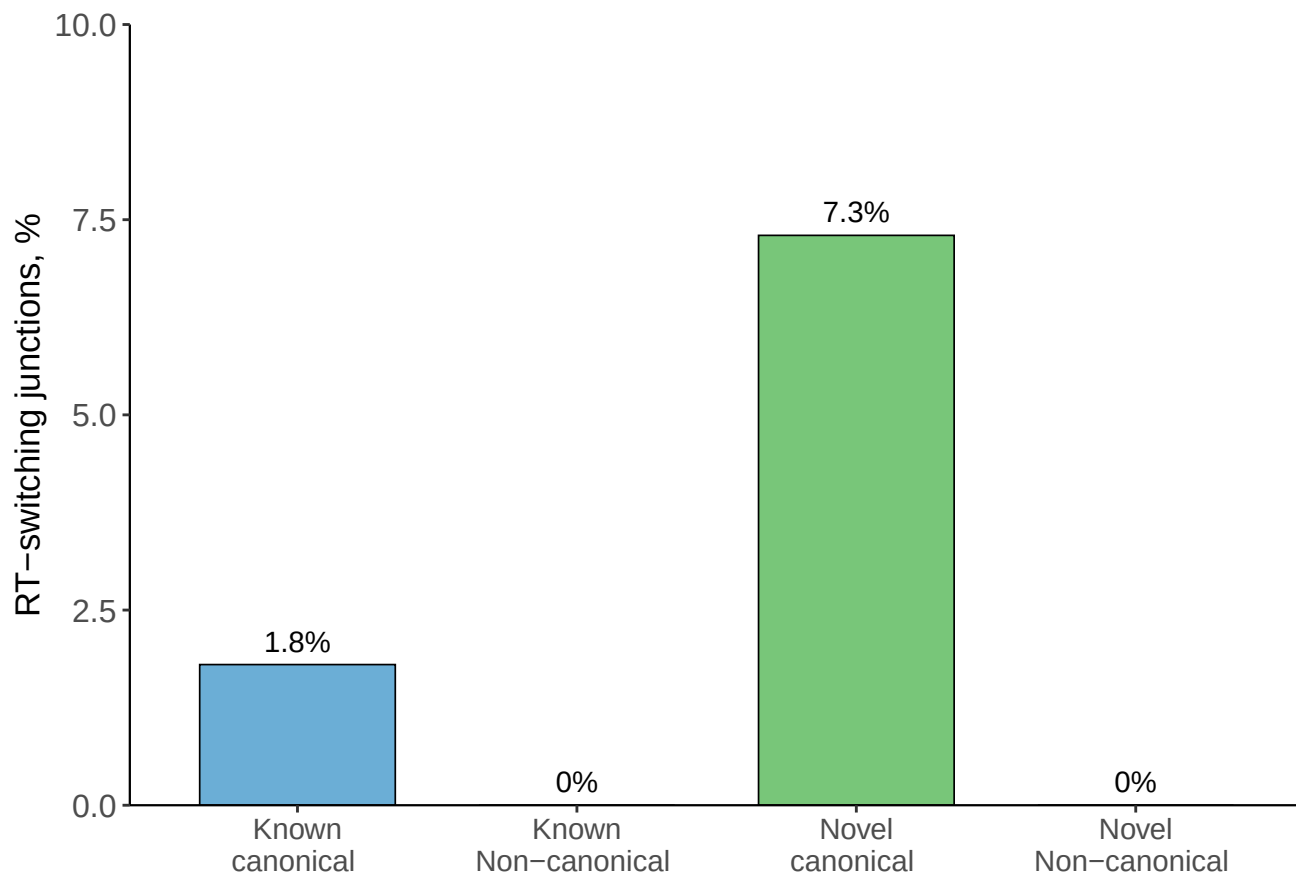
Distribution of Splice Junctions by Structural Classification



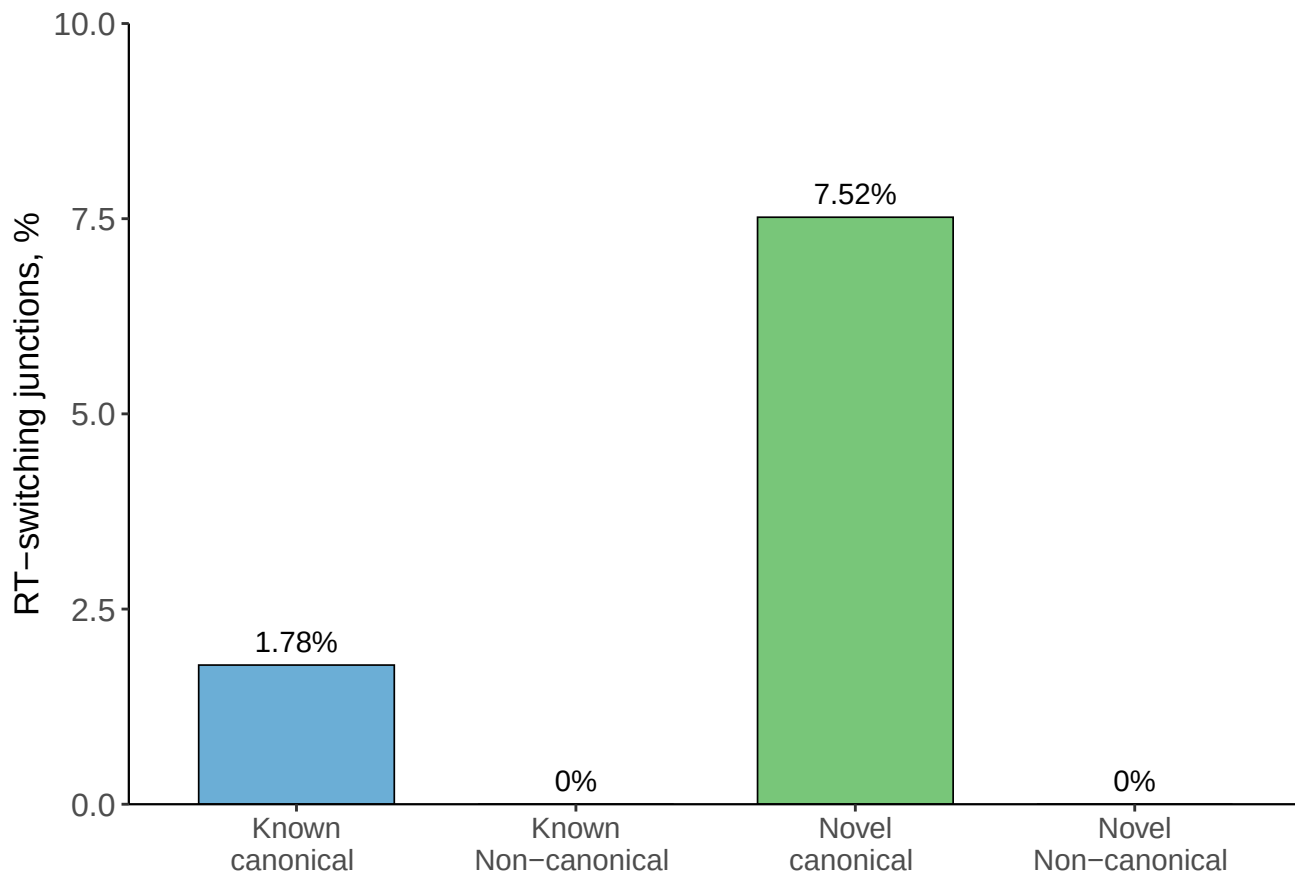
Distribution of Transcripts by Splice Junctions



RT-Switching All Junctions



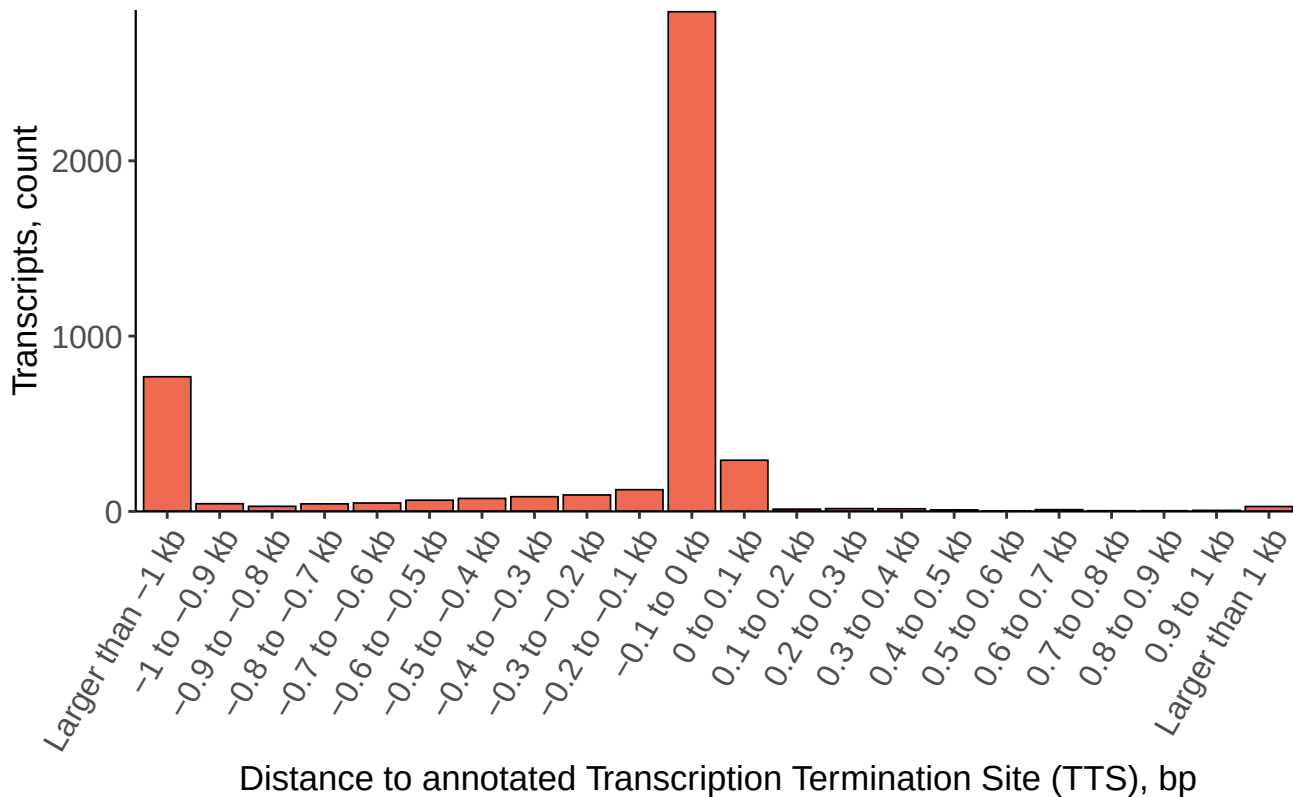
Unique Junctions RT-switching



Comparison With Annotated TSS and TTS

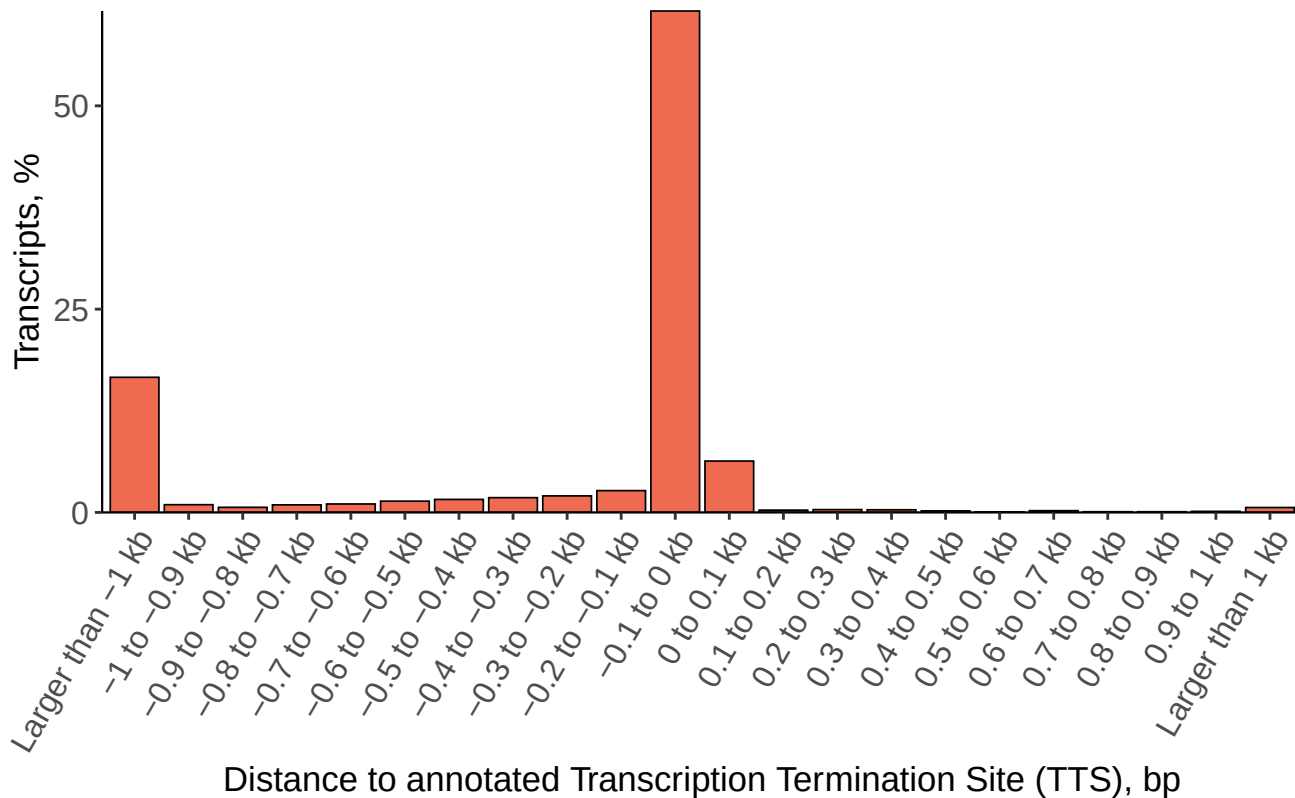
Distance to annotated Transcription Termination Site (TTS) FSM

Negative values indicate upstream of annotated termination site



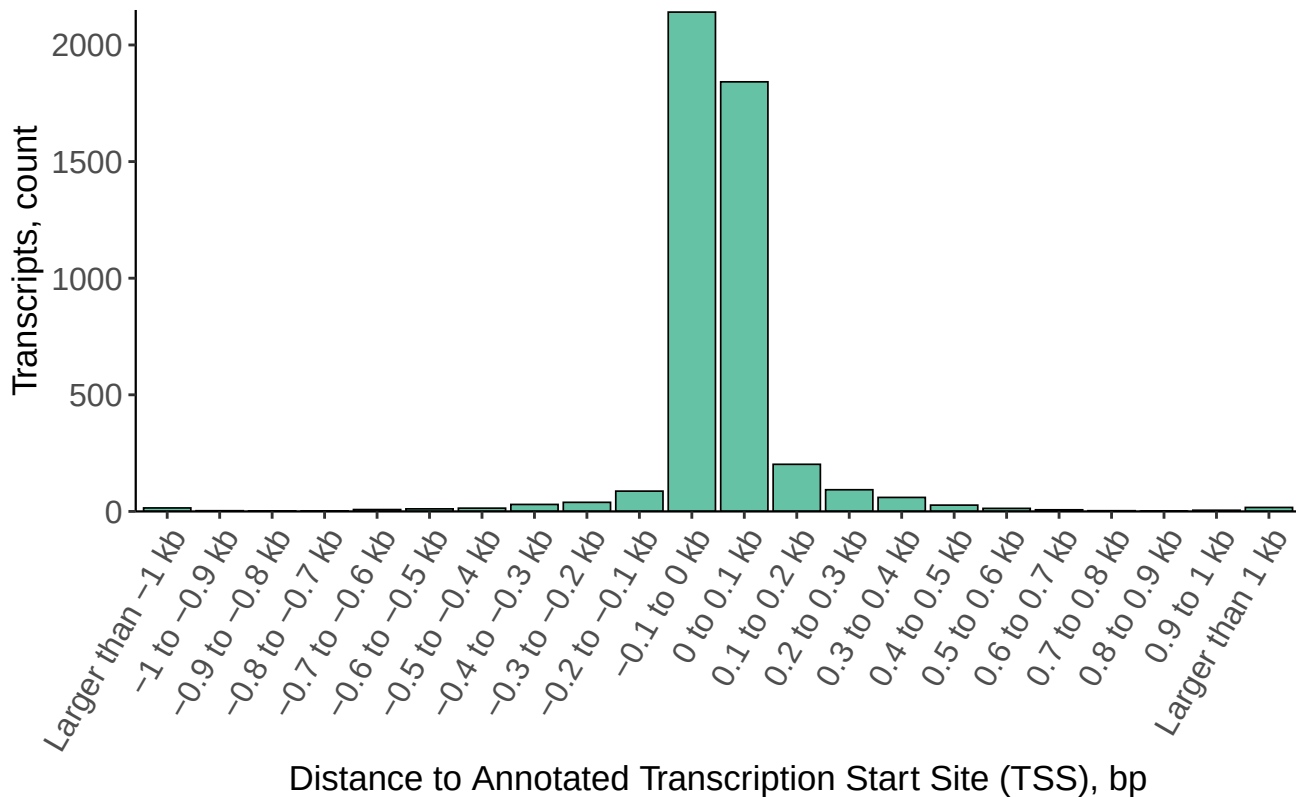
Distance to annotated Transcription Termination Site (TTS) FSM

Negative values indicate upstream of annotated termination site



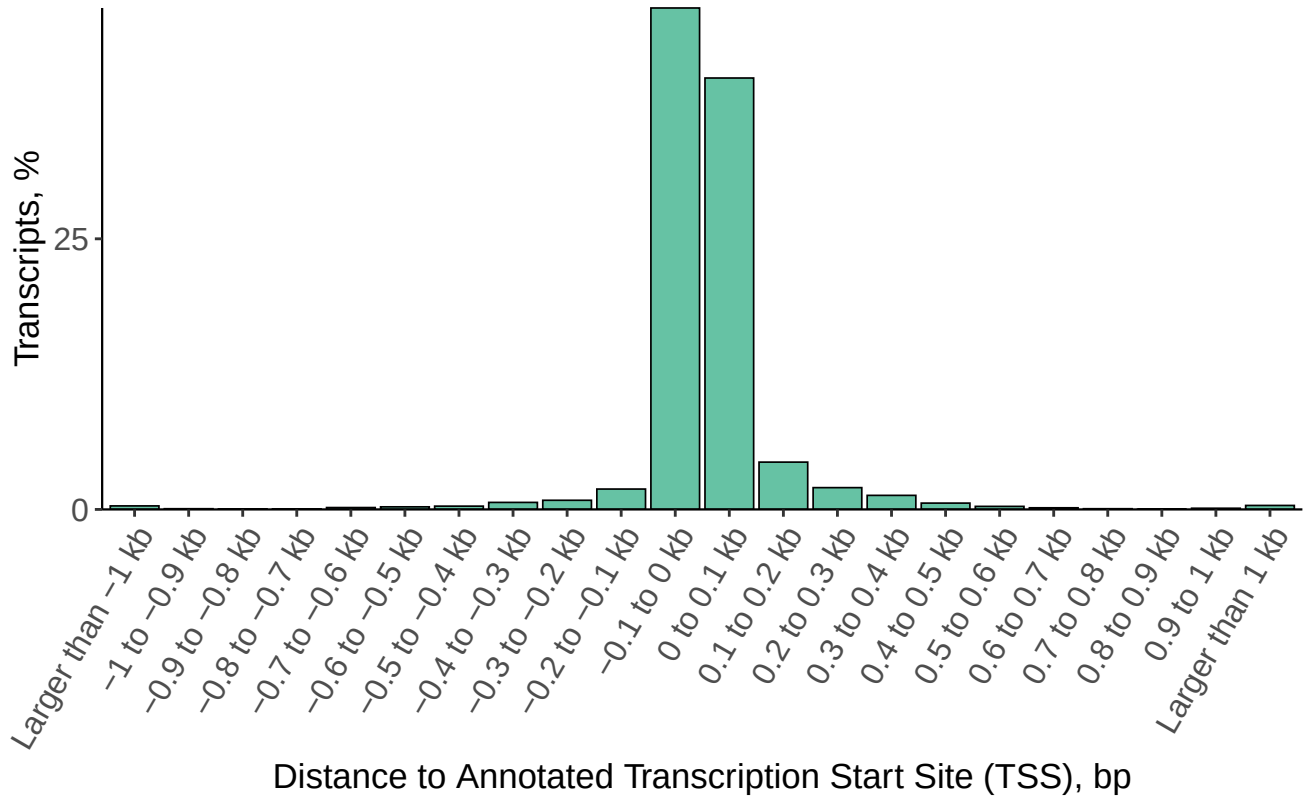
Distance to Annotated Transcription Start Site for FSM

Negative values indicate downstream of annotated TSS



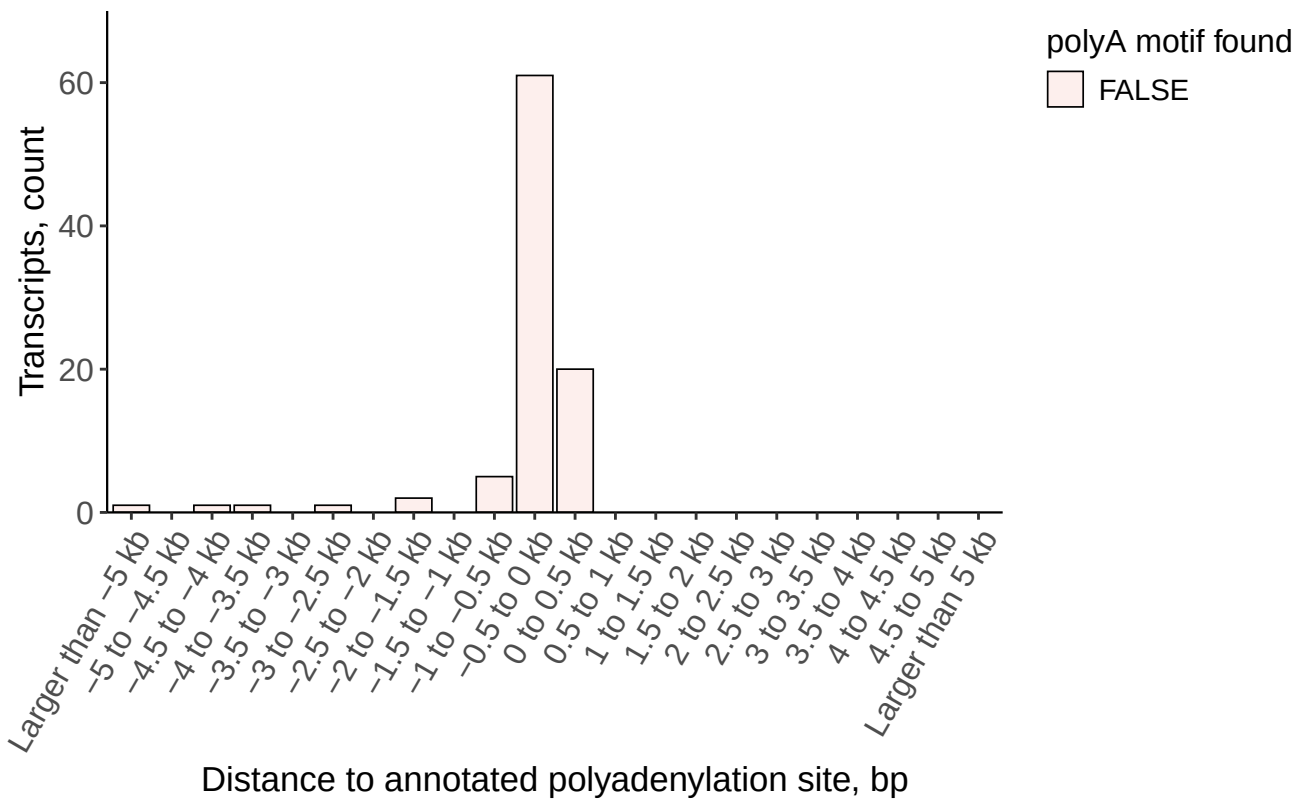
Distance to Annotated Transcription Start Site for FSM

Negative values indicate downstream of annotated TSS



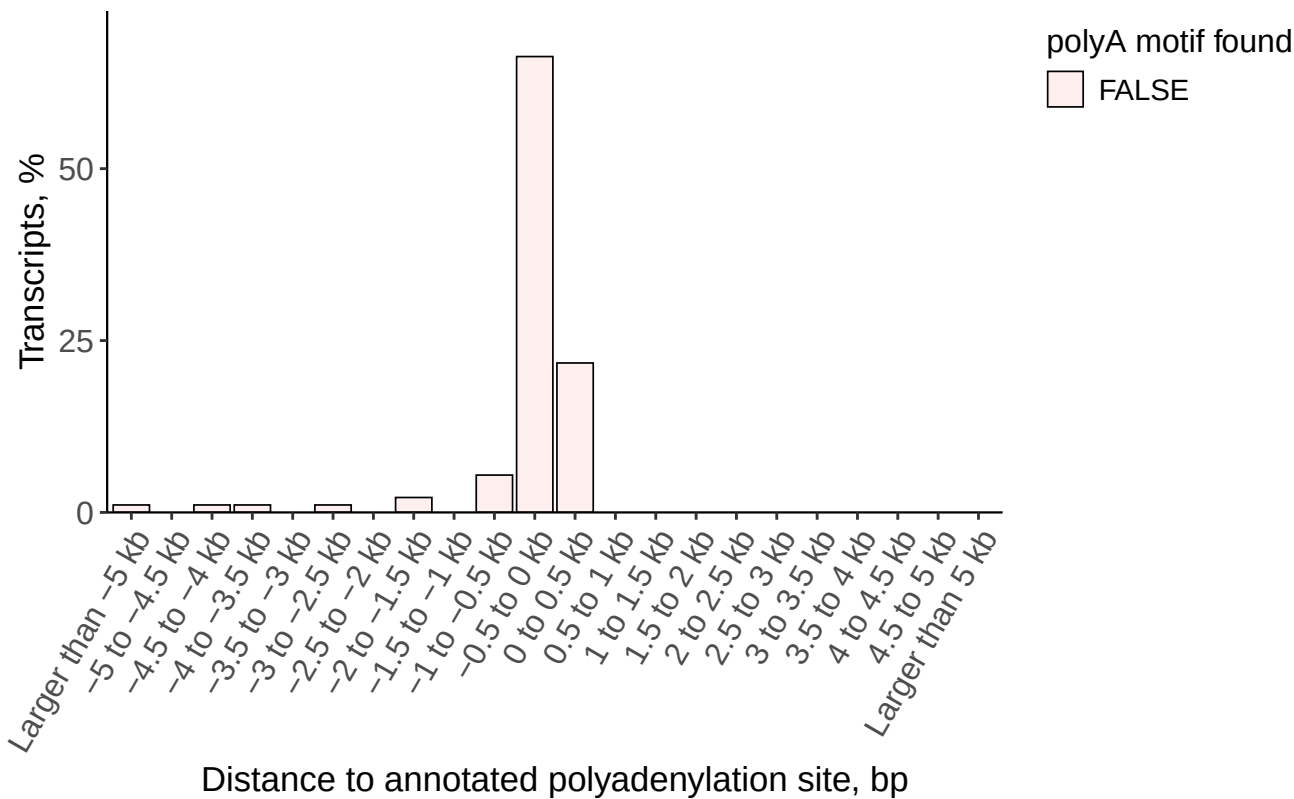
Distance to Annotated Transcription Termination Site (TTS)

Negative values indicate upstream of annotated TTS



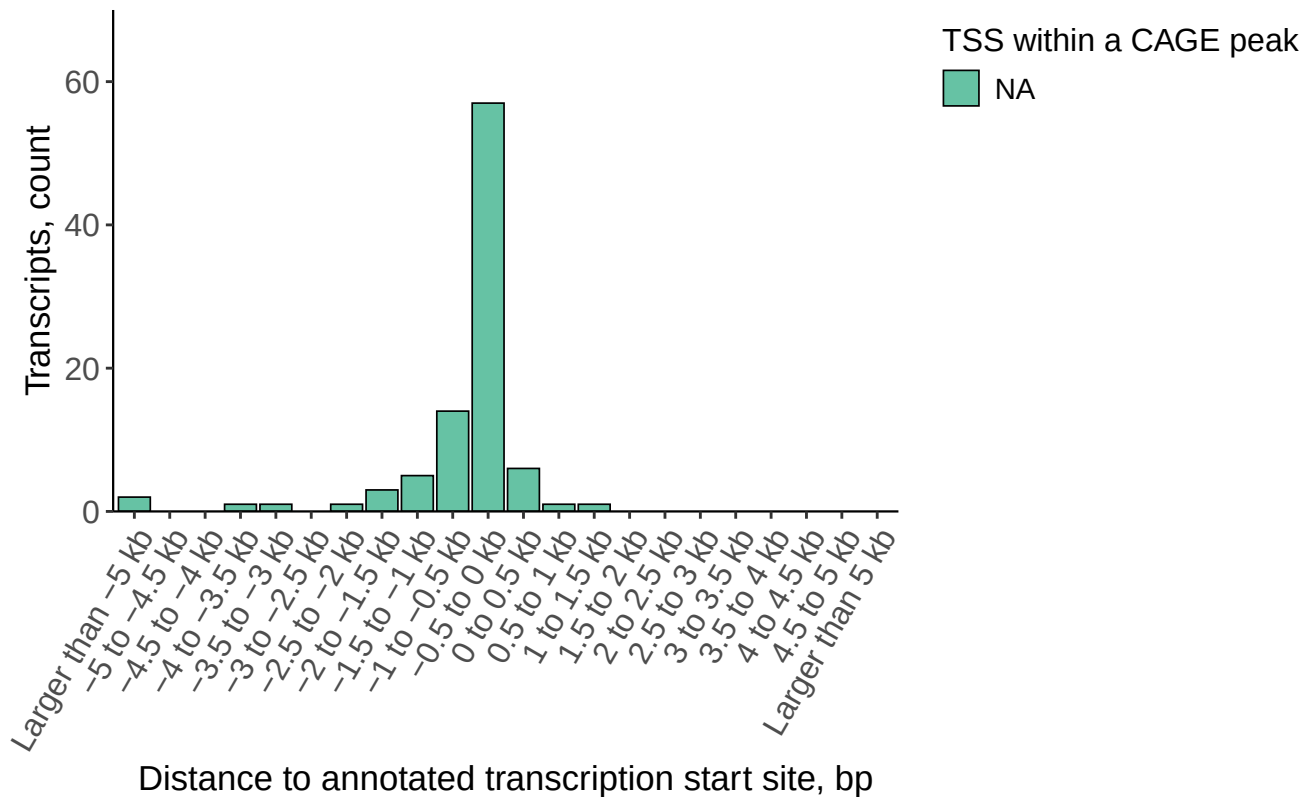
Distance to Annotated Transcription Termination Site (TTS)

Negative values indicate upstream of annotated TTS



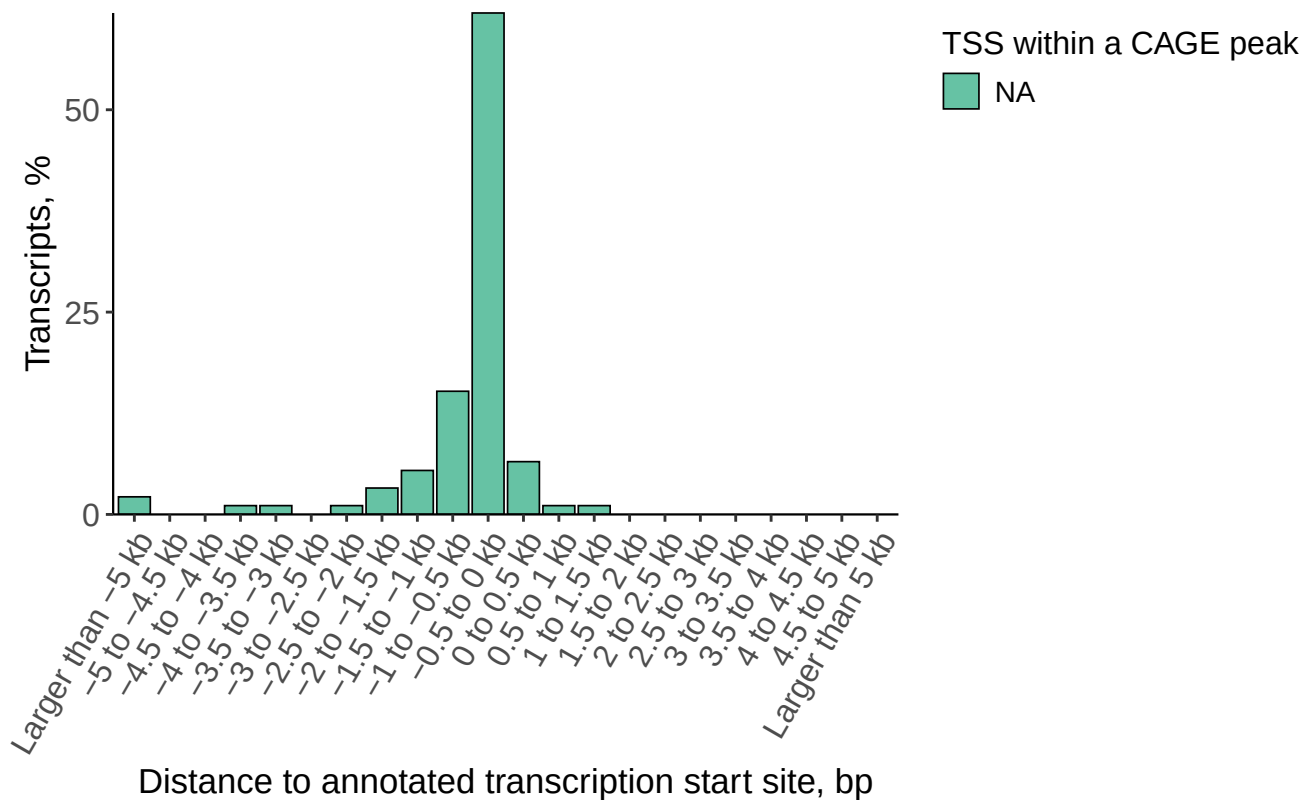
Distance to Annotated Transcription Start Site for ISM

Negative values indicate downstream of annotated TSS



Distance to Annotated Transcription Start Site for ISM

Negative values indicate downstream of annotated TSS

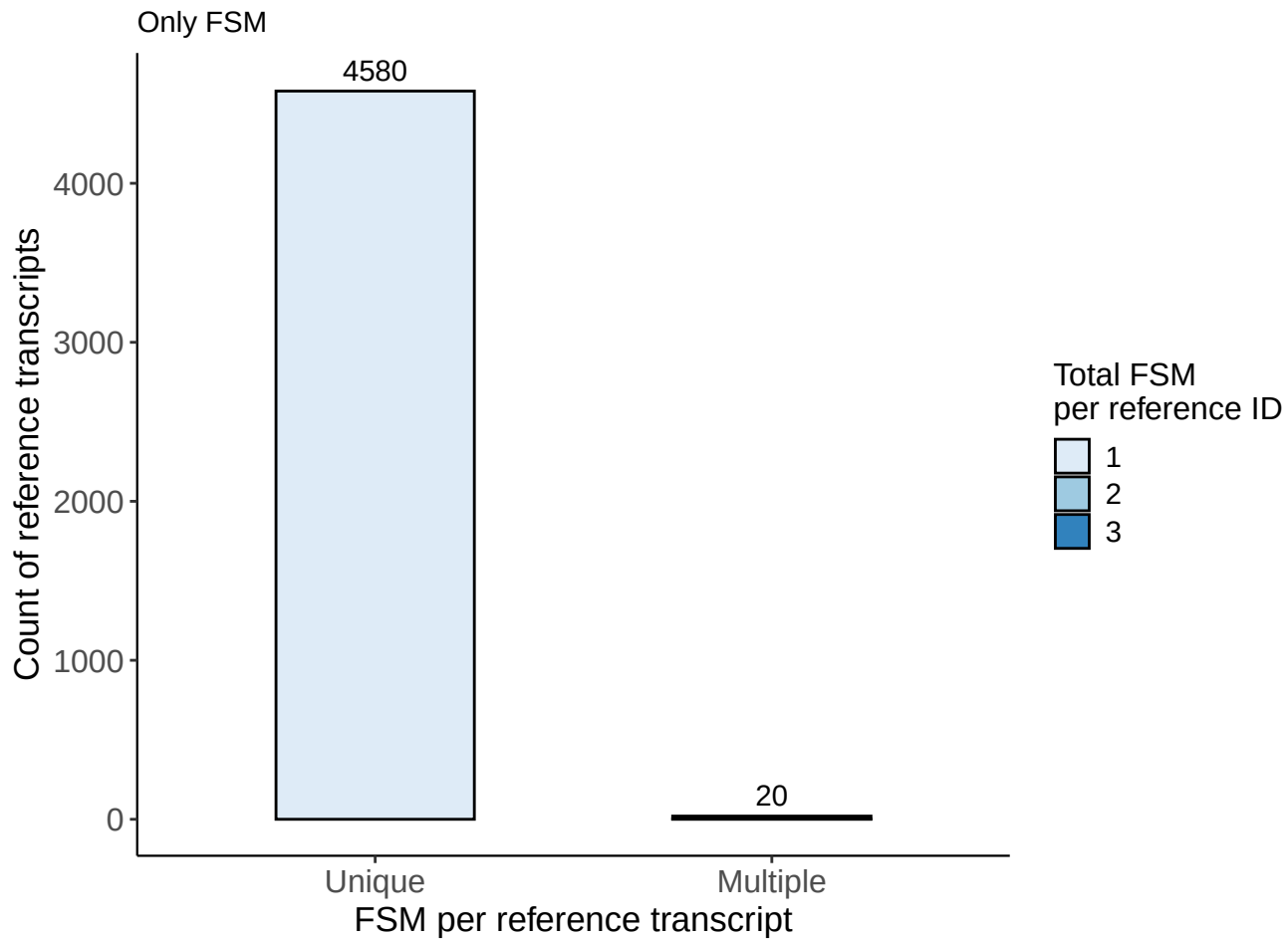


*Comparison With Annotated TSS and TTS
by Subcategories*

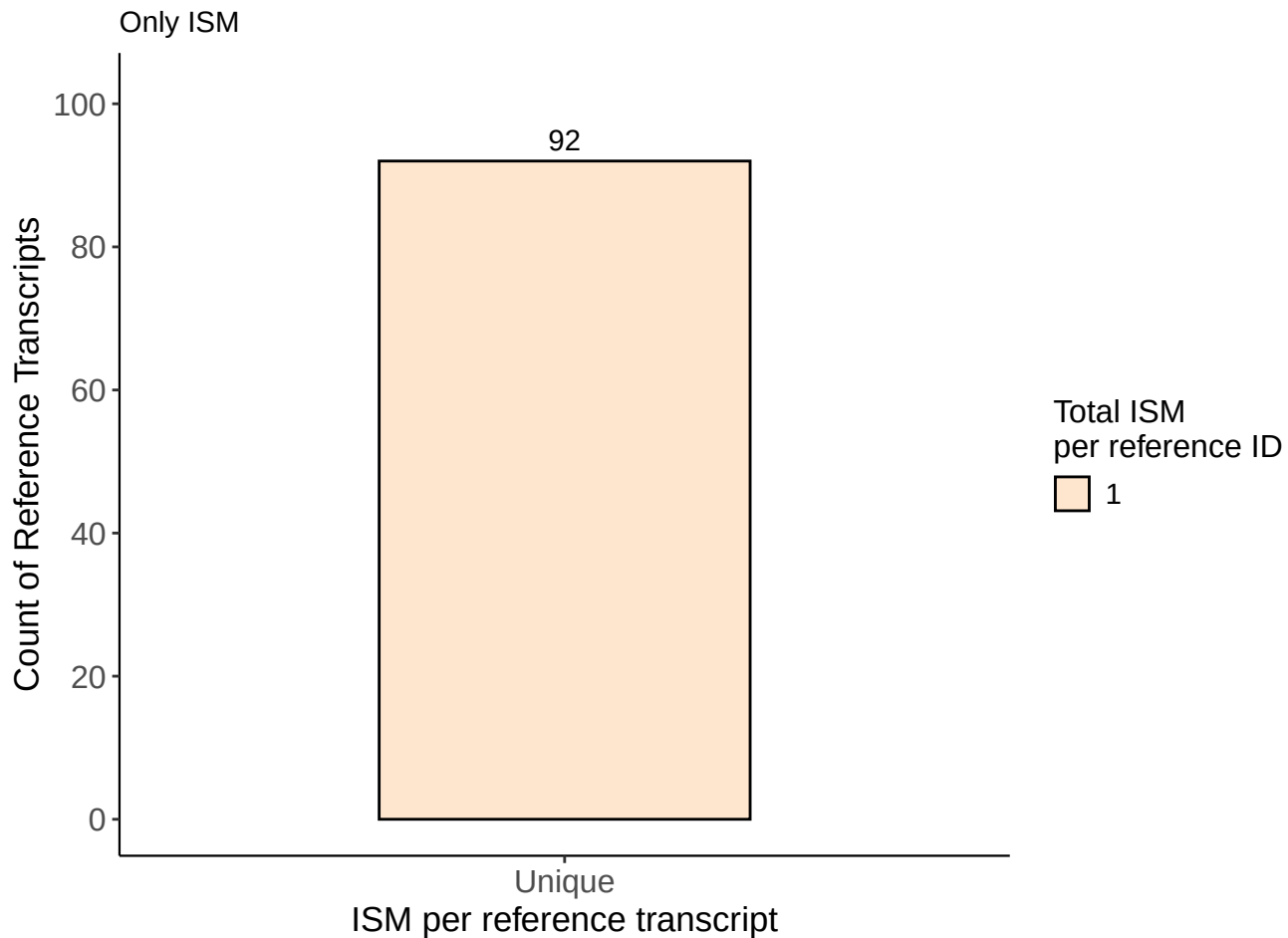
PolyA Distance Analysis

Redundancy Analysis

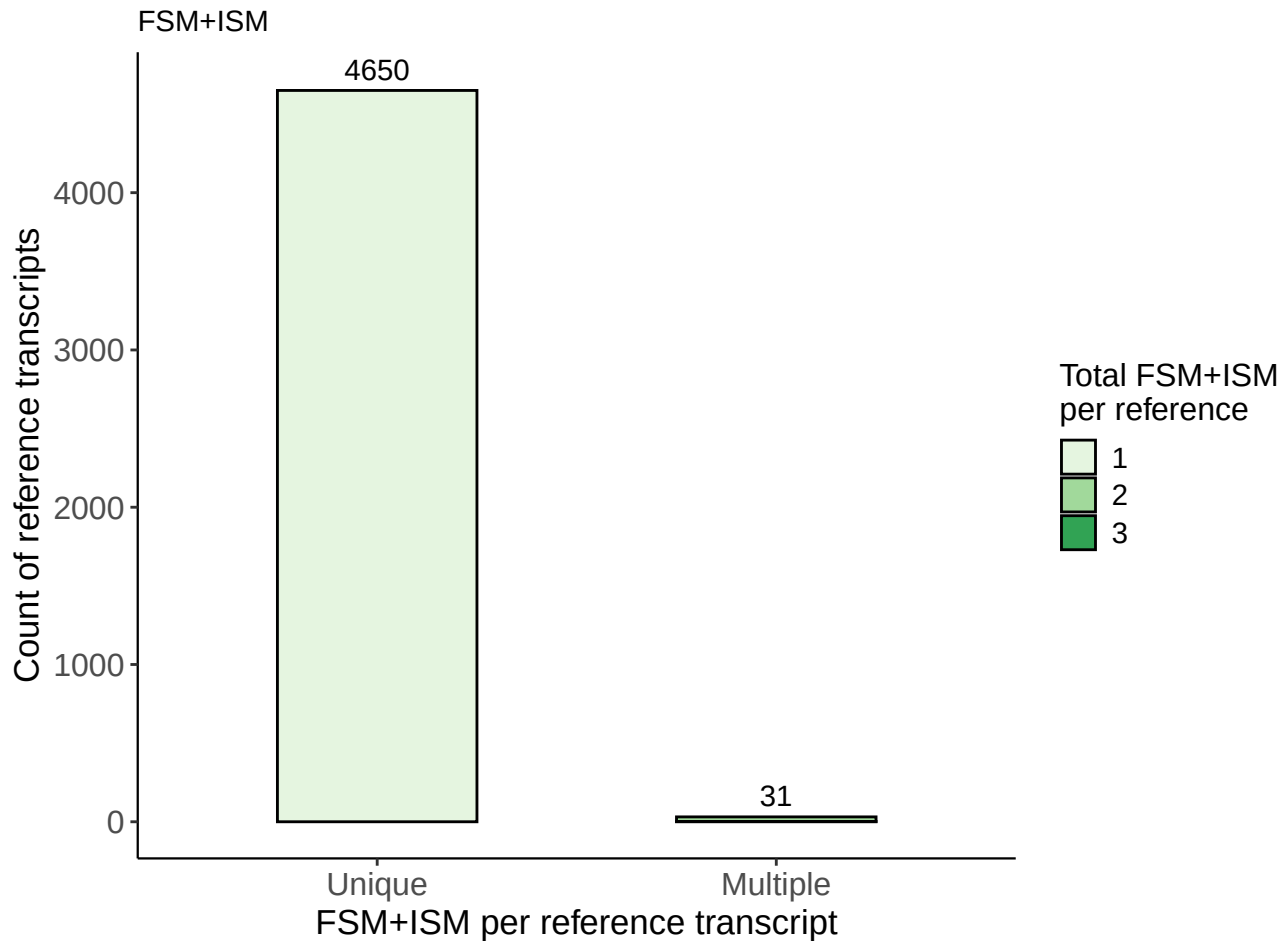
Reference Transcript Redundancy



Reference Transcript Redundancy



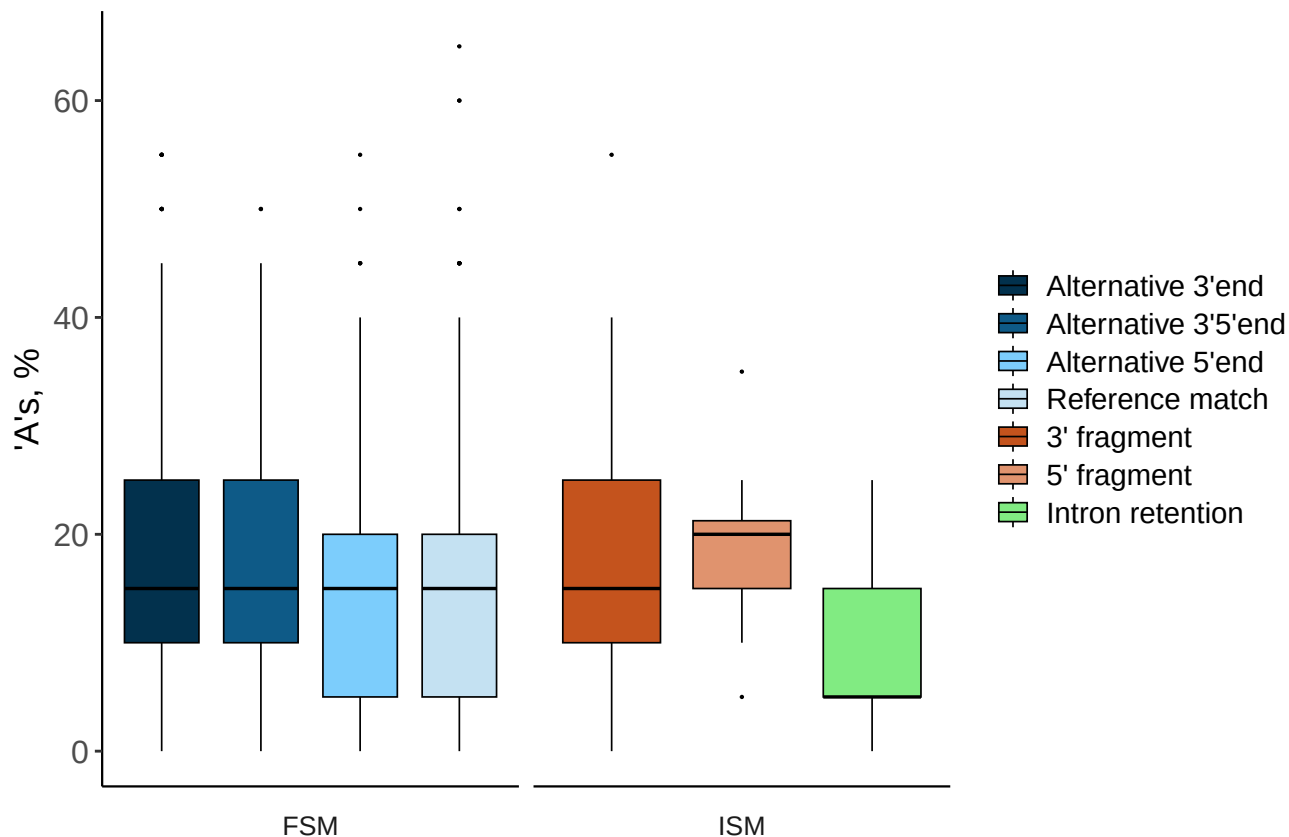
Reference Transcript Redundancy



Intra-Priming Quality Check

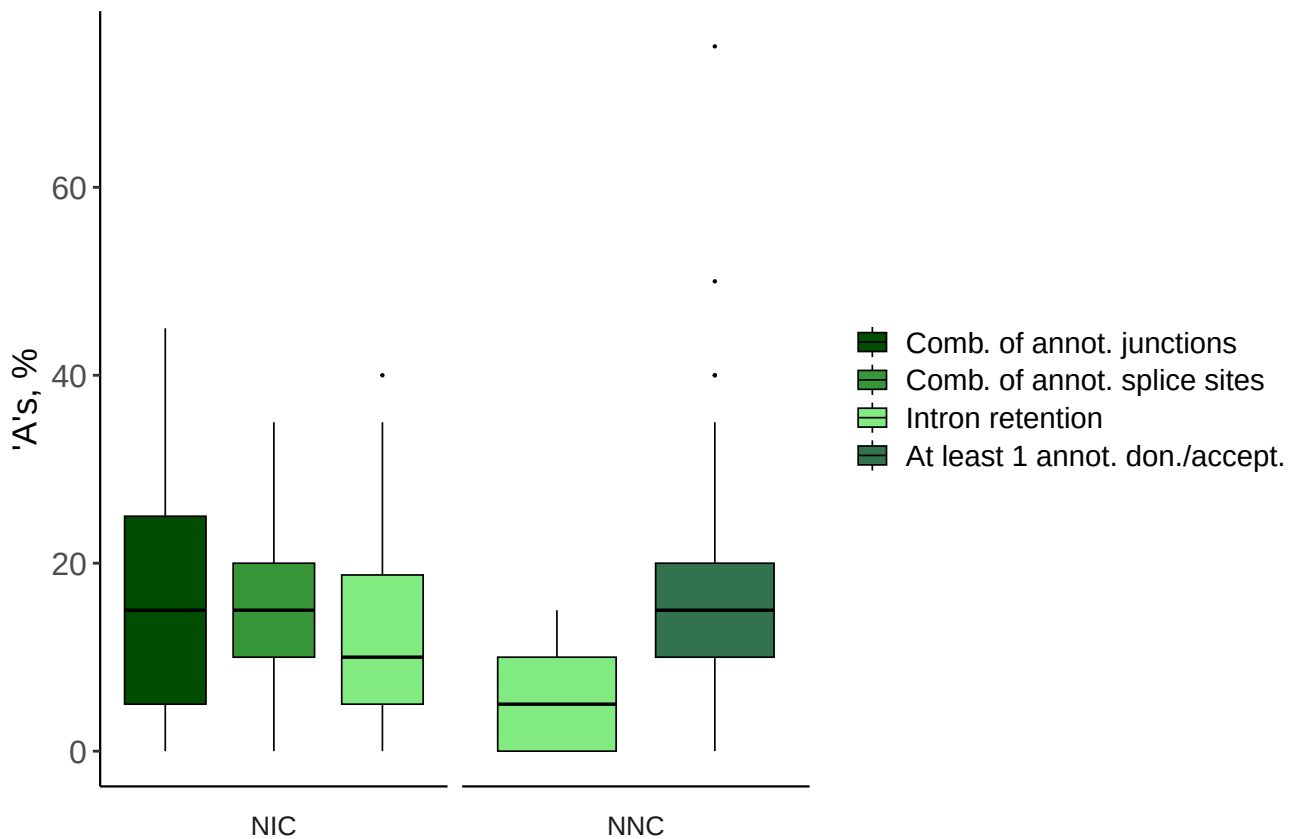
Possible Intra-Priming by Structural Category

Percent of genomic 'A's in downstream 20 bp



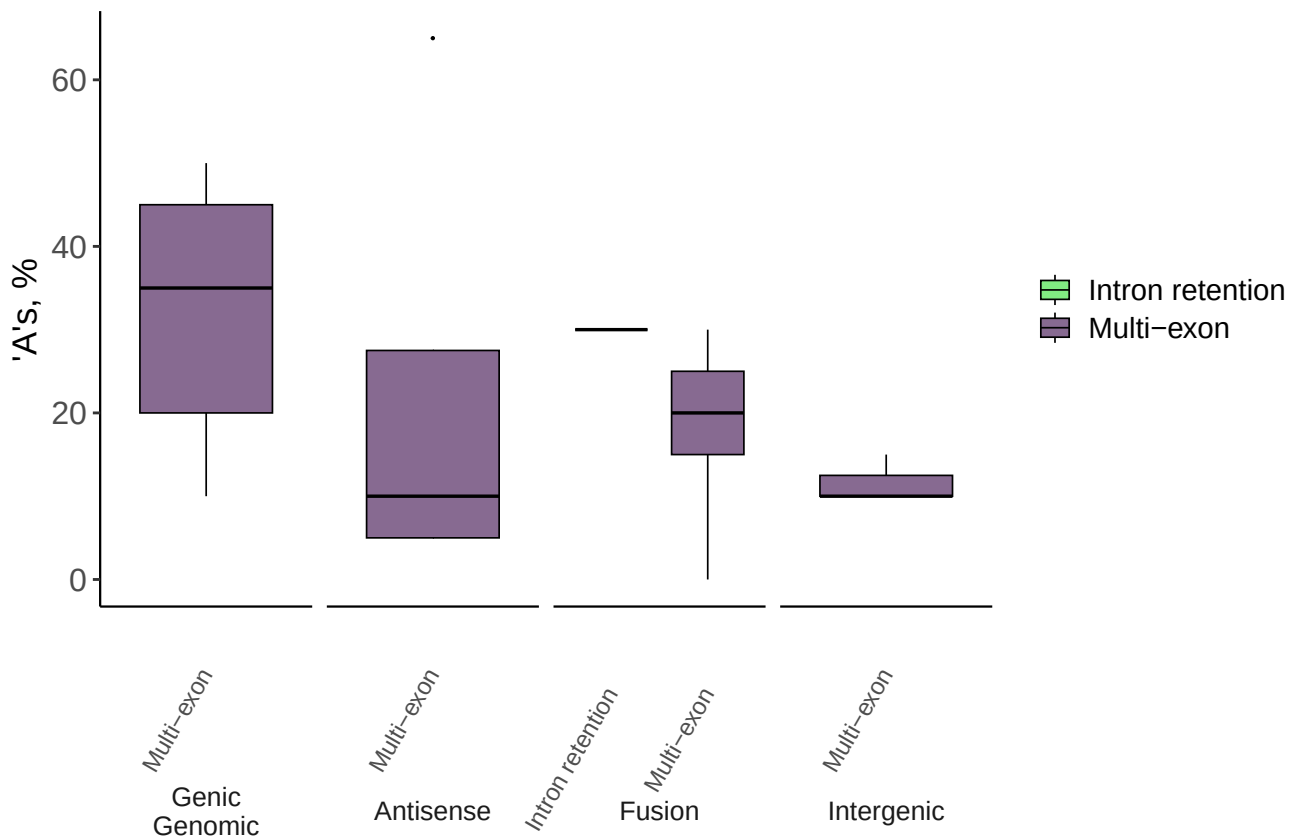
Possible Intra-Priming by Structural Category

Percent of genomic 'A's in downstream 20 bp



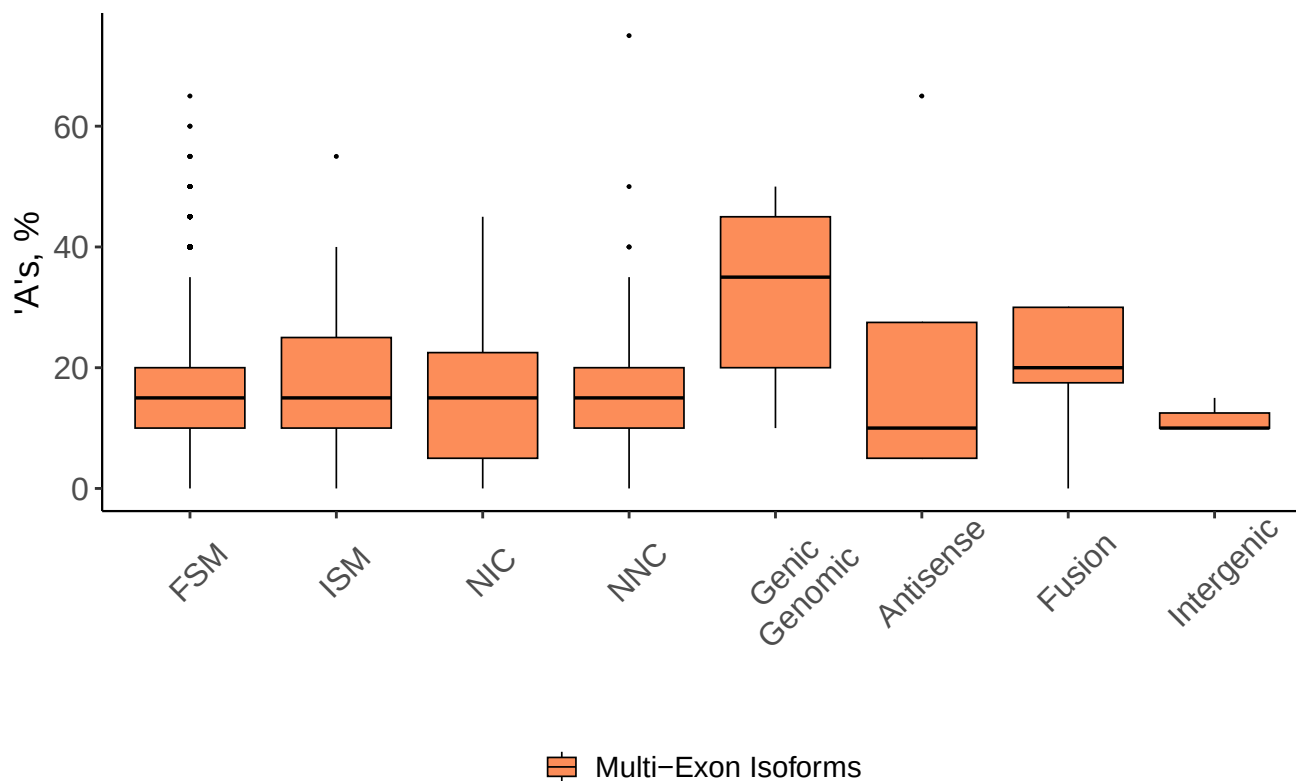
Possible Intra-Priming by Structural Category

Percent of genomic 'A's in downstream 20 bp



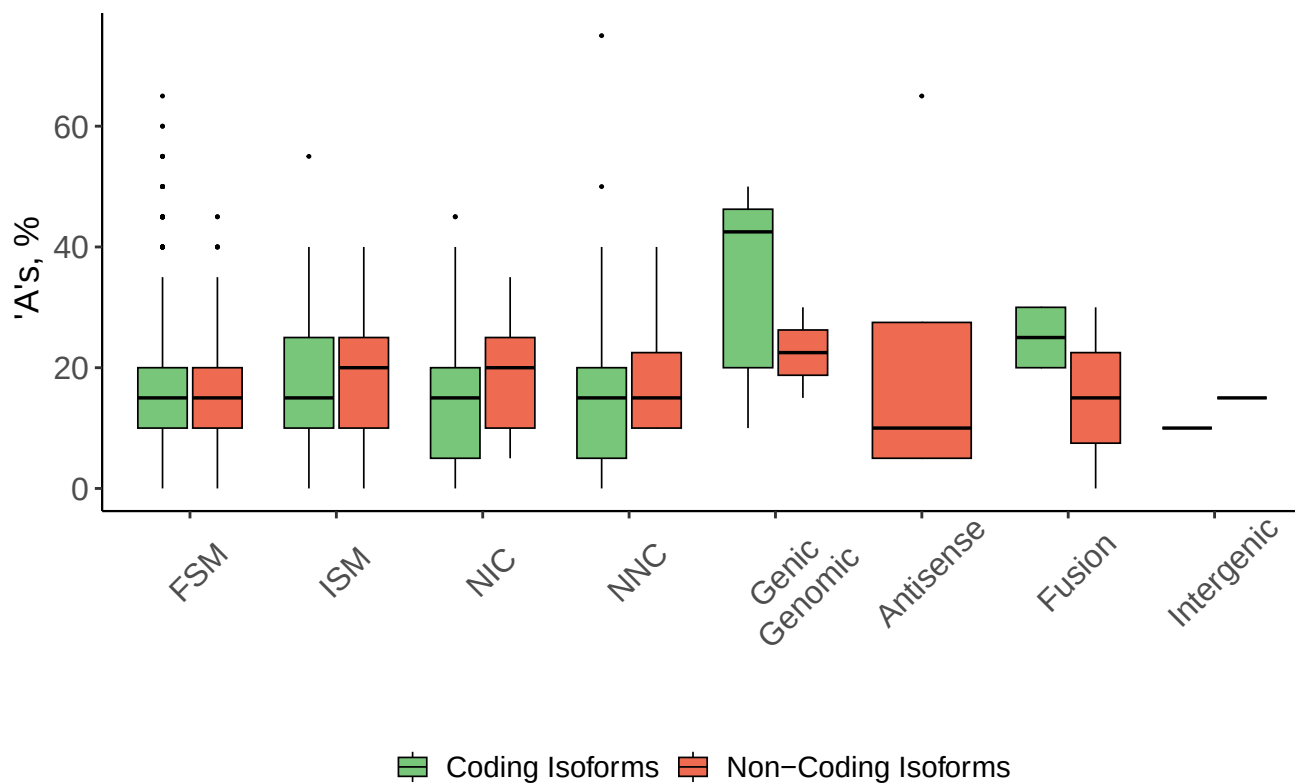
Mono- vs Multi-Exon Possible Intra-Priming

Percent of genomic 'A's in downstream 20 bp



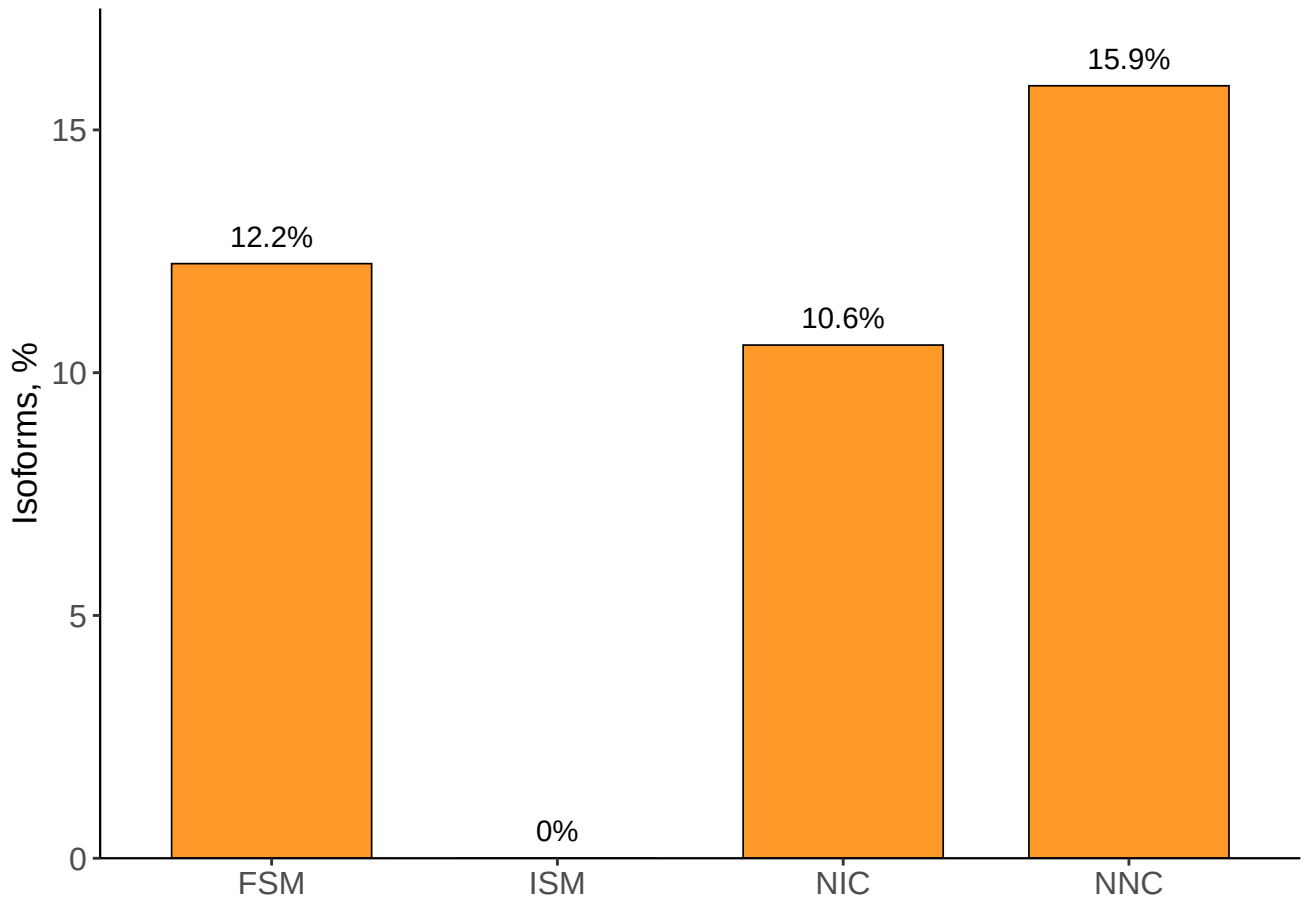
Coding vs Non-Coding Possible Intra-Priming

Percent of genomic 'A's in downstream 20 bp

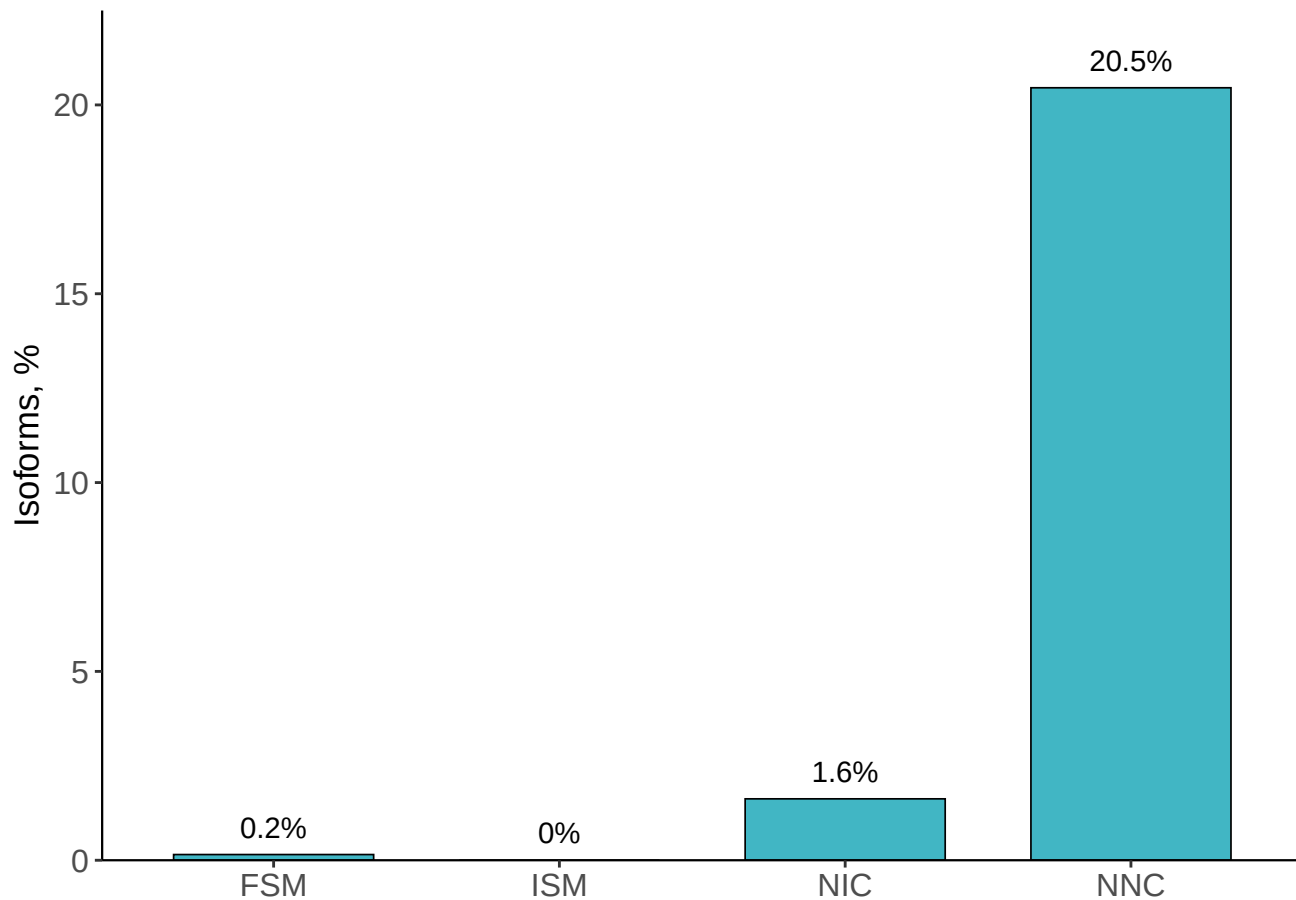


Features of Bad Quality

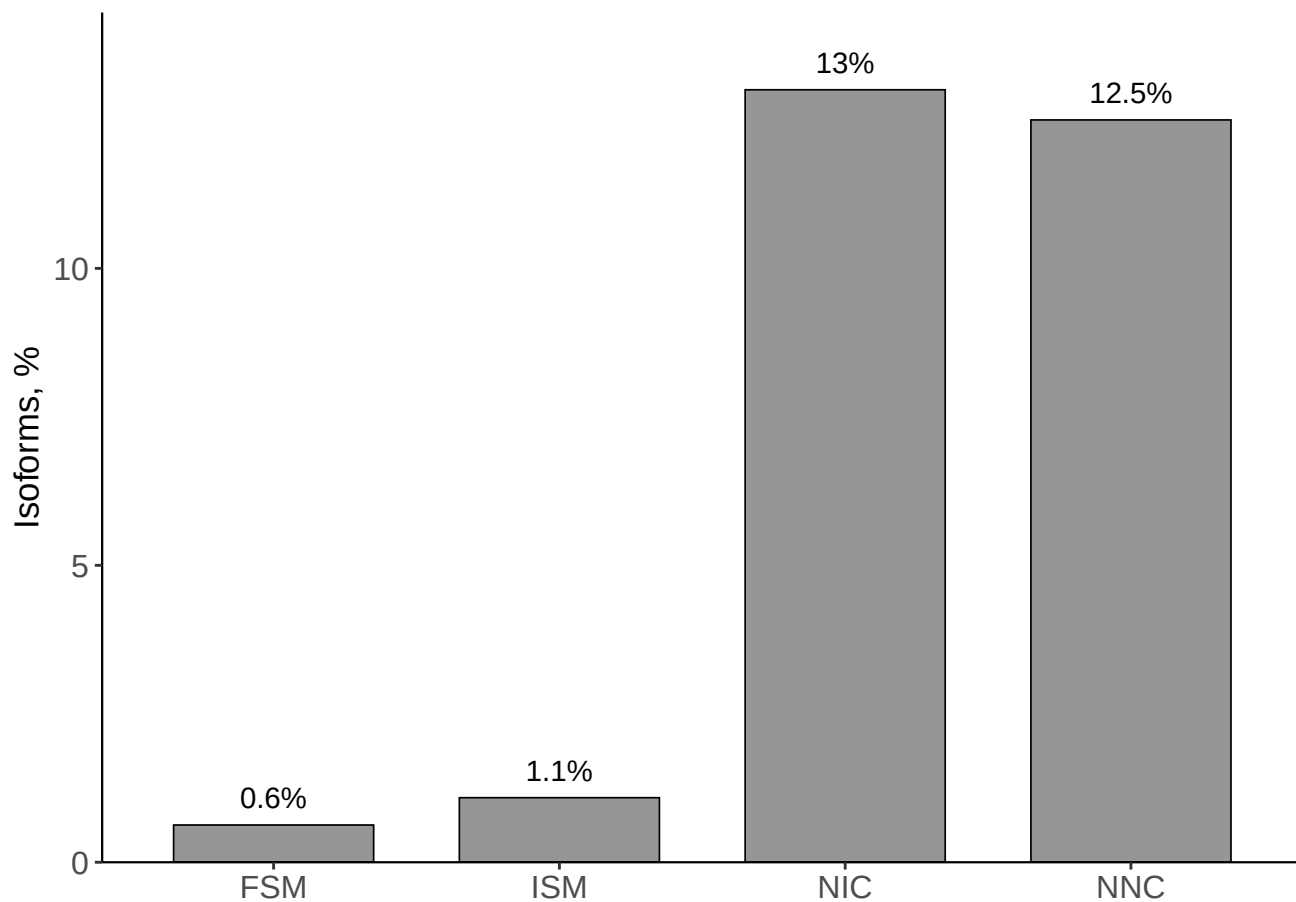
RT-switching



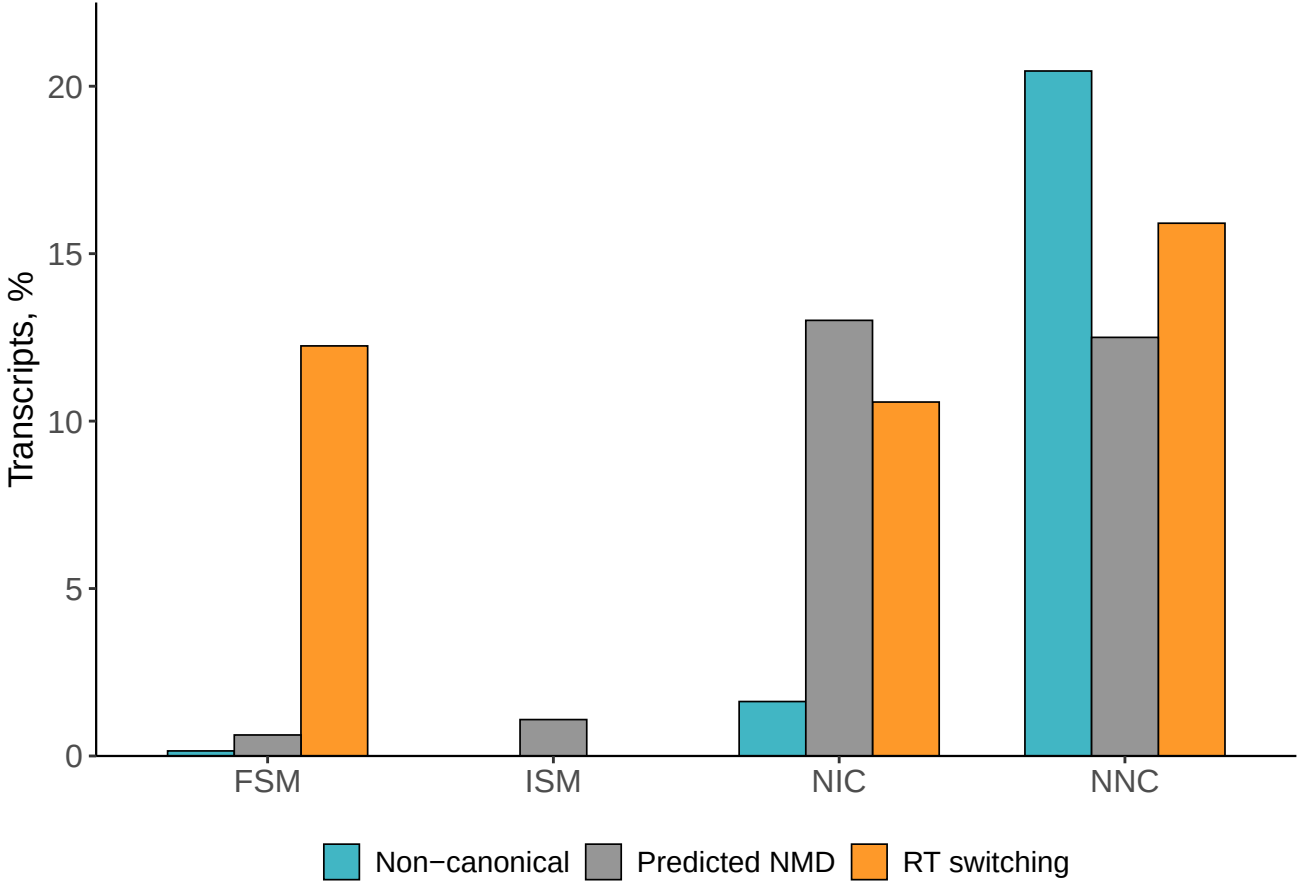
Non-Canonical Junctions



Nonsense-Mediated Decay by Structural Category

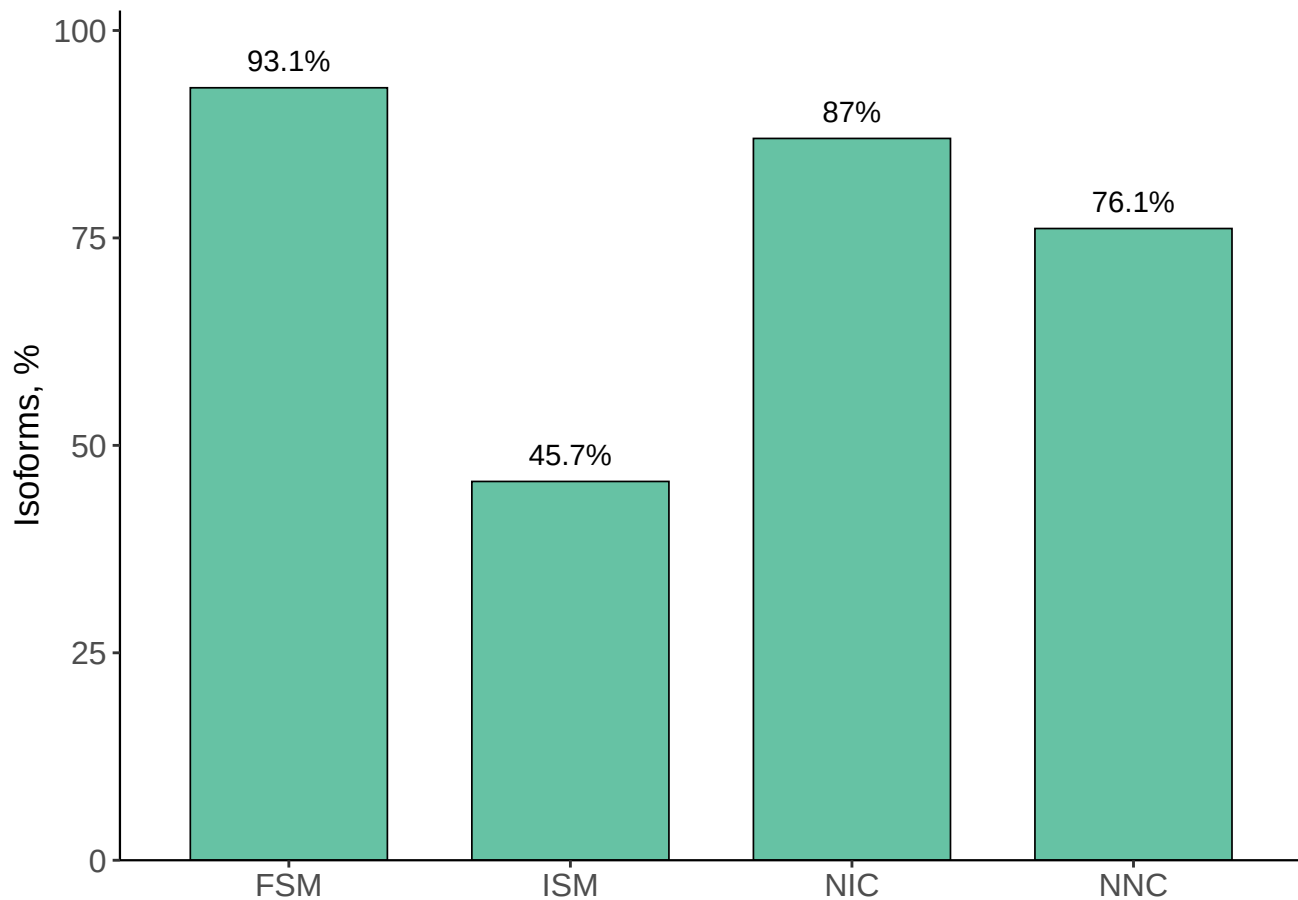


Quality Control Attributes Across Structural Categories

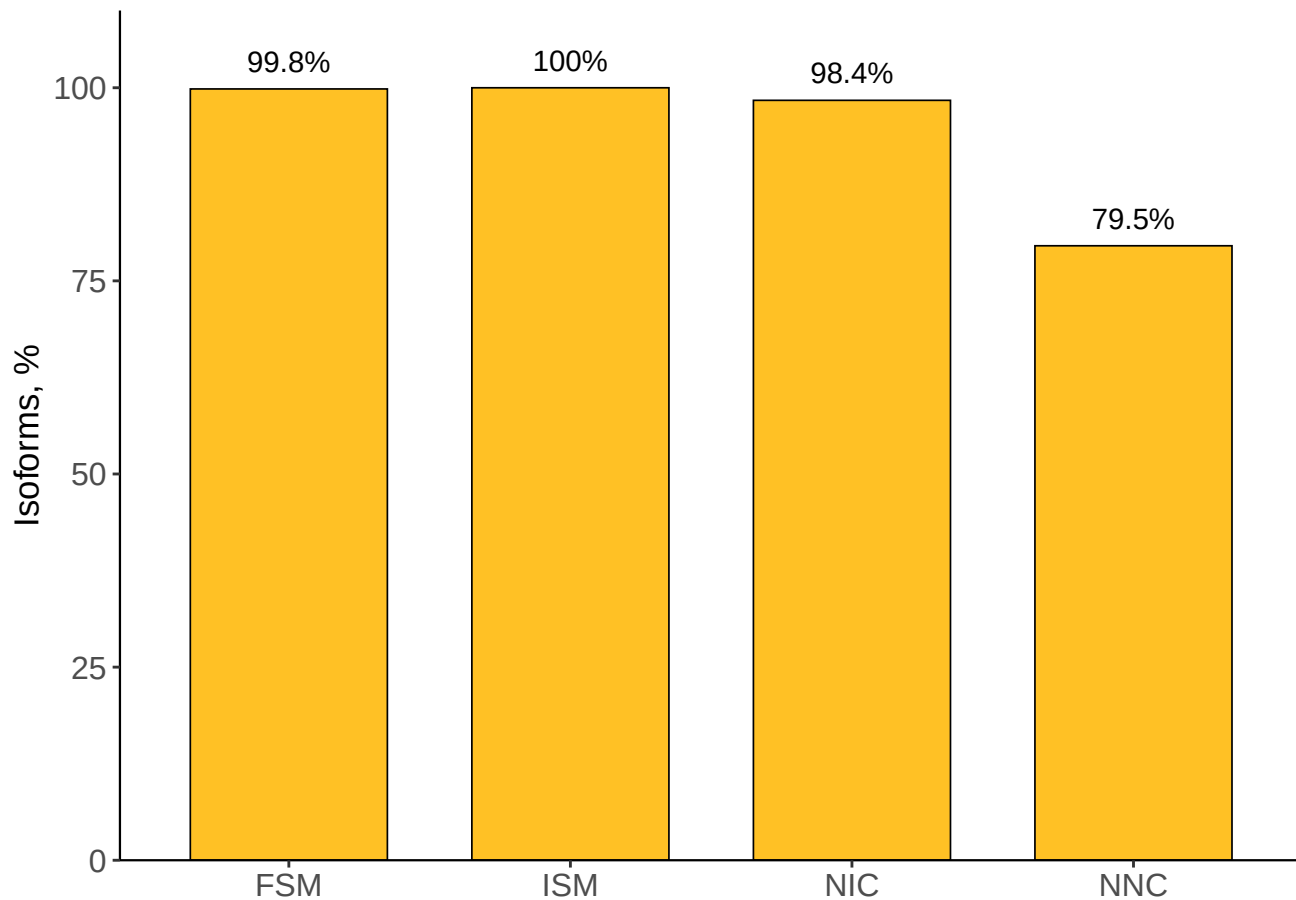


Features of Good Quality

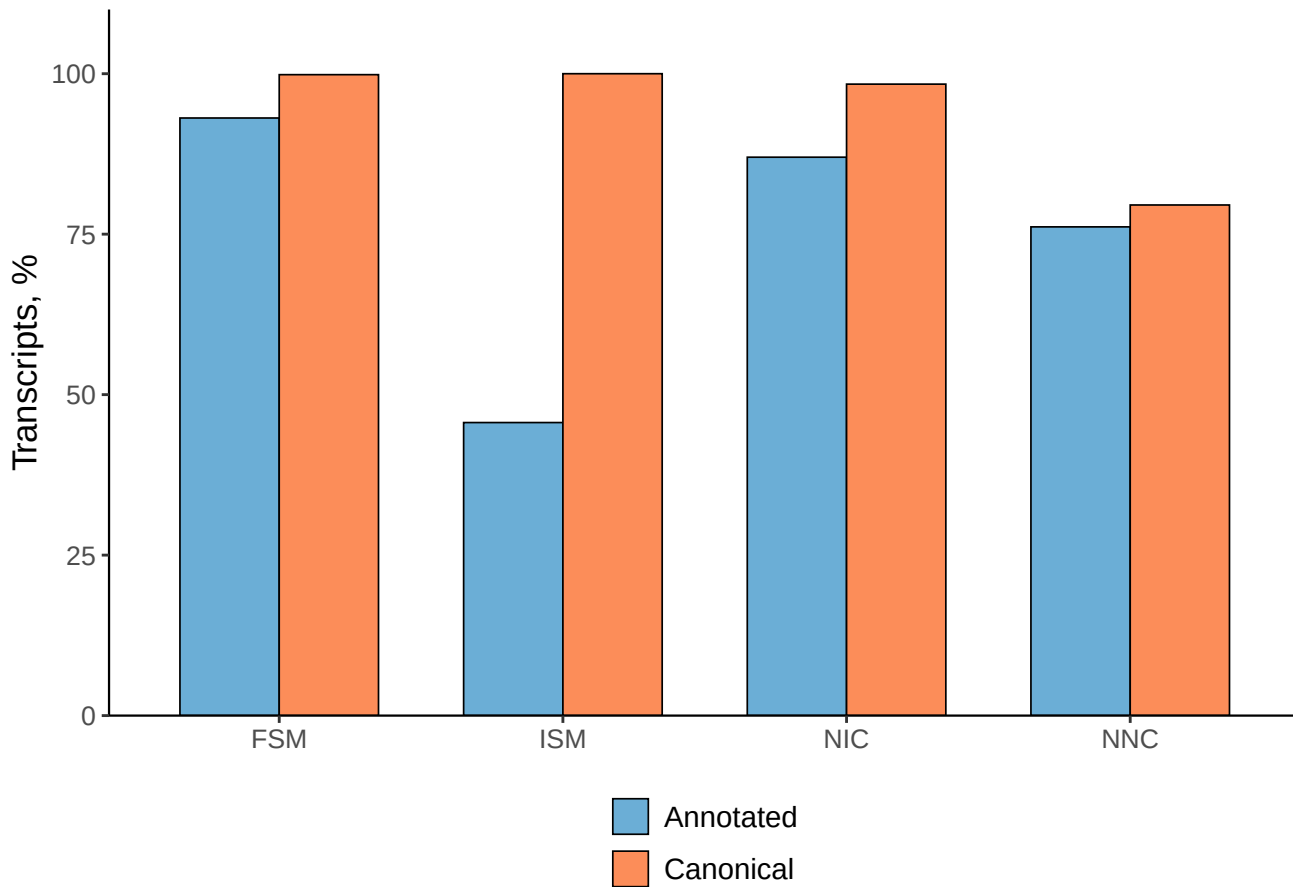
Annotation Support



All Canonical Junctions

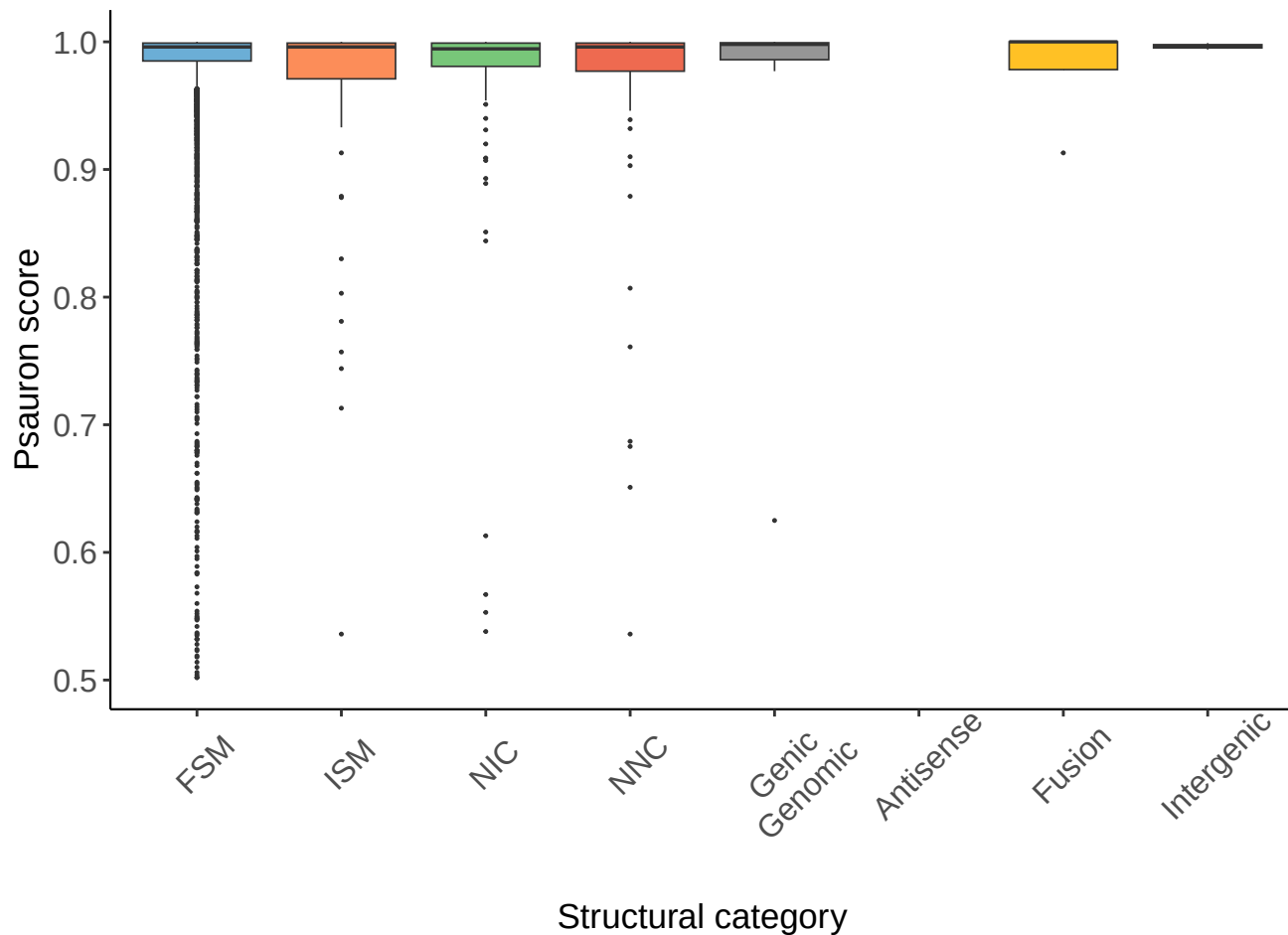


Good Quality Control Attributes Across Structural Categories

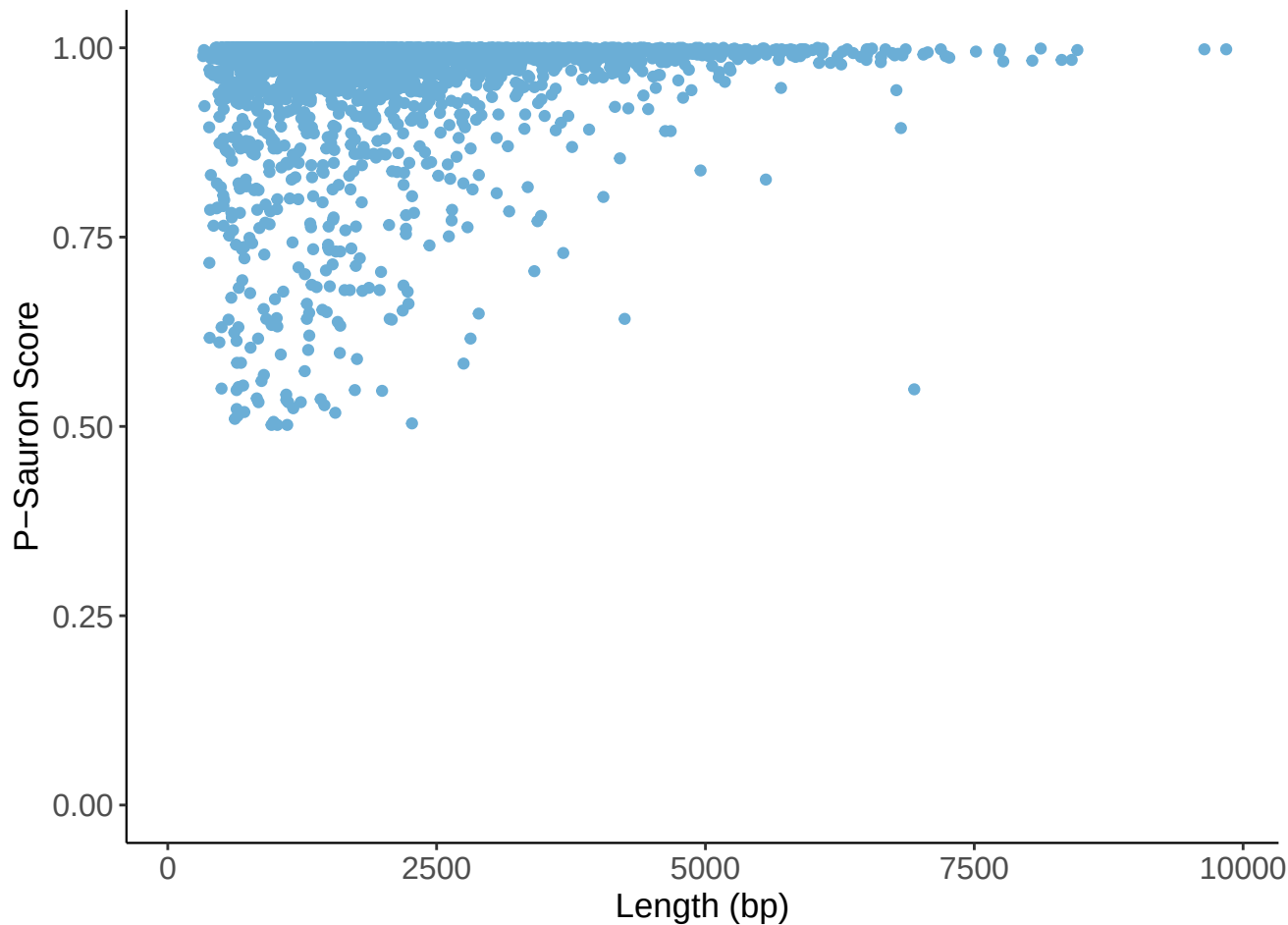


TransDecoder2 CDS prediction statistics

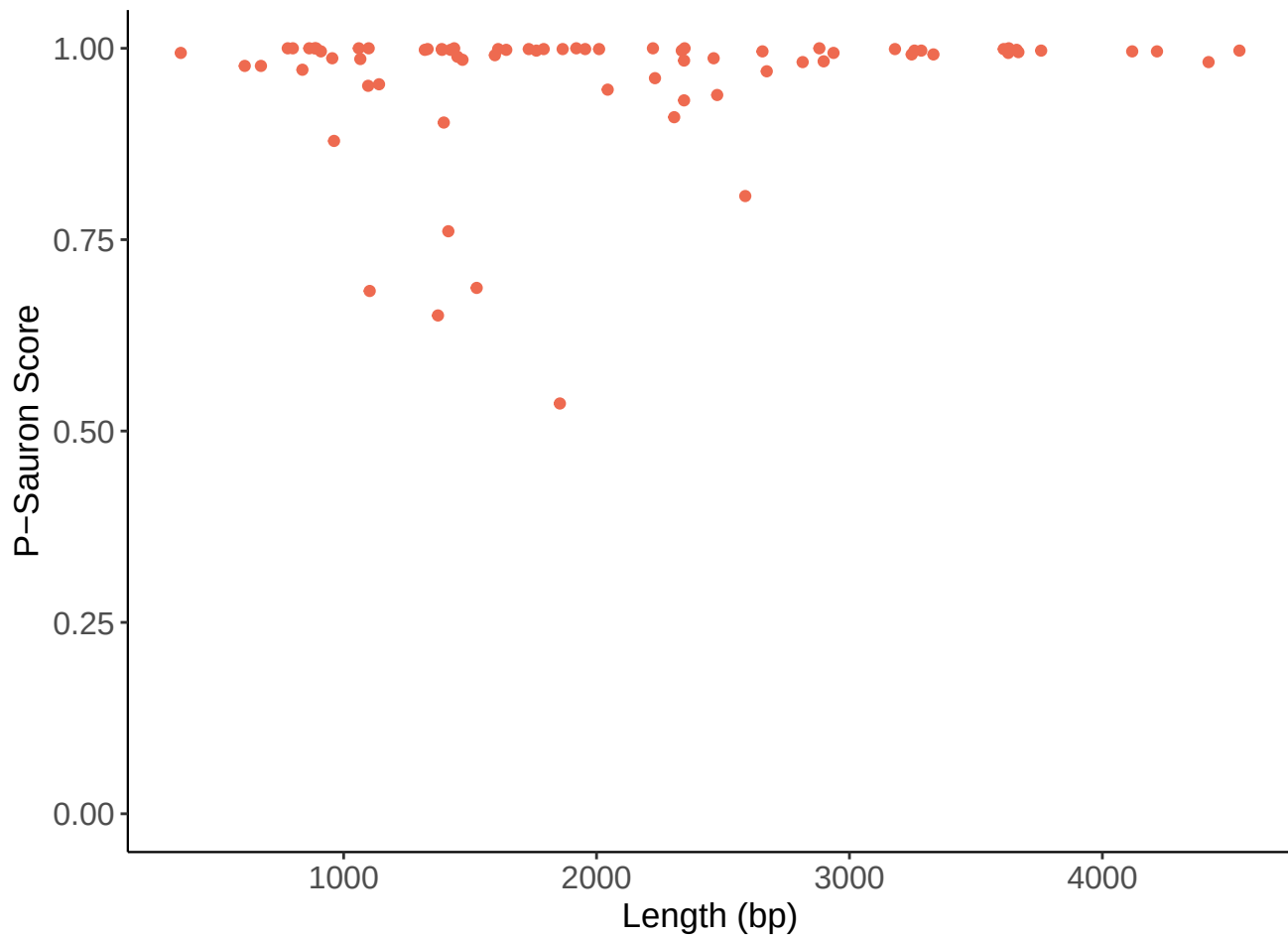
Psauron score per structural category



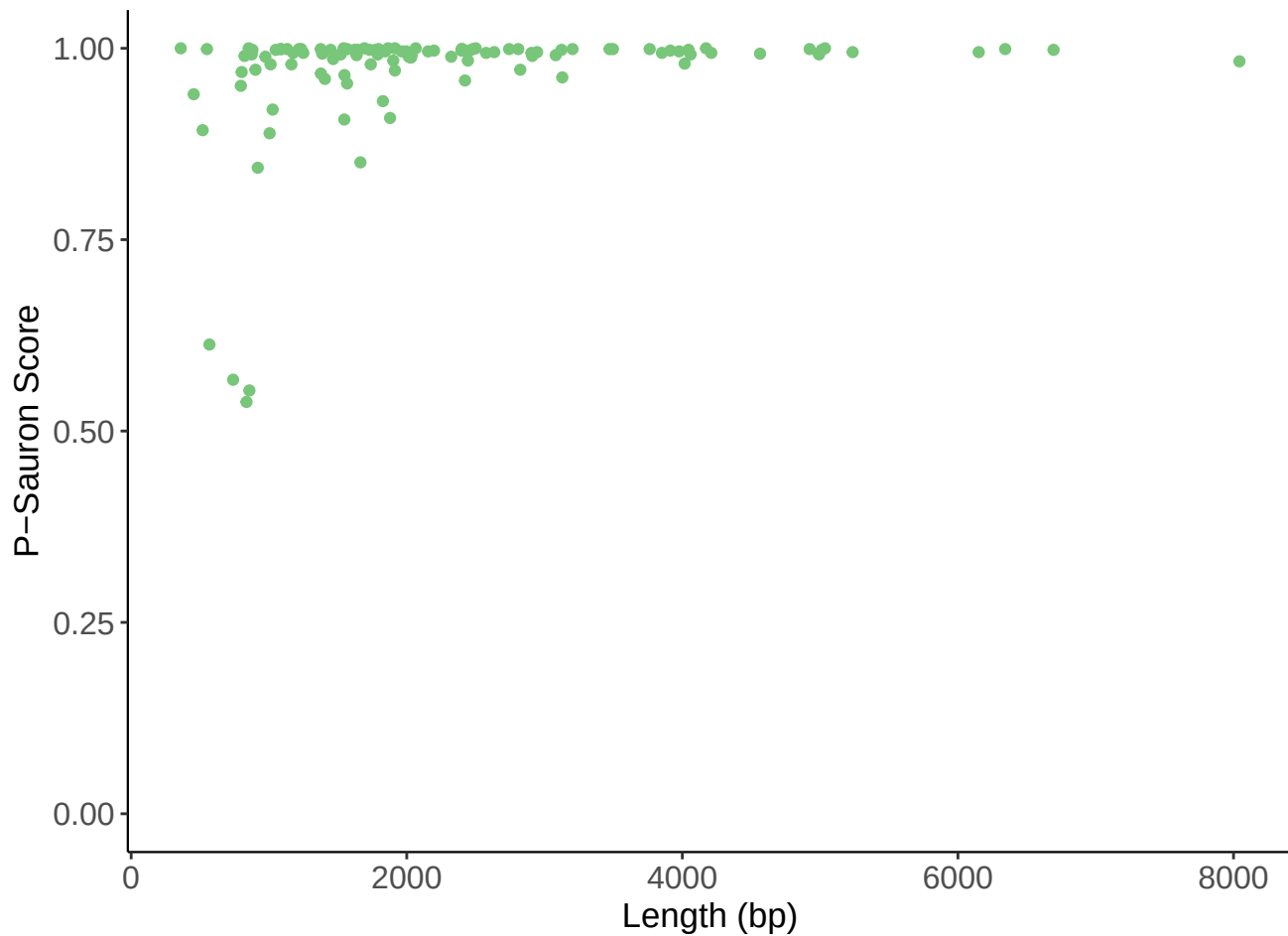
P-Sauron Score vs Length (FSM)



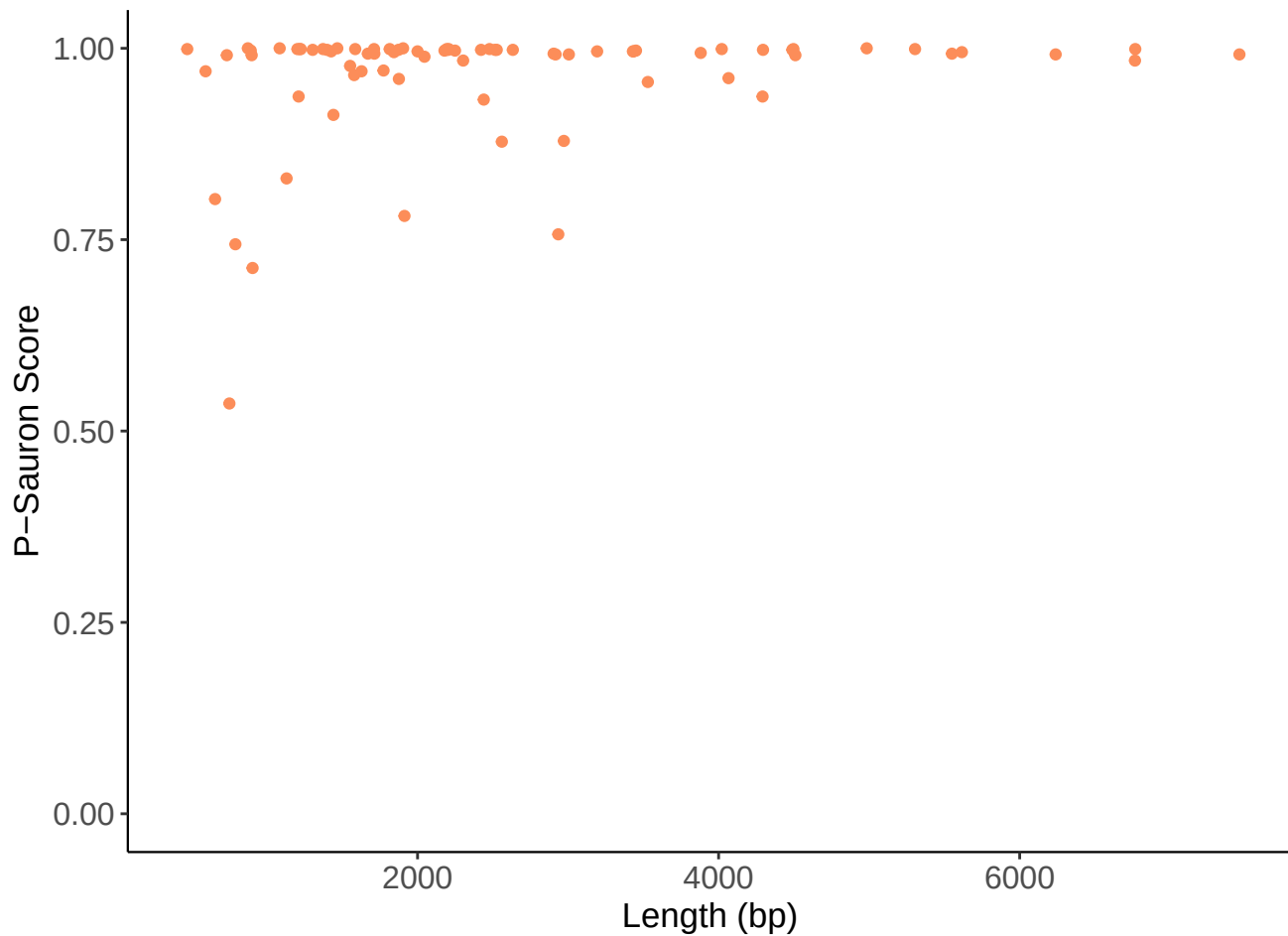
P-Sauron Score vs Length (NNC)



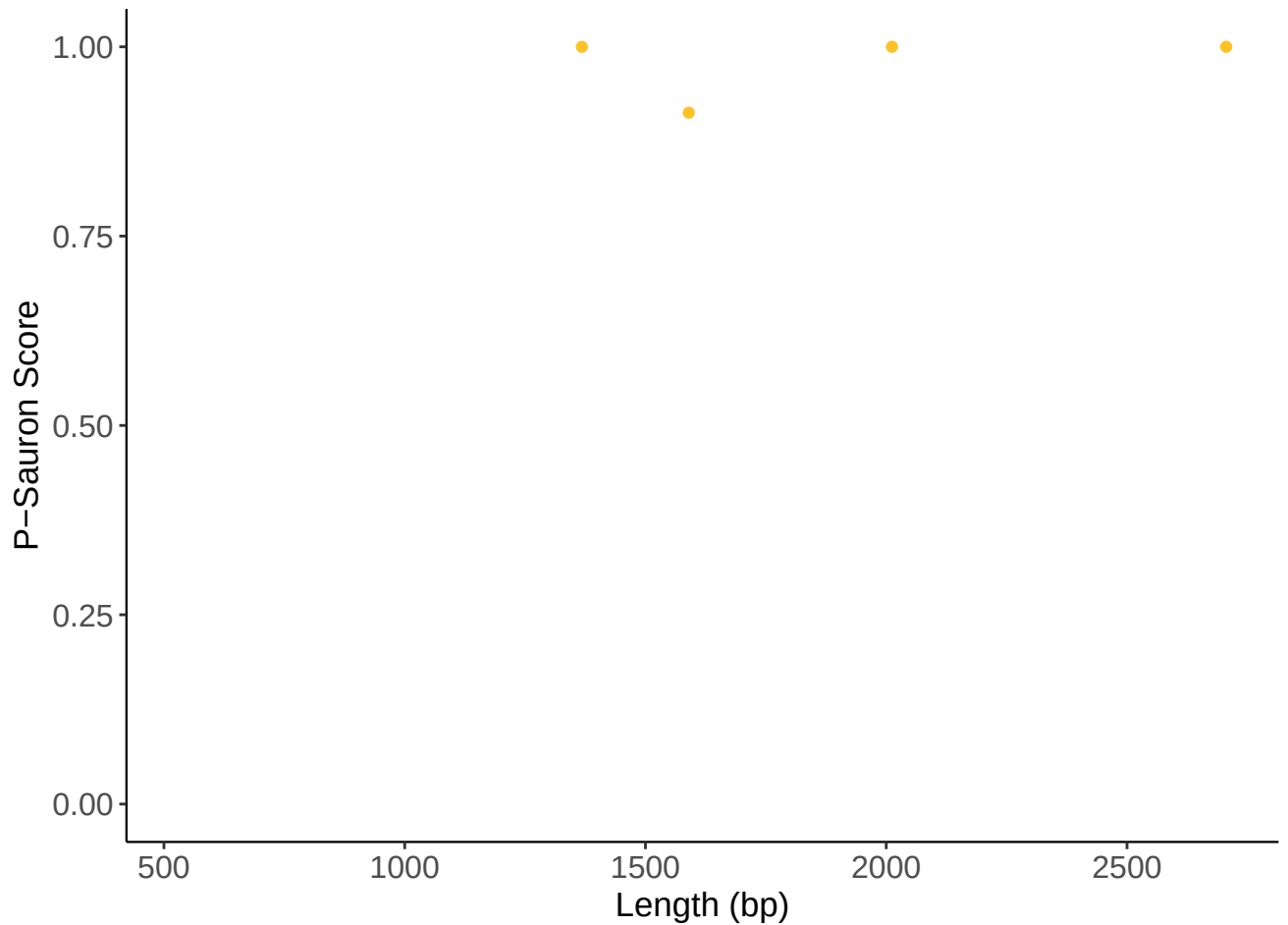
P-Sauron Score vs Length (NIC)



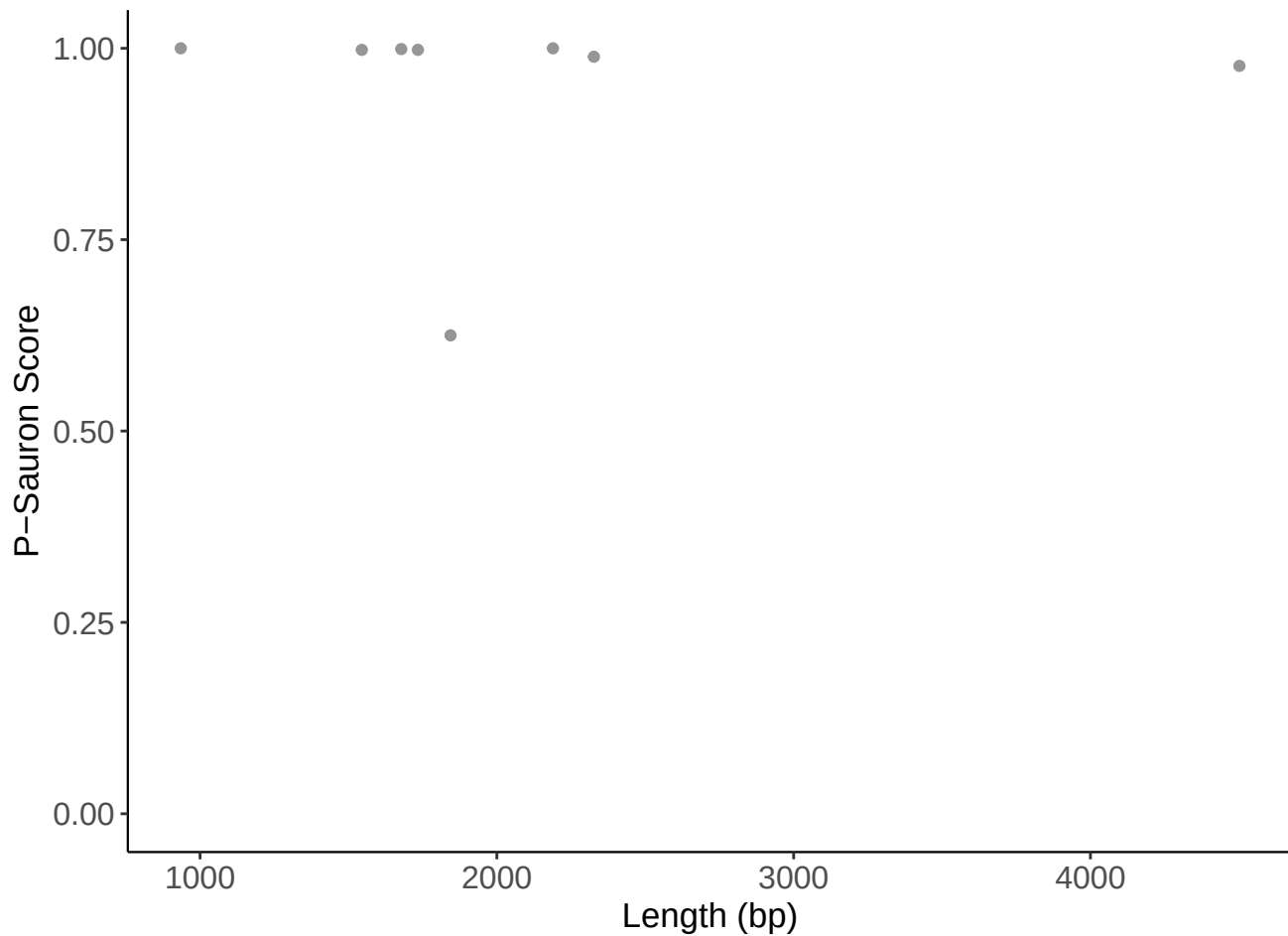
P-Sauron Score vs Length (ISM)



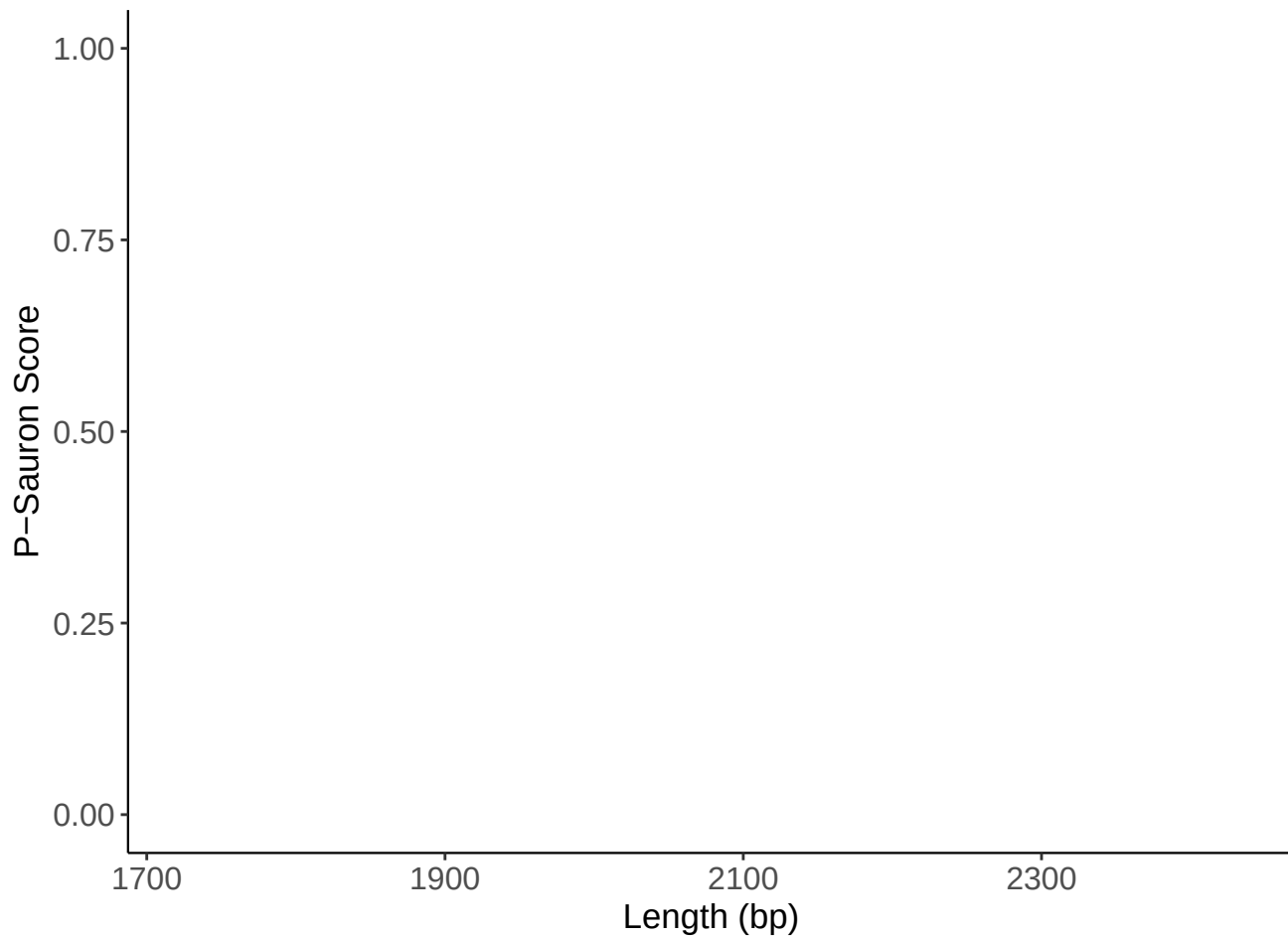
P-Sauron Score vs Length (Fusion)



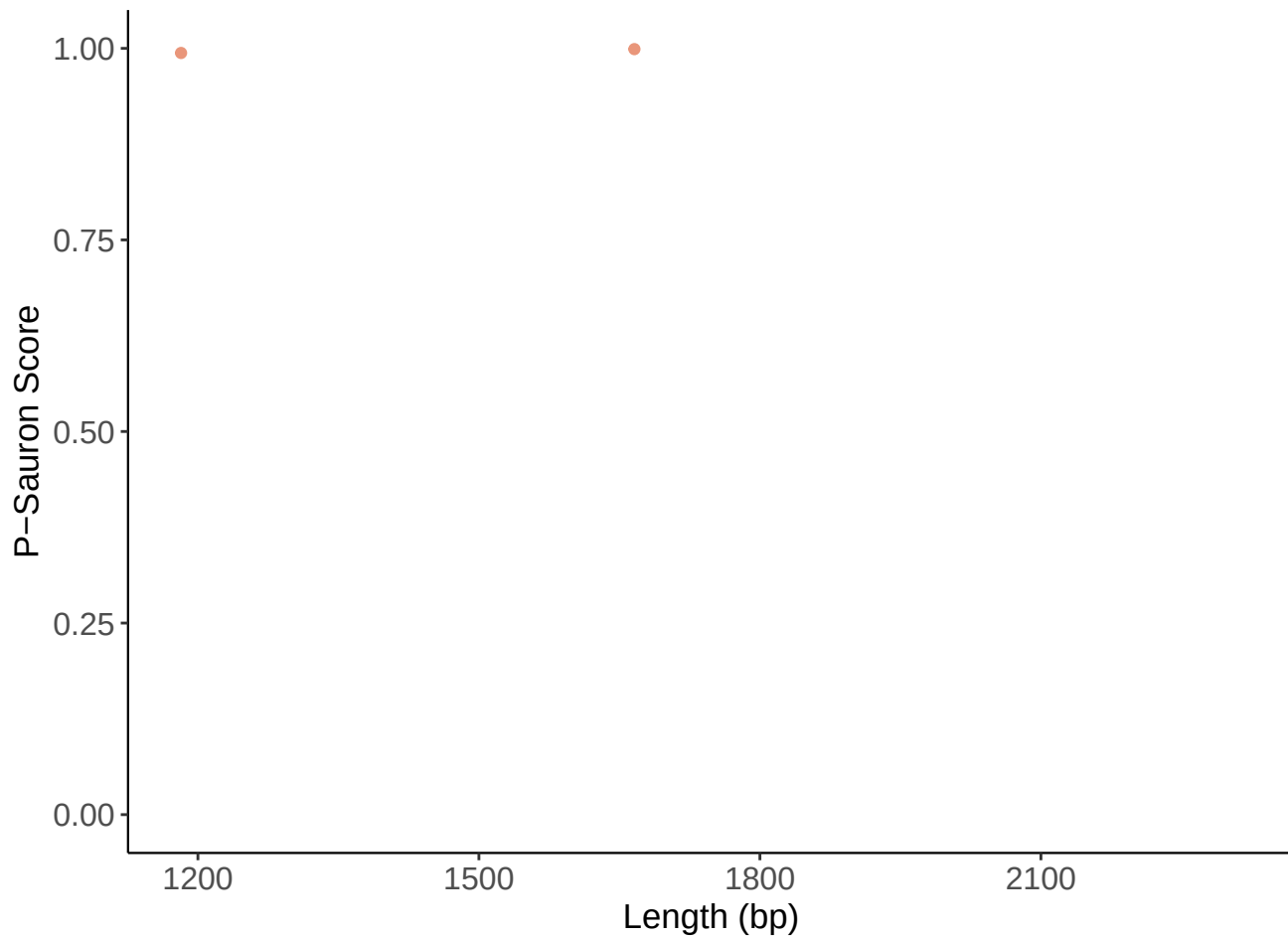
P-Sauron Score vs Length (Genic Genomic)



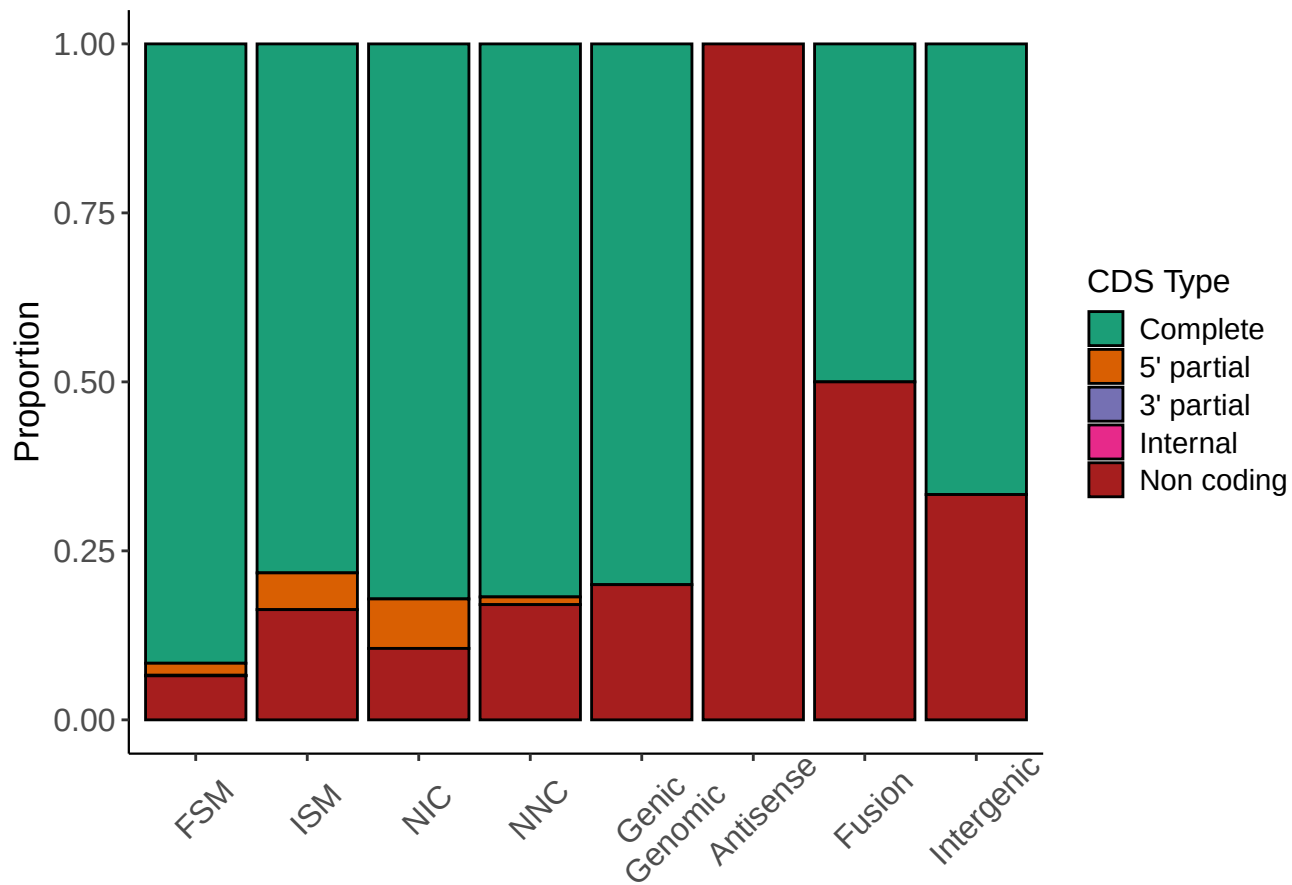
P-Sauron Score vs Length (Antisense)



P-Sauron Score vs Length (Intergenic)



CDS type by subcategory



Psaaron score per ORF type

